1 AWS Certified Solutions Architect – Associate (SAA-C03)

1.1 Complete Exam Preparation Study Guide

Prepared from the supplied AWS SAA-C03 Exam Guide, current AWS documentation, and the supplementary migration/hybrid presentation.

> This guide is a preparation aid, not a substitute for the current AWS exam guide, AWS documentation, hands-on practice, or the official practice exam. AWS service behavior, names, quotas, and pricing can change; verify current details before exam day.

1.2 How to use this guide

Read the orientation and domain notes once, then work through the service decision chapters and architecture patterns. Use the lab checklists to turn memorized facts into working intuition. Finally, take the practice questions under time pressure, record the requirement you missed, and revisit the relevant section. The exam rewards choosing the best design under constraints rather than naming the most services.

1.3 Contents

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2 AWS Certified Solutions Architect – Associate (SAA-C03)

2.1 Exam structure, Well-Architected Framework, and study strategy

Scope of this section. This report covers the SAA-C03 exam contract, how the AWS Well-Architected Framework (the Framework) is used as the design lens, scenario-reasoning patterns, and a practical study strategy. It uses the supplied official exam guide plus current AWS documentation. It is intentionally not a service-by-service study guide and does not claim exhaustive coverage of the exam.

2.2 1. What the exam is testing

SAA-C03 is a solutions-architecture judgment exam. AWS describes it as validating the ability to design solutions based on the AWS Well-Architected Framework, design for current and projected business needs,

produce secure/resilient/high-performing/cost-optimized architectures, and review existing designs for improvements. The target candidate description recommends at least one year of hands-on experience designing cloud solutions with AWS services. 1 3

The key implication is that the exam is not primarily a memorization test of service definitions. A correct answer must satisfy the stated business and technical constraints, and it must usually be the **best fit** among several plausible AWS designs. A service can be technically capable of a task and still be the wrong answer because it is too operationally heavy, violates a security boundary, cannot meet an RTO/RPO, creates a single point of failure, or costs more than the requirement justifies.

The exam's current official overview lists a 130-minute duration, 65 questions, multiple-choice or multiple-response format, and a USD 150 price (with market and foreign-exchange details handled by AWS's pricing information). 2 The supplied exam guide and current SAA-C03 documentation identify 50 scored questions and 15 unscored questions; unscored questions are not identified during the exam. 1 3

2.2.1 Exam contract at a glance

- **Question count:** 65 total; 50 affect the score and 15 are unscored evaluation items. 1 2 3
- **Time:** 130 minutes. 2
- **Response types:** Multiple choice has one correct response and three distractors. Multiple response has two or more correct responses among five or more options. 1 3
- **Scoring:** Reported on a 100–1,000 scaled score; the minimum passing score is 720. The exam uses compensatory scoring, so a candidate does not have to pass every domain separately. 1 3
- **Unanswered questions:** Count as incorrect; AWS states there is no penalty for guessing. 1 3
- **Language and service names:** AWS's current exam page notes that short service names may be used and that a list of short names and corresponding full names is available through the exam Help button. 2
- **Experience expectation:** At least one year of hands-on AWS solution-design experience is recommended, but AWS describes the certification as a starting point for people with cloud or strong on-premises IT experience. 2 3

The 130 minutes average to about two minutes per question. That is a planning aid, not a rule: short recall questions should be answered quickly so that long scenario questions receive enough reading and comparison time. Because unanswered items are incorrect, leave time to select an answer for every question.

2.2.2 Scored domains and weights

The official exam guide assigns the scored content as follows. 1

Domain	Weight	What to reason about
1. Design Secure Architectures	**30%**	Identity and access, network segmentation, application/data protection, encryption, shared responsibility, and secure connectivity
2. Design Resilient Architectures	**26%**	Loose coupling, scaling, high availability, fault tolerance, backups, disaster recovery,

Domain	Weight	What to reason about
		failover, and recovery objectives
3. Design High-Performing Architectures	**24%**	Storage, compute, databases, networking, ingestion, transformation, caching, and elastic capacity
4. Design Cost-Optimized Architectures	**20%**	Cost-aware storage, compute, databases, networking, purchasing, lifecycle, transfer, and right-sizing

The weights describe scored-content proportions; they are not a promise of an exact number of questions on a particular form. The four domains are related but not interchangeable. For example, a multi-AZ database design may be resilient, but the answer must still respect the requested write/read pattern, performance, security, and cost constraints.

2.2.3 Current scope and terminology discipline

Use the current exam guide and current AWS service documentation as the authority when a training slide or older course uses a legacy name. For example, the current SAA-C03 domain material uses **AWS IAM Identity Center**, whereas older material may say **AWS Single Sign-On (AWS SSO)**. 4 The exam guide's service examples are examples of knowledge and skills, not a complete list of every possible question or service. The current exam-guide site also separates in-scope and out-of-scope service lists; use those lists when deciding how far to expand study beyond the core architecture patterns. 3

The supplied supplementary slide deck is primarily a migration and hybrid-architecture lecture. It is useful for background examples such as hybrid connectivity and migration choices, but it should not override the current exam guide or live AWS documentation for names, availability, or exam mechanics. Avoid carrying forward retired or renamed service claims without checking current AWS documentation.

2.3 2. AWS Well-Architected Framework as the design lens

AWS defines the Well-Architected Framework as a consistent way to evaluate architectures against cloud best practices, understand the trade-offs of decisions, and identify improvements. AWS describes a workload as a set of components that together deliver business value; architecture is how those components work together. 8 9

A useful exam translation is:

> Start with business outcomes and constraints, identify the failure and security consequences, then choose the simplest managed AWS design that meets them.

The framework is not a checklist that produces one universally correct architecture. AWS explicitly describes trade-offs between pillars based on business context. For example, a mission-critical workload may justify higher cost to improve reliability, while a development environment may optimize cost or

sustainability differently. AWS also cautions that security and operational excellence are generally not traded off against other pillars. 9

2.3.1 The six pillars

The current framework uses six pillars: **operational excellence, security, reliability, performance efficiency, cost optimization, and sustainability**. 8 9 10

2.3.1.1 Operational excellence

Operational excellence is the ability to support development and run workloads effectively, gain insight into operations, and continuously improve processes and procedures to deliver business value. 9

Exam-relevant signals include observability, automation, infrastructure as code, repeatable deployments, safe rollback, runbooks, and learning from operational events. AWS's operational-excellence design principles include actionable observability, safe automation, frequent small reversible changes, anticipating failure, learning from metrics and events, and using managed services where possible. 13

A design that is highly available on paper but cannot be monitored, tested, deployed consistently, or recovered by the operating team is not a strong Well-Architected answer. When two answers meet the direct requirement, favor the one that reduces undifferentiated operational work and makes failure or change visible.

2.3.1.2 Security

Security is protecting data, systems, and assets while improving the security posture. 9 Current AWS security design principles include a strong identity foundation and least privilege, traceability, defense in depth, automated security controls, protection of data in transit and at rest, reducing unnecessary human access to data, and preparation for security events. 12

For SAA-C03 scenarios, read security requirements literally:

- "Users" may imply federation or application identity, while "workloads" may imply IAM roles for AWS API access. Do not assume long-lived access keys are required.
- "Private" means more than placing a resource in a private subnet. Check route paths, security groups, network ACLs, endpoints, public exposure, and the identity/resource policies that control access.
- "Encrypt" requires distinguishing data at rest from data in transit, and may require key-policy or IAM-policy reasoning.
- "Only this account," "central security account," or "multiple environments" may point to cross-account roles, resource policies, AWS Organizations guardrails, or IAM Identity Center rather than a larger set of IAM users.

2.3.1.3 Reliability

Reliability is a workload's ability to perform its intended function correctly and consistently when expected, including operating and testing it throughout its lifecycle. 9 Reliability questions commonly test fault isolation, redundancy, automatic recovery, backup and restore, failure testing, and disaster recovery. AWS's reliability guidance states that failures are expected in systems of reasonable complexity. Reliable workloads should detect failures, withstand them, and automatically repair where possible. The guidance ties backups and recovery to **RTO** (recovery time objective: how quickly service must be restored) and **RPO** (recovery point objective: how much data loss measured in time is acceptable). It also emphasizes fault-isolation boundaries, testing recovery, and regularly verifying backups. 14

Do not equate “highly available” with “multi-Region” by default. A multi-AZ design may satisfy the requirement with less complexity and cost. A multi-Region design becomes more plausible when the scenario requires protection from a regional disruption, geographic latency, data sovereignty, or a stated cross-Region recovery objective.

2.3.1.4 Performance efficiency

Performance efficiency is using computing resources efficiently to meet requirements and maintain efficiency as demand and technologies change. 9 Scenario cues include throughput, latency, concurrency, read/write patterns, burst behavior, data size, access frequency, and scale variability.

The exam expects the candidate to select an appropriate resource type and size, decouple components so they can scale independently, and use purpose-built services. Typical performance reasoning includes caching hot reads, selecting block/file/object storage by access semantics, choosing an appropriate database model, using a CDN for cacheable global content, and scaling horizontally when a workload can be distributed.

Do not optimize a component in isolation. A faster database may not fix application latency if requests are serialized, cross-Region, or repeatedly fetch the same immutable objects. Conversely, a cache can reduce database load but introduces invalidation, freshness, and failure behavior that must fit the requirement.

2.3.1.5 Cost optimization

Cost optimization is running systems to deliver business value at the lowest price point. 9 The cost pillar is not “always choose the cheapest service.” It is a requirement-aware discipline: measure use, right-size, select suitable purchasing and pricing models, remove idle resources, use lifecycle policies, and avoid unnecessary transfer or duplicated processing.

Exam cues include “least expensive,” “cost-effective,” “infrequent access,” “unpredictable demand,” “steady state,” “non-production,” “minimum operational overhead,” and “avoid data transfer charges.” Evaluate the whole workload cost and the cost of operating it. A managed service may cost more per raw unit but reduce administration, overprovisioning, and failure risk. A single NAT gateway may reduce gateway cost but weaken AZ isolation and create cross-AZ dependency or transfer cost; the correct choice depends on the scenario’s availability and cost priorities.

2.3.1.6 Sustainability

Sustainability in the current framework focuses on continually improving environmental impact by reducing energy consumption and increasing efficiency across workload components, maximizing the benefits from provisioned resources, and minimizing total resources required. 9 The sustainability pillar documentation describes environmental sustainability and stresses efficient architectures and reducing waste. 15

Sustainability is not a separate scored SAA-C03 domain in the supplied guide. It is still useful as a cross-cutting trade-off lens: right-size resources, avoid idle capacity, use managed/serverless options where the workload fits, improve utilization, and select storage lifecycles. Do not choose a sustainability optimization that violates a stated security, reliability, latency, or compliance requirement.

2.3.2 Applying the framework in reviews

AWS describes a Well-Architected review as a consistent, blame-free, lightweight conversation rather than an audit. The outcome is an action list that improves the customer’s experience. Reviews are recommended

at important lifecycle milestones: early in design, before go-live, after major changes, and continually as the workload evolves. 11

That review behavior is also a good exam method. When given an existing architecture, do not redesign everything immediately. Identify the stated business outcome, find the largest risk or single point of failure, and select the targeted change that removes it while preserving the rest of the design. If the prompt asks for "the best improvement," answer the improvement that most directly addresses the named risk.

2.4 3. A scenario-reasoning method

Use the following sequence on long questions. It prevents attractive service names from replacing requirements analysis.

2.4.1 Step 1: Extract the hard constraints

Write or mentally classify the requirements:

- **Security:** who can access what, account boundaries, public/private exposure, encryption, compliance, auditability.
- **Reliability:** availability target, failure scope, backup requirement, RTO/RPO, failover behavior, data durability.
- **Performance:** latency, throughput, concurrency, read/write ratio, burst versus steady load, payload size, geographic distribution.
- **Cost and operations:** budget, variable demand, licensing, management burden, acceptable manual work, and whether the workload is production or non-production.
- **Change and compatibility:** legacy application constraints, migration downtime, operating-system constraints, or whether the application can be refactored.

Words such as **must**, **only**, **without changing the application**, **near real time**, **automatically**, **least operational overhead**, and **lowest cost** are usually answer-selection constraints rather than decorative prose.

2.4.2 Step 2: Identify the workload shape

Classify the main workload before choosing services:

- Request/response web or API workload.
- Batch or scheduled processing.
- Event-driven asynchronous processing.
- Streaming ingestion and near-real-time processing.
- Transactional relational workload.
- Key-value/document workload with high scale or predictable access keys.
- File share or shared filesystem workload.
- Object archive, static content, data lake, or backup workload.
- Hybrid or migration workload.

This classification eliminates many distractors. For example, a shared POSIX filesystem is not replaced by object storage merely because S3 is scalable; a relational transaction workload is not automatically a DynamoDB workload because it needs high availability.

2.4.3 Step 3: Find the failure boundary and bottleneck

Ask what fails first and what the design must do when it fails:

- A single EC2 instance: use redundancy and load balancing, often with Auto Scaling.
- A single AZ dependency: distribute stateless tiers and stateful data appropriately across AZs.
- A synchronous downstream dependency: consider a queue, retry policy, idempotency, or event-driven fan-out.
- A database read bottleneck: consider read replicas, caching, or a database model that matches the access pattern; do not use a read replica to solve writes.
- A regional dependency: consider cross-Region replication and a tested DR/failover plan only when the requirement supports it.
- A network egress path: inspect NAT, endpoints, routing, bandwidth, and transfer-cost implications.

The answer should explain the mechanism, not only name the product.

2.4.4 Step 4: Choose the least complex architecture that satisfies the constraints

Prefer managed services when they meet the requirement and reduce operational burden, but do not turn "managed" into an unconditional rule. A managed relational database may be appropriate where transactions and SQL compatibility matter; a serverless key-value database may be better where the access pattern is known and massive elasticity is required. Containers may be appropriate for packaging and runtime control, while Lambda may be appropriate for short event-driven functions with variable demand and no server management requirement.

"Most features" is not a selection criterion. The best option normally has the fewest unnecessary moving parts while still meeting the stated security, availability, performance, and cost objectives.

2.4.5 Step 5: Compare every answer against the negative requirements

A distractor often meets the positive requirement but violates a negative one. Check:

- Does it introduce a public endpoint where private access was required?
- Does it add a single point of failure?
- Does it require application changes that the prompt forbids?
- Does it create cross-AZ or cross-Region traffic that the cost or latency requirement rejects?
- Does it use synchronous coupling where the producer must continue when the consumer is unavailable?
- Does it protect the instance but not the data, or encrypt at rest but not in transit?
- Does it restore a service but not the required data state?
- Does it scale compute but leave the database or queue as the bottleneck?

2.4.6 Step 6: Confirm the scope of the answer

If the question asks for one service, answer the service and configuration that solves the requirement. If it asks for a complete architecture, include the relevant supporting components: identity, networking, observability, data protection, scaling, and recovery. Avoid adding unrelated services merely to make a diagram look sophisticated.

2.5 4. Common architectures and the decisions they teach

These are reasoning patterns, not claims that one pattern is correct for every workload.

2.5.1 Highly available three-tier web application

A common design uses Route 53 for DNS, a public load-balancer tier, private application instances or containers across multiple AZs, and a database tier with an appropriate multi-AZ or managed

configuration. Static or cacheable content can be served through Amazon S3 and Amazon CloudFront; the database may use caching or read replicas when the access pattern justifies it.

The exam-relevant decisions are:

- **Public versus private subnets:** the load balancer or edge entry point may be public, while application and database resources are private.
- **Horizontal scaling:** stateless application instances can be distributed across AZs and scaled independently.
- **State location:** sessions should be externalized when instances are replaceable; do not rely on local instance state for a horizontally scaled tier.
- **Database semantics:** multi-AZ standby/failover is different from read replicas. A standby or failover design protects availability; a read replica primarily scales reads and may have replication lag.
- **Failure isolation:** eliminating one instance failure does not eliminate every dependency failure. Check load balancer, NAT, database, DNS, and deployment paths.

2.5.2 Asynchronous order or media-processing pipeline

A producer accepts work and places messages on Amazon SQS. Consumers running on Lambda, containers, or EC2 process the messages. A dead-letter queue captures messages that cannot be processed after the configured retry behavior. The worker writes results to a durable store and should be idempotent so a retry does not duplicate the business effect.

This pattern is strong when the producer should remain available while workers are slow or temporarily unavailable, when workloads need independent scaling, or when burst absorption matters. It is not automatically correct when the caller requires an immediate transactional response or strict ordering that the selected queue and consumer design cannot guarantee.

For publish/subscribe fan-out, use an event or notification pattern that lets multiple consumers receive the event independently. Do not force unrelated consumers into a single synchronous chain. For workflows with explicit steps, branching, retries, and state transitions, a workflow-orchestration service may be more appropriate than hand-coded coordination.

2.5.3 Global static-content and API delivery

For globally distributed users, place cacheable objects close to users through a CDN and use an origin such as S3 or an application endpoint. Use HTTPS/TLS and apply edge/application protections where required. For dynamic APIs, reason separately about caching, origin latency, connection behavior, and global routing or acceleration.

The key pitfall is treating a CDN as a general database or as a substitute for origin correctness. Cache behavior must match freshness and authorization requirements. Dynamic, user-specific responses may need different cache keys or no caching. An accelerator and a CDN solve different problems; select based on whether the requirement is primarily global network path performance for dynamic traffic or edge caching of content.

2.5.4 Hybrid connectivity and migration

A hybrid scenario commonly combines an on-premises environment with VPCs through AWS Site-to-Site VPN or AWS Direct Connect. The choice depends on required bandwidth, predictability, private connectivity, setup time, and resiliency. A migration question then adds a strategy constraint: rehost when

application change must be minimized, replatform when targeted cloud optimizations are allowed, or refactor when a cloud-native redesign is justified.

Avoid assuming that a physical dedicated connection is automatically the best answer. It may need redundancy and lead time; a VPN may be the faster or lower-cost fit. Also distinguish data transfer from server migration and database migration: the correct service depends on source/target type, data volume, downtime, continuous replication, and whether schema conversion is required.

2.5.5 Backup, pilot-light, warm-standby, and active-active DR

Use the stated RTO/RPO and business impact to compare DR strategies:

- **Backup and restore:** lowest ongoing standby cost, but the longest recovery time and dependence on successful restore procedures.
- **Pilot light:** core data or minimal foundational components are prepared; capacity is added during recovery.
- **Warm standby:** a scaled-down but running environment reduces recovery time compared with pilot light, at higher cost.
- **Active-active:** users can be served by multiple live environments and failover can be fastest, but data consistency, conflict handling, operations, and cost are more complex.

These labels are decision aids, not substitutes for a recovery design. The answer must also cover backups, replication, DNS or routing failover, capacity and service quotas, secrets/configuration, observability, and a tested runbook. AWS specifically emphasizes testing backup recovery and failure behavior rather than assuming that a backup exists and therefore DR works. 14

2.6 5. High-yield trade-offs and pitfalls

2.6.1 Availability versus cost

Redundancy across AZs usually improves availability but adds resources and sometimes transfer or NAT cost. Redundancy must be applied to the actual failure boundary. Two instances in one AZ do not satisfy an AZ-failure requirement. Conversely, adding a second Region without a stated regional objective can add replication, consistency, and operational complexity without being the best answer.

2.6.2 Horizontal versus vertical scaling

Vertical scaling increases the size of one resource and can be simple, but has a ceiling and may require disruption. Horizontal scaling adds instances or workers and supports elasticity and fault isolation, but requires statelessness, shared/externalized state, load balancing, and coordination. Prefer the method that fits the component's behavior and the prompt's scaling pattern.

2.6.3 Synchronous versus asynchronous coupling

Synchronous calls are appropriate when the caller needs an immediate result and latency is acceptable. They also propagate downstream failure and latency. Queues and event-driven patterns absorb bursts and isolate failures, but add eventual consistency, retry, duplicate-delivery, ordering, and observability considerations. Choose asynchronous communication when the business process can tolerate those semantics and benefits from decoupling.

2.6.4 Multi-AZ versus multi-Region

Multi-AZ is normally the first availability design for a regional workload. Multi-Region is justified by the stated disaster scope, global user experience, sovereignty, or RTO/RPO requirement. Do not confuse a

cross-Region replica with a complete DR solution; the application, networking, identity, secrets, routing, capacity, and runbooks also need a recovery path.

2.6.5 Read replicas versus failover replicas

A read replica primarily serves read scaling and may have replication lag. A standby/failover replica is intended to improve availability and failover. The prompt's words "read-heavy," "scale reads," "reporting," "automatic failover," and "survive primary failure" point to different decisions.

2.6.6 Security group versus network ACL

Security groups are stateful controls associated with network interfaces; network ACLs are subnet-level controls and are stateless. The exam may test which layer is appropriate or whether return traffic must be explicitly allowed. Neither control is a complete application-security strategy: also consider IAM, resource policies, encryption, WAF/Shield use cases, secrets, logging, and patching responsibilities.

2.6.7 Managed service versus self-managed compute

Managed services generally reduce operations and expose service-specific scaling and durability behavior. Self-managed EC2 or containers may be necessary for operating-system control, specialized software, licensing, or workload compatibility. The best answer balances control against patching, scaling, availability, and maintenance work rather than selecting one category by habit.

2.6.8 Cheapest component versus lowest total cost

A low unit price can be offset by idle capacity, operations, data transfer, backups, licensing, or support. Cost questions often reward right-sizing, lifecycle policies, serverless or managed services for variable demand, appropriate purchasing models for steady workloads, and reduced unnecessary network transfer. Keep the availability requirement intact while optimizing cost.

2.6.9 "Private" does not mean "no network path"

A private subnet can still reach the internet through a NAT gateway, reach AWS services through endpoints, or reach on-premises networks through VPN or Direct Connect. Check the intended route, endpoint policy, security controls, and whether a public IP or public service endpoint is actually present. The least-cost or simplest answer may be an appropriate VPC endpoint rather than routing private service traffic through NAT.

2.6.10 Encryption is not authorization

Encryption protects confidentiality in transit or at rest, but it does not grant a principal permission to read data. IAM/resource policies, key policies, network controls, and application authorization still matter. Conversely, authorization without encryption may fail a data-protection requirement. Read whether the question asks for TLS, KMS-backed encryption, key rotation, access control, or the full combination.

2.7 6. Study strategy aligned to the exam contract

AWS's official preparation guidance uses a four-step sequence: understand the exam with exam-style questions, refresh knowledge and skills, review and practice by domain, and assess readiness with the Official Practice Exam. AWS provides free Official Practice Question Sets and Exam Prep courses; subscription offerings may add labs, additional questions, and full-length practice exams. 2 16

The following plan adapts that official sequence into a repeatable study workflow.

2.7.1 Phase 1: Baseline and scope (one session)

1. Read the current exam guide end to end, especially the response types, scoring, domain weights, task statements, and service references. 1 3
2. Take a small official practice set without looking up answers first. Record the **reason** for each miss: missing service behavior, wrong requirement extraction, confused trade-off, arithmetic/time issue, or careless reading.
3. Create a four-domain gap map. Weight study time roughly by the domain percentages, but give extra attention to repeated reasoning failures rather than blindly studying the largest service list.
4. Build a current-name glossary. Include IAM Identity Center and any other service renamings encountered in current AWS documentation. Do not memorize legacy names as if they were current product names.

2.7.2 Phase 2: Build architecture primitives

Study by architectural decision, not by isolated service alphabet. For each topic, answer four questions: what problem does it solve, what requirement makes it a fit, what does it not solve, and what is the nearest distractor?

A strong sequence is:

- **Identity and boundaries:** IAM users, groups, roles, policies, federation/IAM Identity Center, STS, cross-account access, resource policies, Organizations guardrails, and shared responsibility.
- **Network foundation:** Regions/AZs, VPCs, subnet tiers, route tables, security groups, network ACLs, NAT, VPC endpoints, load balancers, Route 53, VPN, Direct Connect, PrivateLink, CDN, and global acceleration.
- **Storage semantics:** object, block, and file; performance, durability, access patterns, lifecycle, backups, and hybrid transfer.
- **Compute and scaling:** EC2 families and Auto Scaling, containers/ECS/EKS/Fargate, Lambda, batch, placement, horizontal versus vertical scaling, and workload statelessness.
- **Databases:** relational versus non-relational, RDS/Aurora versus DynamoDB, replicas/failover, caching, proxies, serverless capacity, backups, and connection limits as described in current service documentation.
- **Integration and resilience:** SQS, pub/sub, event-driven patterns, retries, dead-letter queues, idempotency, Step Functions, observability, and DR strategies.
- **Cost reasoning:** right-sizing, lifecycle, pricing models, transfer paths, NAT/endpoints, utilization, and managed-service operations.

Use small hands-on builds where possible: a private application tier behind a load balancer, a queue-backed worker, an S3 lifecycle policy, a multi-AZ database or failover exercise, an IAM role and cross-account policy, and a monitored deployment. The purpose is to see the behavior and failure mode, not to build a production platform.

2.7.3 Phase 3: Learn the framework as a decision checklist

For every architecture you study, ask:

1. **Operational excellence:** Can the team observe, deploy, roll back, and improve it?
2. **Security:** Are identities least-privileged? Are boundaries, data, traffic, and credentials protected?

3. **Reliability:** What happens when an instance, AZ, dependency, or Region fails? How are RTO/RPO met and tested?
4. **Performance efficiency:** What is the bottleneck, access pattern, latency target, and scaling signal?
5. **Cost optimization:** What is paid for continuously, what scales with demand, and what transfer or idle costs exist?
6. **Sustainability:** Can waste and overprovisioning be reduced without violating the stated requirements?

AWS's framework is explicitly about understanding pros and cons, so write down the trade-off rather than memorizing only a service-to-use-case pair. 8 9

2.7.4 Phase 4: Practice in exam mode

Use official question sets first to learn AWS's wording and explanation style. AWS says its 20-question Official Practice Question Sets demonstrate exam style and include feedback and recommended resources. A full-length Official Practice Exam provides exam-style scoring and is recommended by AWS before the real exam. 16

For each question, use this review template:

- What was the single business outcome?
- Which constraints were hard constraints?
- What architecture pattern was being tested?
- Why is the correct answer sufficient?
- Which requirement does each distractor violate?
- What assumption did I make that the prompt did not state?
- Which AWS documentation page should I read next?

Do not count a question as mastered merely because the letter was correct. Count it as mastered when you can explain why the closest distractors fail and can reproduce the decision on a changed scenario.

2.7.5 Phase 5: Final readiness and exam execution

In the final preparation window, prioritize current exam-domain gaps and timed mixed sets over collecting more unrelated service trivia. Revisit the framework, common architectures, and failure boundaries. Check volatile details directly in current AWS documentation rather than relying on old notes; avoid spending scarce time memorizing quotas that are not stable or not relevant to the scenario.

During the exam:

- Read the final sentence and explicit constraints before analyzing the service list.
- Separate "must," "should," and background context.
- Eliminate options that violate a hard constraint immediately.
- For multiple response, select all options that independently satisfy the requirement; do not choose an option merely because it is generally good practice.
- Mark genuinely uncertain questions and return after completing the rest.
- Use the Help/service-name reference if a short name is unfamiliar. 2
- Answer every question before time expires because unanswered questions are scored incorrect and guessing has no penalty. 1 3

2.8 7. Compact revision checklist

Before considering this section complete, be able to explain without notes:

- Why SAA-C03 has 65 questions but only 50 scored questions, what the 720 score means, and why domain percentages do not create separate pass thresholds.
- The difference between multiple-choice and multiple-response questions and the no-penalty guessing rule.
- The four scored domains and their weights: 30%, 26%, 24%, and 20%.
- The six current Well-Architected pillars and why the four exam domains are not a one-to-one replacement for all six pillars.
- The definitions of workload, component, architecture, RTO, and RPO.
- How to reason from requirements to a managed, secure, resilient, performant, and cost-aware design.
- When multi-AZ is sufficient and when multi-Region DR is justified.
- When to prefer synchronous communication, queues, pub/sub, workflows, or direct service integration.
- Why read replicas, standby/failover replicas, backups, and cross-Region replicas solve different problems.
- Why security groups, network ACLs, IAM/resource policies, encryption, and WAF/Shield controls are different layers.
- How object, block, and file storage differ in access semantics.
- How horizontal scaling, caching, decoupling, and right-sizing interact.
- How to explain not only why the chosen answer works, but why each distractor violates a stated constraint.

2.9 References

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3 Domain 1 — Design Secure Architectures (SAA-C03)

Exam scope. Domain 1 is **30%** of SAA-C03. AWS organizes it into three tasks: (1) secure access to AWS resources, (2) secure workloads and applications, and (3) appropriate data security controls. The official outline is a scope guide rather than an exhaustive service list; questions test choosing a design that satisfies requirements and tradeoffs, not memorizing every security feature. [1] [2]

The governing mental model is **defense in depth plus least privilege**. Start with the identity that makes a request, the resource being accessed, the network path, the data sensitivity, and the failure or compromise that must be contained. Then select the smallest set of controls that directly addresses those requirements. Security controls should also preserve availability, operability, and cost objectives; a control that cannot be operated or prevents required traffic is not a good architecture.

3.1 1. Shared responsibility: decide what AWS secures and what you configure

AWS secures **of** the cloud: the facilities, physical hardware, networking, and virtualization layer. The customer secures **in** the cloud: configuration and data, identities and permissions, network controls, guest operating systems and patches where applicable, applications, and service-specific settings. Responsibility varies by service. For example, AWS operates the managed infrastructure for Amazon RDS, but the

customer still controls database users, network access, encryption choices, and data. With Amazon EC2, the customer also patches and hardens the guest operating system. 3

Scenario rule: never answer "AWS handles security" without identifying the service boundary. Managed services reduce the customer's patching and host burden; they do not automatically make a bucket private, an IAM policy least privilege, a security group narrow, or data classified and retained correctly.

3.2 2. Task 1.1 — Design secure access to AWS resources

3.2.1 2.1 Accounts, Organizations, OUs, and guardrails

Use separate AWS accounts to create administrative and blast-radius boundaries, for example:

- a management account used for organization administration and billing;
- security/log-archive accounts with tightly restricted write paths;
- workload accounts separated by environment, application, or sensitivity; and
- network/shared-services accounts where central connectivity is justified.

AWS Organizations Service Control Policies (**SCPs**) provide organization-wide guardrails. An SCP sets the **maximum available permissions** for IAM users and roles in member accounts; it does not grant any permission by itself. Effective access still needs an identity-based or resource-based allow. An explicit deny wins, and a missing allow at the SCP layer can also prevent an action. SCPs affect member accounts, including their root users, but do not restrict users or roles in the management account and do not restrict service-linked roles. 4

Use a deny-style SCP for controls such as "deny actions outside approved Regions," "deny disabling CloudTrail," or "deny leaving the organization," but test it in a small OU before attaching it broadly. An allow-list SCP is more difficult to maintain because every required AWS service and dependent action must remain allowed. Do not remove the default `FullAWSAccess` SCP unless a replacement allow strategy is intentional and tested; otherwise member-account actions can fail. 4

AWS Control Tower can help establish and govern a multi-account landing zone, but it is an operating model and governance layer, not a replacement for IAM policies, SCPs, resource policies, encryption, or workload network controls. Choose it when the requirement is repeatable account provisioning and preventive/detective controls at organization scale; do not select it merely because an individual application needs an IAM role.

3.2.2 2.2 Human access: federation, IAM Identity Center, and IAM users

For workforce access across accounts, prefer **IAM Identity Center** (formerly AWS Single Sign-On) integrated with AWS Organizations and, where appropriate, an external identity provider. Permission sets define the access level and can be assigned to groups or users across multiple accounts. The AWS access portal provides console access and temporary programmatic credentials. This centralizes onboarding, offboarding, MFA and role assignment instead of duplicating IAM users in every account. 5

Prefer federated, temporary access for people. Create an IAM user only for a specific compatibility or emergency use case that cannot use federation, and protect it with MFA and a narrow policy. Protect the root user with a strong credential and phishing-resistant MFA; do not use root for routine administration. AWS IAM guidance recommends temporary role credentials for human users and workloads, least privilege, regular removal of unused credentials, and IAM Access Analyzer to validate or generate policies from observed activity. 6

Do not confuse authentication with authorization:

- **Authentication** establishes who or what is making a request.
- **Authorization** determines which actions are allowed on which resources under which conditions.
- Federation or IAM Identity Center improves authentication and workforce lifecycle; permission sets and IAM policies determine authorization.

3.2.3 2.3 Workload identities: roles and temporary credentials

An IAM role has a **trust policy** that says who or what may assume it, and permissions policies that say what the resulting session may do. A role has no ordinary long-lived password or access key; assuming it returns temporary credentials. Attach roles to EC2 instances through instance profiles, to Lambda functions through execution roles, and to other AWS services through service roles. For workloads outside AWS, consider STS federation, IAM Roles Anywhere, or a service-native mechanism rather than embedding an IAM access key. 7 6

A secure role design answers both questions:

1. **Who can assume the role?** Constrain the principal in the trust policy and add conditions such as organization, source account, external ID, or session attributes where appropriate.
2. **What can the role do?** Use a narrow permissions policy with specific actions, resources, and conditions.

For cross-account access, the trusting account owns the role and its trust policy; the trusted principal also needs permission to call `sts:AssumeRole`. A role is usually the clearest cross-account pattern because the resource-owning account controls the permissions and the session is temporary. Some services also support direct resource policies, such as S3 buckets, SNS topics, and SQS queues; use those when the service and use case make a resource-centric grant simpler. 7

For a third-party SaaS or monitoring provider, use a dedicated cross-account role and require the provider's unique `sts:ExternalId` in the trust policy. The external ID prevents a confused-deputy scenario in which one customer tricks the provider into assuming another customer's role. For AWS service principals writing to a resource, constrain the resource policy with `aws:SourceArn`, `aws:SourceAccount`, `aws:SourceOrgID`, or `aws:SourceOrgPaths` as applicable. 8

Role-chaining pitfall: role chaining through temporary credentials limits the chained console/API/CLI session to one hour, even when the target role has a longer maximum session duration. Avoid unnecessary chains for long-running automation; instead federate directly to the needed role or use a workload role. 7

3.2.4 2.4 Policy types and policy evaluation

Use the following distinctions when reading a scenario:

- **Identity-based policies** attach to users, groups, or roles and grant actions on resources.
- **Resource-based policies** attach to resources and name the principals allowed to access them. They are common for S3, SNS, SQS, KMS keys, and some other services.
- **Trust policies** are resource-based policies on roles; they govern who may assume the role, not what the role may do after assumption.
- **Permissions boundaries** cap the maximum permissions an IAM user or role's identity policies can grant. A boundary does not grant access on its own.
- **Session policies** further limit a role session or federated session.
- **SCPs/RCPs** set organization-level maximums; they do not grant access.

An explicit deny in an applicable policy overrides an allow. In a straightforward same-account case, identity- and resource-based allows combine, while a permissions boundary and an SCP constrain the effective result. Think of effective permissions as "a required allow in each limiting layer, with any explicit deny winning," while remembering that direct grants to role sessions and some resource-policy cases have special IAM evaluation behavior. 9 10

Common distractor: attaching `AdministratorAccess` to a role does not overcome an SCP deny or a permissions boundary. Conversely, an SCP that allows an action does not make the action usable if the role has no identity/resource-based allow.

Use IAM Access Analyzer for external-access analysis, policy validation, and policy generation from CloudTrail access activity. Treat generated policies as a tested starting point: exercise all legitimate code paths and operational procedures before removing broader permissions.

3.3 3. Task 1.2 — Design secure workloads and applications

3.3.1 3.1 A common secure VPC shape

A general internet-facing application pattern is:

1. Route users through Route 53 and, when appropriate, CloudFront and AWS WAF.
2. Place an internet-facing Application Load Balancer in public subnets in at least two Availability Zones.
3. Place application instances, containers, or Lambda integrations in private subnets with no direct inbound Internet route.
4. Place databases and internal data stores in isolated/private subnets; allow access only from the application security group.
5. Give private workloads only the egress they need: NAT gateways for required Internet destinations, or VPC endpoints for supported AWS services.
6. Apply IAM roles, Secrets Manager, TLS, logging, and detective controls to the workload.

"Private subnet" means the subnet's route table has no route directly to an Internet Gateway; it is not an automatic security guarantee. A private subnet can still reach the Internet through a NAT gateway, and a permissive security group, route, endpoint policy, or application listener can still expose data. A database should normally have no public IP and no public load balancer path.

3.3.2 3.2 Security groups, NACLs, route tables, and segmentation

A **security group** is a stateful virtual firewall attached to supported resource network interfaces. It allows traffic by source/destination, protocol, and port. Return traffic for an allowed flow is automatically allowed, so you do not create a separate ephemeral-port return rule. Prefer references to peer security groups over broad CIDR ranges for tier-to-tier traffic. Keep ingress narrow and explicitly restrict administrative ports such as SSH (22) and RDP (3389). 11

A **network ACL (NACL)** is a stateless, subnet-level filter with ordered numbered allow/deny rules. Because it is stateless, the reverse direction must be explicitly allowed, including required ephemeral response ports. NACLs are useful as a coarse additional boundary or emergency deny (for example, blocking a known CIDR across a subnet), but they are more error-prone for application flows than stateful security groups. NACLs do not filter some VPC-provided traffic such as DNS and instance metadata; use the appropriate service controls for those paths. 12

Exam comparison:

- Need instance/ENI-level, state-aware allow rules → security groups.

- Need subnet-wide stateless allow/deny, including an explicit deny → NACL.
- Need determine whether a packet can reach a destination → route tables first, then IGW/NAT/VPN/TGW/endpoint path, then security groups/NACLs and the destination service policy.
- Need to prevent an instance from reaching the Instance Metadata Service → configure EC2 metadata options; a security group or NACL is not the control for that path.

Use separate security groups by tier (edge, application, database, endpoints), not one permissive group reused everywhere. A typical rule is `ALB-SG → App-SG` on the application port and `App-SG → DB-SG` on the database port; do not allow `0.0.0.0/0` to the database.

3.3.3 3.3 VPC endpoints, NAT, and service endpoints

Use an **S3 or DynamoDB gateway VPC endpoint** when private-subnet workloads need those services without a NAT gateway or Internet Gateway. AWS adds a prefix-list route to selected route tables; the endpoint is free of an additional hourly endpoint charge. Endpoint policy and S3/DynamoDB resource policy still matter. Gateway endpoints are Regional; traffic to a different Region does not use the local Region's endpoint route. 13

Use an **interface VPC endpoint** for many AWS APIs and PrivateLink services. AWS creates a private endpoint network interface in each selected subnet/AZ, and clients reach the service through private IP addresses. Attach a security group that allows the expected HTTPS traffic from clients and use an endpoint policy when the service supports it. Interface endpoints incur hourly and data-processing charges, so place them in the required AZs and avoid assuming they are free substitutes for every NAT path. 14

Frequent endpoint traps:

- A private subnet still needs DNS support/hostnames configured for private endpoint DNS behavior.
- The endpoint security group and subnet NACL must allow the client-to-endpoint flow.
- A VPC endpoint does not replace IAM or the target service resource policy.
- A gateway endpoint route does not automatically cover every route table; associate the required route tables.
- NAT is for outbound connections initiated by private resources; it is not an inbound exposure mechanism and does not make a database public.

3.3.4 3.4 Hybrid and external network connections

AWS Site-to-Site VPN uses IPsec-encrypted tunnels between an on-premises device and a VPC, virtual private gateway, or Transit Gateway. Each connection includes two tunnels that can be used for high availability. Use VPN when encrypted connectivity over the Internet is acceptable or rapid deployment is important. 15

AWS Direct Connect provides a private connection from the customer network to AWS through a Direct Connect location, with private, public, or transit virtual interfaces. It can offer more predictable latency and bandwidth than Internet VPN, but Direct Connect is not by itself application-layer encryption; add encryption such as a VPN over the path when the requirement is confidentiality in transit. Use redundant connections/locations for resilience. The supplied supplementary slides correctly emphasize VPN and Direct Connect as the standard hybrid building blocks, but the AWS documentation is authoritative for current behavior. 16

Use Transit Gateway when many VPCs and on-premises networks need a central routing hub. Preserve segmentation with separate route tables/attachments and explicit propagation decisions; a hub without route isolation can become an unintended lateral-movement path.

3.3.5 3.5 Edge and application security services

Amazon CloudFront reduces origin exposure and provides a natural attachment point for AWS WAF. **AWS WAF** evaluates HTTP(S) requests using web ACL rules, managed rule groups, rate-based rules, IP sets, and application-specific conditions. Use it for layer-7 threats such as SQL injection patterns, cross-site scripting patterns, malicious bots, and request-rate abuse. WAF is not a replacement for parameterized queries, output encoding, authentication, patching, or least-privilege database roles.

AWS Shield Standard provides baseline DDoS protection for AWS services. **Shield Advanced** is a subscription for expanded protection and response features on protected resources; its protection varies with architecture and configuration. Shield/WAF address edge and application-layer threats, while security groups and NACLs control network paths. 17

Amazon Cognito solves customer identity, not general AWS workforce administration. A **user pool** is a user directory and OIDC/OAuth authorization service that issues JWTs to an app or API. An **identity pool** exchanges trusted identity claims for temporary AWS credentials, optionally for authenticated or guest users, and can select IAM roles or use attributes for fine-grained access. Use a user pool when the application needs sign-in and tokens; add an identity pool only when the client genuinely needs scoped direct AWS API access. 18

Detection and classification services:

- **Amazon GuardDuty** continuously analyzes supported AWS data sources and logs to produce threat findings, such as compromised credentials, cryptomining, suspicious network behavior, and data-exfiltration indicators. It detects; it is not a preventative firewall. 19
- **Amazon Macie** inventories S3 and discovers sensitive data using machine learning, managed identifiers, pattern matching, and custom identifiers. It helps answer "where is PII or credentials?" and "is this bucket exposed?"; it does not replace the bucket policy or encryption control. 20
- **Amazon Inspector** continuously discovers and scans eligible EC2 instances, ECR images, and Lambda functions for software vulnerabilities and unintended network exposure. It is vulnerability management, not a WAF or IAM authorization service. 21
- **CloudTrail** records API activity for audit and investigation. Event history provides 90 days of management events; trails can deliver ongoing events to S3 and optionally CloudWatch Logs/ EventBridge. Configure an organization trail and protect the log destination against alteration. 22

3.4 4. Task 1.3 — Determine appropriate data security controls

3.4.1 4.1 Classify first, then select controls

Classify data by impact and requirements such as public, internal, confidential, regulated, credentials, or cryptographic material. For each class, document:

- who may read, write, administer, and share it;
- allowed Regions and accounts;
- retention and deletion requirements;
- recovery point and recovery time objectives;
- audit and evidence requirements;

- encryption at rest and in transit requirements; and
- whether AWS-managed convenience is acceptable or customer-controlled keys are required.

A control can fail through misconfiguration even when the underlying service is secure. Use preventive controls (SCPs, Block Public Access, endpoint policies, encryption requirements), detective controls (CloudTrail, Config, Macie, GuardDuty, Inspector, Security Hub), and recovery controls (versioning, immutable retention, backups, and cross-account/cross-Region copies).

3.4.2 4.2 Encryption at rest and KMS design

AWS KMS provides key-encryption operations and policy-controlled key use. AWS services commonly use envelope encryption: a data key encrypts the data, and KMS protects the data key. Select key type according to control requirements:

- **AWS owned keys:** AWS owns and manages the keys outside the customer account. They are convenient and free to the customer, but the customer cannot inspect or change their key policy or lifecycle.
- **AWS managed keys:** service-created keys in the customer account (often aliases such as `aws/service`). They reduce management effort, but the customer cannot change their policy, rotate them manually, or share resources across accounts using that key in the same way as a customer managed key.
- **Customer managed keys:** customer-created keys with customer-controlled key policies, IAM grants/policies, lifecycle, and audit visibility. They support requirements for separation of duties, explicit key access, cross-account sharing, or customer-controlled deletion/rotation, but create operational and KMS-cost overhead.

The exact service integration matters: verify whether the target service supports the chosen key type, multi-Region keys, grants, and cross-account use. 23

A KMS authorization commonly requires both the key policy and IAM policy to permit the operation, depending on the key policy's account-principal delegation. Use a key policy that limits administrators separately from cryptographic users, restricts service principals with conditions where possible, and avoids broad `kms:*` grants to application roles. For cross-account use, configure both the key policy in the key-owning account and the IAM/resource permissions in the caller's account.

KMS rotation is not a magic re-encryption control. Automatic rotation changes the current cryptographic material while retaining old material so existing ciphertext can still decrypt; it does not rotate the data keys or re-encrypt existing data, and it does not remediate a compromised data key. Use a new KMS key and a data migration/re-encryption plan when policy, ownership, algorithm, or compromise requirements demand it. 24

3.4.3 4.3 S3 data protection

S3 encrypts new object uploads at rest by default with SSE-S3. Choose SSE-KMS when you need customer-controlled key policies, CloudTrail key-use auditing, grants, or cross-account key administration. Customer-managed SSE-KMS can introduce KMS permissions and request-cost considerations. Changing a bucket's default encryption does not retroactively re-encrypt existing objects; use an explicit migration such as S3 Batch Operations when needed. 25

For a sensitive bucket:

- enable S3 Block Public Access at the strongest organization/account/bucket level that is compatible with the use case;
- use bucket policies scoped to specific principals, prefixes, and actions;

- require TLS with an explicit deny on `aws:SecureTransport=false`;
- use IAM roles for applications rather than embedded access keys;
- consider S3 Versioning for accidental overwrite/deletion recovery;
- use S3 Object Lock with a retention mode when immutability/WORM evidence is required; and
- use lifecycle policies to manage retention and cost without deleting records before the required retention period.

S3 Block Public Access is a safety control, not a substitute for least-privilege policies. A static public website is a deliberate exception with a different architecture and must not be confused with a private data bucket. S3 guidance also warns that versioning alone is not immutable protection; Object Lock addresses deletion/overwrite resistance. 26

3.4.4 4.4 Credentials, secrets, and configuration

Do not put database passwords, API keys, OAuth secrets, or private credentials in source code, AMIs, container images, user data, or plaintext environment variables when a managed secret is appropriate. **AWS Secrets Manager** stores, retrieves, encrypts, and can automatically rotate database credentials and other secrets; applications retrieve values at runtime using an IAM role. Use a customer managed KMS key when secret access or key ownership requires it. 27

Use Systems Manager Parameter Store for configuration and, where suitable, lower-cost parameter storage. Do not choose Parameter Store as a blanket substitute for Secrets Manager when automatic rotation or secret lifecycle workflows are required. IAM credentials are managed in IAM/STS, encryption keys in KMS, and certificates in ACM; Secrets Manager's own guidance distinguishes these use cases. 27

3.4.5 4.5 Encryption in transit and certificates

Use TLS for client-to-load-balancer, service-to-service, administrative, API, and hybrid links whenever confidentiality and integrity are required. **AWS Certificate Manager (ACM)** issues, stores, and renews public/private X.509 certificates for integrated AWS services. ACM certificates are Regional; request/import a certificate in each Region where it is used, while a CloudFront certificate must be in `us-east-1`. For an EC2 server that is not integrated with an ACM-managed service, use an appropriate certificate automation approach or export an eligible private certificate. 28

Use **AWS Private CA** when the organization needs an internal CA hierarchy for private TLS, device/user authentication, or private service certificates. Private CAs have their own lifecycle and Regional considerations; they are not the cost-free answer for ordinary public web certificates. 29

Do not confuse TLS termination with end-to-end encryption. If a load balancer terminates TLS and forwards plaintext to an application tier, the segment after termination is not encrypted. For sensitive workloads, use TLS again from the load balancer to the target and validate certificate/trust requirements.

3.4.6 4.6 Backup, replication, retention, and recovery

Security includes availability and recoverability. Use the service-native backup/replication feature that matches the RPO/RTO and compromise model:

- S3 Versioning recovers prior object versions from accidental overwrite/deletion; Object Lock adds retention enforcement against deletion or overwrite.
- Database automated backups and cross-Region backup copies address data loss and Regional failure; read replicas are primarily an availability/read-scaling mechanism and are not a substitute for independent backups.

- Cross-account backup copies reduce the chance that a workload administrator or ransomware event can delete both production data and its backups. Use a tightly controlled backup account and restrict restore/delete permissions.
- Cross-Region copies address Regional loss but add cost, key-policy design, replication lag, and residency/compliance decisions.
- Test restores. A successful backup job proves that a copy was made, not that the application can meet its recovery objective.

The exam's design choice is driven by RPO/RTO, blast radius, retention immutability, and data sovereignty—not simply by selecting the most expensive replication mode. S3's documented versioning/Object Lock guidance is a concrete example of the difference between recovery from user error and protection against deliberate deletion. 26

3.5 5. Reference architectures and scenario reasoning

3.5.1 5.1 Internet-facing three-tier application

Requirement: public HTTPS application, private compute and database, protection from common web attacks, and no long-lived application keys.

Design: Route 53 → CloudFront/WAF (optional but appropriate for global edge and layer-7 filtering) → public ALB in two or more AZs → private application tier with an IAM execution role → private database subnets. Use security-group references between tiers, no database public IP, Secrets Manager for database credentials, ACM certificate for TLS, CloudTrail and GuardDuty for visibility, and a VPC endpoint for S3/DynamoDB access where applicable. Use NAT only for private workloads that need outbound Internet access. The key reasoning is to match each threat to a layer: WAF for HTTP patterns, SGs for network paths, IAM for API actions, Secrets Manager for credentials, and database authorization for data operations.

3.5.2 5.2 Multi-account workforce and cross-account operations

Requirement: employees need console/CLI access to many accounts, developers may deploy only within a project boundary, and central security must prevent disabling logs.

Design: AWS Organizations with OUs; IAM Identity Center backed by the corporate directory; group-to-permission-set assignments; dedicated workload roles; SCP guardrails at the organization/OU layer; organization CloudTrail delivered to a protected log archive account; security services delegated to a security account. Add permissions boundaries when delegating the ability to create roles within a workload account. The identity policies grant useful work; permission sets and roles provide access; SCPs cap it. The guardrail must not be mistaken for an authorization grant.

3.5.3 5.3 Private application with no Internet egress

Requirement: private compute pulls images and writes to AWS services without traversing the public Internet.

Design: private subnets, no NAT if all dependencies support private endpoints, interface endpoints for needed APIs/ECR/Secrets Manager/CloudWatch-style services, gateway endpoints for S3/DynamoDB, endpoint security groups and endpoint policies, and explicit service/resource policies. Verify DNS attributes, route-table association, NACLs, endpoint SGs, and KMS permissions. If a required third-party

endpoint is public-only, a controlled NAT egress path or private connectivity service may still be needed; "no NAT" is not a universal security requirement.

3.5.4 5.4 Sensitive S3 data with audit and recovery requirements

Requirement: confidential objects must not be public, require TLS and customer-controlled key access, be recoverable from accidental deletion, and be retained immutably for a period.

Design: S3 Block Public Access, narrow bucket and KMS key policies, SSE-KMS with a customer managed key, explicit TLS-only deny, Versioning, Object Lock retention, lifecycle rules aligned to legal retention, CloudTrail data-event logging where access-level audit is required, Macie discovery, and a separately controlled backup/replication strategy. Ensure the writer role can use the key and bucket but cannot alter retention or disable audit controls. The answer is not merely "enable default encryption," because default encryption does not provide public-access prevention, key governance, or immutable retention.

3.5.5 5.5 Hybrid regulated workload

Requirement: an on-premises system must reach private AWS resources with encrypted traffic and controlled routes.

Design: Site-to-Site VPN for IPsec encryption over the Internet, or Direct Connect for predictable private connectivity; use redundant paths and Transit Gateway for multi-VPC routing. If Direct Connect carries sensitive data and the requirement says "encrypted," add encryption rather than assuming private transport equals cryptographic confidentiality. Apply VPC route-table segmentation, security groups, NACLs where justified, and private service endpoints. The supplied slide deck's hybrid guidance supports the VPN/Direct Connect choice; current AWS docs should be used for implementation details. 15 16

3.6 6. High-frequency pitfalls and how to eliminate distractors

1. **"SCP grants access."** False. SCPs limit maximum permissions; IAM/resource policies still grant access. 4
2. **"A role trust policy grants S3 access."** False. Trust controls assumption; the role's permissions policy controls actions after assumption. 7
3. **"A permissions boundary grants the listed actions."** False. It only caps identity-policy permissions. 10
4. **"Security groups are stateless."** False. Security groups are stateful; NACLs are stateless and subnet-level. 11 12
5. **"A private subnet cannot reach the Internet."** False. It can use a NAT gateway; private means no direct route to an Internet Gateway.
6. **"NAT accepts inbound Internet connections to private instances."** False. NAT is for return traffic to outbound connections initiated from private resources; it is not an inbound publishing mechanism.
7. **"A VPC endpoint makes all access safe."** False. Endpoint SGs, endpoint policies, IAM, target resource policies, KMS permissions, and route/DNS configuration still apply. 13 14
8. **"Direct Connect is automatically encrypted."** False. It is private transport; use VPN or another encryption layer when cryptographic confidentiality is required. 16
9. **"GuardDuty blocks the attack."** Usually false. GuardDuty detects and reports; remediation may be automated with EventBridge/Lambda or handled by operations. 19
10. **"Macie encrypts sensitive S3 data."** False. Macie discovers/classifies and reports; use S3/KMS policies and encryption for protection. 20

11. **"Cognito user pools issue IAM credentials."** Not by themselves. User pools issue app tokens; identity pools can issue temporary AWS credentials. 18
12. **"S3 default encryption re-encrypts old objects."** False. Existing objects are unchanged when default encryption changes. 25
13. **"KMS rotation re-encrypts all data."** False. It changes key material for the logical key and does not re-encrypt existing data keys/data. 24
14. **"Versioning is immutable retention."** False. Versioning helps recovery; Object Lock is the control for WORM-style retention. 26
15. **"ACM certificates are global."** False. ACM certificates are Regional; CloudFront certificates are requested/imported in `us-east-1`. 28
16. **"CloudTrail event history is an organization's long-term archive."** False. Event history is a 90-day management-event view. Configure trails or CloudTrail Lake with protected retention for ongoing audit. 22

3.7 7. A repeatable exam method

For each scenario, write a one-line security contract before looking at the answer choices:

1. **Principal:** human, AWS workload, customer app user, service principal, or third party?
2. **Resource and action:** which API action and which exact data/resource?
3. **Boundary:** same account, cross-account, cross-Region, organization, on-premises, or Internet?
4. **Threat:** credential theft, excessive privilege, public exposure, lateral movement, DDoS, injection, data loss, or audit gap?
5. **Control layer:** IAM/STS, role trust, resource policy, SCP, SG/NACL/routes, endpoint/NAT, WAF/Shield, KMS/TLS, secret store, detective service, or backup/retention?
6. **Operational tradeoff:** managed versus customer control, availability, cost, Regional scope, rotation, recovery time, and ongoing maintenance.

Prefer the option that is secure by default, uses temporary credentials, limits blast radius, keeps data private by network and policy, encrypts with the required ownership model, and addresses the stated threat without adding unrelated complexity. If two choices both work, select the one that best matches the explicit requirement—especially whether the question asks for prevention versus detection, encryption versus private routing, customer-managed keys versus AWS-managed convenience, or recovery versus immutability.

3.8 References

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4 Domain 2: Design Resilient Architectures (SAA-C03)

AWS defines this domain as **26% of the SAA-C03 exam**. The official guide divides it into two task statements:

1. **Design scalable and loosely coupled architectures.** Topics include APIs, managed services, caching, microservices, event-driven design, horizontal and vertical scaling, edge acceleration, containers, load balancing, multi-tier and multi-Region architectures, queues and messaging, serverless patterns, storage types, container orchestration, read replicas, and workflow orchestration.

2. **Design highly available and/or fault-tolerant architectures.** Topics include AWS global infrastructure, managed services, basic networking, disaster recovery strategies, distributed design, failover, immutable infrastructure, load balancing, proxy concepts, service quotas and throttling, storage options, and workload visibility.

These task statements and the domain weighting come from the official SAA-C03 exam guide 1. This report explains the design reasoning behind those topics rather than treating any one topology as universally correct.

4.1 1. Core definitions and exam mental model

4.1.1 Availability, reliability, resilience, and fault tolerance

- **Availability** is the proportion of time a workload is usable. It is an outcome measured against a target, not merely the number of servers deployed.
- **Reliability** is the ability of a workload to perform its intended function correctly and consistently, including the ability to recover from failures. AWS's Reliability Pillar emphasizes strong foundations, resilient architecture, consistent change management, and tested recovery processes 2.
- **Resilience** is the ability to withstand and recover from failures while continuing to provide an acceptable level of service. A resilient design anticipates failure instead of assuming that every component stays healthy.
- **Fault tolerance** means the workload can continue operating when a defined fault occurs, often without visible interruption. Fault tolerance is relative to the failure scope: an architecture may tolerate an EC2 instance failure but not an entire Region failure.
- **Durability** is the ability to preserve data. Durability does not automatically mean the application is available. For example, replicated data may remain safe while the application is temporarily unable to serve requests.

The central exam question is usually: **What can fail, what must continue, how much data loss is acceptable, and how quickly must service return?** Start with the business requirement, then choose failure isolation, replication, routing, and recovery mechanisms.

4.1.2 Failure domains and blast radius

An **Availability Zone (AZ)** is an isolated location within an AWS Region. AWS describes AZs as physically separated, independently powered and connected, while still close enough for low-latency networking. Deploying compute and data across multiple AZs limits the blast radius of an AZ-level incident 3. A **Region** is a separate geographic and administrative failure domain. Multi-Region designs can protect against a Regional outage, but they add replication, routing, data-consistency, deployment, and operational complexity.

A common exam distinction is:

- **Multi-AZ**: usually the default answer for high availability inside one Region. It gives low-latency, fault-isolated redundancy and is generally simpler and cheaper than multi-Region.
- **Multi-Region**: justified when requirements include Regional disaster recovery, global user latency, data-residency constraints, or unusually stringent continuity targets. AWS explicitly warns against adding multi-Region dependencies when multi-AZ already satisfies the requirement 3.

A design is not resilient merely because it names two AZs. Every critical dependency must be evaluated: compute, load balancing, database, cache, queues, storage, DNS, secrets/configuration, deployment

pipeline, and observability. A single-AZ NAT gateway, database, proxy, or self-managed appliance can reintroduce a single point of failure.

4.1.3 RTO and RPO

- **Recovery Time Objective (RTO)** is the maximum acceptable time to restore service after an outage.
- **Recovery Point Objective (RPO)** is the maximum acceptable amount of data loss measured in time.

RTO answers "how long can the business be down?" RPO answers "how much recent data can the business lose?" They are independent. A system can have a short RTO but a non-zero RPO, or retain data accurately while requiring a long restoration process.

AWS recommends selecting a defined recovery strategy based on business objectives rather than choosing the most expensive option by default. In its current Reliability Pillar guidance, AWS presents approximate strategy profiles: backup and restore has the highest RTO/RPO; pilot light improves them; warm standby is faster; and multi-Region active/active can approach zero RTO/RPO for some failure types, but with substantially greater complexity and cost 4. These figures are design guidance, not guarantees for every workload.

4.2 2. Scalable and loosely coupled architecture

4.2.1 Horizontal versus vertical scaling

Vertical scaling (scale up/down) changes the size of a resource, such as moving to a larger EC2 instance or database instance. It is often simple but has an upper limit, may require downtime or a restart, and leaves a larger single failure unit.

Horizontal scaling (scale out/in) adds or removes independent instances, tasks, or functions. It supports elasticity and fault isolation, but requires the application to be designed for distribution. Stateless web tiers, queues, and partitioned data stores are natural fits.

For EC2, use an **EC2 Auto Scaling group** with a launch template, minimum/desired/maximum capacity, health checks, and subnets in multiple AZs. EC2 Auto Scaling replaces failed instances, balances capacity across selected AZs, and scales within configured bounds 5. Pair it with Elastic Load Balancing so new instances register automatically and unhealthy targets stop receiving traffic 6.

Scale on a signal that represents user demand or backlog, not only CPU. Examples include request count per target, latency, queue depth, or the age of the oldest message. The scaling policy must also account for warm-up time, cooldown behavior, quotas, and whether the downstream database can absorb the new load.

4.2.2 Stateless application tiers

A stateless tier stores durable session and business state outside an individual instance. Requests can therefore be routed to any healthy target, allowing instances to be replaced, scaled, or moved across AZs. Store state in an appropriate managed service, such as:

- Amazon DynamoDB for key-value/document access patterns;
- Amazon ElastiCache for a low-latency cache or session store when loss or reconstruction of cached state is acceptable;
- Amazon S3 for objects and durable artifacts; or
- a relational database for transactional state.

Sticky sessions can make a legacy application easier to run, but they reduce flexibility and can create uneven load or session loss when a target fails. If the requirement is high availability and horizontal scaling, prefer externalized session state over relying on local disk or in-memory state on one instance.

4.2.3 Loose coupling and asynchronous buffering

Loose coupling limits how much one component knows about another and prevents a slow or failed component from immediately taking down its caller. Common patterns include:

- **Synchronous request/response** for operations that need an immediate result. Use timeouts, retries with backoff, idempotency, and circuit-breaking behavior.
- **Queue-based load leveling** for work that can be processed later. Producers place jobs on Amazon SQS; consumers poll and process at their own rate. A queue absorbs bursts and lets workers scale independently.
- **Fanout** when one event must reach multiple consumers. Amazon SNS publishes to subscribers such as SQS queues, Lambda, or HTTP endpoints.
- **Event routing** when many event sources and targets need content-based routing. Amazon EventBridge event buses receive events and route them to zero or more targets using rules; EventBridge Pipes are designed for point-to-point source-to-target integrations with transformation or enrichment 7 8.

The AWS decision guide summarizes the practical distinction: SQS is pull-based and persistent until consumed or expired; SNS is push-based pub/sub; EventBridge is rule-based event routing. SQS and SNS provide at-least-once delivery in common modes, so consumers should be idempotent and tolerate duplicates. FIFO queues/topics are for ordering requirements, but ordering and throughput tradeoffs must be checked against the scenario 7. SQS long polling can wait up to 20 seconds, reducing empty polls and cost 7.

A queue is not a complete reliability design by itself. Configure a visibility timeout long enough for normal processing, a dead-letter queue for repeatedly failing messages, retention suitable for the recovery window, and alarms for backlog age and depth. Do not acknowledge a message before durable processing completes. Use a poison-message strategy so one malformed item does not block useful work.

4.2.4 Microservices, APIs, and managed services

Split services around independently changeable business capabilities, not arbitrary technical layers. Microservices can scale and deploy independently, but they add network calls, distributed tracing, versioning, failure handling, and data-consistency problems. A monolith may be the more resilient choice when the domain is small and operational simplicity matters.

For an HTTP API, Amazon API Gateway can provide the managed front door, throttling, authentication integration, and routing. Put an ALB behind it when the workload needs HTTP layer-7 routing to EC2, ECS, or IP targets. Use Amazon SQS or EventBridge to decouple long-running or retryable work instead of keeping client connections open.

Managed services are usually preferred when they meet the requirement because AWS operates their underlying infrastructure and many availability mechanisms. That does not remove application responsibility: configure the service correctly, distribute resources across AZs or Regions where appropriate, plan for quotas, and test failure paths. A managed service can still be a bottleneck if limits, partitions, connection pools, or throttling are ignored.

4.2.5 Serverless and containers

AWS Lambda is a good fit for event-driven, short-lived, independently scalable functions. Design handlers to be idempotent, keep initialization efficient, and move durable state to managed stores. Lambda functions not configured for a VPC are managed for availability by AWS; VPC-connected functions should use subnets across multiple AZs when they need private resources 3. Use reserved or provisioned concurrency only when the requirement justifies capacity control or predictable startup behavior; these features do not replace downstream resilience.

Amazon ECS services maintain a desired number of tasks and replace failed or unhealthy tasks. Use Fargate when you want managed container compute without managing EC2 worker instances; use ECS on EC2 when instance-level control or specialized capacity is required. Place service tasks across multiple AZs and put user-facing tasks behind an ALB. ECS service scheduling and load-balancer health checks help replace failed tasks, but the desired count, placement constraints, subnet capacity, and container startup time still matter 9.

Use **Amazon EKS** when Kubernetes compatibility and its ecosystem are requirements; do not choose it only because containers are involved. More control brings more operational responsibility. For a simple stateless service, ECS/Fargate can be the lower-complexity answer.

4.2.6 Workflow orchestration

Use **AWS Step Functions** when a distributed process needs explicit state, retries, branching, parallelism, timeouts, auditability, or human approval. A state machine makes failure handling visible instead of scattering it across custom code. Standard workflows are intended for long-running, auditable workflows; Express workflows suit high-event-rate processing and use at-least-once execution semantics, so task idempotency matters 10.

Do not use Step Functions merely as a synchronous API proxy if a direct call is sufficient. For asynchronous jobs, use a callback or `.sync` integration where supported, and ensure each external side effect is safe to retry.

4.2.7 Caching and read scaling

Caching reduces latency and protects an origin from repeated reads, but introduces freshness, invalidation, and failure behavior. Choose the cache location by access pattern:

- **CloudFront** caches content close to viewers and is the normal choice for globally distributed static or cacheable HTTP content.
- **ElastiCache** provides in-memory caching near application compute for hot database reads, sessions, or transient state.
- **Database read replicas** scale read-heavy relational workloads but do not replace a Multi-AZ standby for automatic write failover.
- **DynamoDB** can scale key-value/document access without managing database servers, but partition-key design and capacity modes still govern behavior.

A cache must have an explicit miss path. The origin should remain correct if the cache is empty or unavailable. Avoid caching personalized or rapidly changing data unless cache keys and invalidation are correct. A cache that is treated as the system of record creates a data-loss risk.

4.3 3. Highly available and fault-tolerant architectures

4.3.1 Canonical multi-tier architecture

A common SAA-C03 baseline is:

1. Route 53 directs the domain to a public entry point.
2. An ALB spans at least two AZs and performs target health checks and HTTP/HTTPS routing.
3. An EC2 Auto Scaling group, ECS service, or Lambda integration runs application capacity in private subnets across multiple AZs.
4. The application uses an AZ-resilient data service, such as RDS Multi-AZ/Aurora, DynamoDB, or S3.
5. CloudWatch metrics, logs, alarms, and traces provide visibility; IaC and automated deployment maintain consistency.
6. SQS, SNS, or EventBridge decouples asynchronous work and allows independent scaling.

An ALB is a layer-7 load balancer. It distributes traffic to healthy targets in one or more AZs, supports listener rules such as host and path routing, and scales as traffic changes 6. Do not use an ALB when the requirement is arbitrary TCP/UDP pass-through; evaluate a Network Load Balancer instead. Do not confuse ALB target health with application correctness: a target can answer a shallow health check while its dependencies are broken. Use health checks that reflect the service's ability to serve safely without making the check itself a load or cascading failure source.

4.3.2 Databases: availability, replicas, and failover

Amazon RDS Multi-AZ is primarily a high-availability and failover feature. A traditional Multi-AZ DB instance deployment has a standby that provides failover support and does not serve read traffic. Current RDS documentation also describes Multi-AZ DB clusters with two standby instances that can serve read traffic 11. Therefore, on an exam, distinguish "automatic failover for the writer" from "scale reads."

Amazon Aurora separates storage and compute. Aurora stores data redundantly across multiple AZs and synchronously replicates writes to storage nodes. Aurora Replicas are read-capable and can be promoted on writer failure; the cluster endpoint follows the current writer, so applications should use it instead of hard-coding an instance endpoint 12. A writer with no reader in another AZ has a slower recovery path than a cluster with appropriately placed replicas 12.

Read replicas use asynchronous replication for read scaling and, depending on the engine and architecture, can support promotion or cross-Region recovery. Asynchronous replication means a replica can lag; it is not equivalent to synchronous Multi-AZ standby. If the scenario says "reduce read load," choose read replicas or caching. If it says "automatic failover with minimal write interruption," choose Multi-AZ/Aurora high availability.

Amazon DynamoDB global tables are fully managed, multi-Region, multi-active tables. Replicas can serve reads and writes, and data is automatically replicated between Regions. The current service supports multi-Region eventual consistency and multi-Region strong consistency modes; multi-Region writes require a conflict strategy, and current global-table documentation describes last-writer-wins behavior for the applicable configurations 13. Use global tables when the application needs multi-Region local reads/writes and can handle its consistency and conflict model. Do not choose it solely because "multi-Region" sounds more available.

Amazon S3 is a Regional service designed for high durability and multi-AZ resilience. Use S3 Versioning to recover from accidental overwrite or deletion, Object Lock for WORM retention, and replication

when copies in another Region are required. Replication protects against some infrastructure and location failures but is not a substitute for point-in-time recovery from corruption or malicious changes. S3's resilience guidance explicitly covers versioning, Object Lock, lifecycle controls, and cross-Region replication 14.

4.3.3 DNS and global traffic routing

Route 53 is a DNS service and routing control, not an application proxy. Its policies include failover, latency, weighted, geolocation, geoproximity, IP-based, and multivalue answer routing 15.

- Use **failover routing** for active/passive primary-secondary endpoints with health checks.
- Use **latency routing** to direct users to the Region with the lowest measured latency.
- Use **weighted routing** for controlled percentages, canary releases, or gradual migration.
- Use **geolocation/geoproximity** for user or resource-location policies, subject to the application's compliance and latency needs.
- Use **multivalue answer** when returning multiple healthy records can improve availability; it is not a full load balancer.

DNS caching means a Route 53 change is not an instantaneous connection switch for every client. Keep TTL and health-check behavior consistent with the RTO requirement, but do not assume a low TTL guarantees zero failover delay.

AWS Global Accelerator provides static anycast IP addresses and sends traffic over the AWS global network to healthy regional endpoints. It can front ALBs, NLBs, EC2 instances, or Elastic IPs in one or more Regions and reacts to health/configuration changes 16. Choose Global Accelerator when you need stable IPs, fast global network entry, rapid endpoint failover, or non-HTTP/TCP/UDP acceleration. Choose CloudFront when the principal need is HTTP content delivery and edge caching. Route 53 and Global Accelerator can be combined, but they solve different layers of the traffic path.

4.3.4 Service quotas and throttling

Every design has finite quotas: API request rates, concurrent Lambda executions, NAT ports, IP addresses, load-balancer targets, database connections, queue throughput, and account-level resource counts. A resilient design identifies quota ceilings before launch, requests increases early where available, and monitors utilization. Quota exhaustion can look like an application outage even when all instances are healthy.

Throttling is normal protective behavior. Use exponential backoff with jitter, bounded retries, idempotency tokens, client-side rate limiting, and a queue to smooth bursts. Do not respond to throttling by blindly increasing concurrency; that can amplify a downstream failure. When a quota is not adjustable, redesign around partitioning, sharding, batching, fanout, or a different service.

4.3.5 Immutable infrastructure and deployments

Immutable infrastructure replaces rather than mutates production instances. Bake an AMI or container image, deploy a new version, validate health, shift traffic, and remove the old version. This limits configuration drift and makes rollback predictable. EC2 Auto Scaling instance refresh supports rolling replacement, and ALB weighted or listener routing can support canary/blue-green patterns 5 6.

A deployment is part of resilience. Use infrastructure as code, versioned configuration, automated rollback, health checks, and staged changes. Deploy the same known-good artifact to every AZ and Region. A multi-

Region topology with manual, inconsistent changes is not resilient if the standby is stale or configuration drift prevents failover.

4.3.6 Observability and recovery testing

Visibility is a design requirement, not an afterthought. Instrument:

- request rate, latency, errors, saturation, and dependency failures;
- Auto Scaling activity and target health;
- queue depth and oldest-message age;
- database replica lag, connections, storage, and failover events;
- cache hit rate and evictions;
- cross-Region replication lag;
- quota utilization and throttling; and
- backup success, restore duration, and recovery validation.

Test the actual recovery path. A backup that has never been restored, a DNS record that has never been failed over, or a standby whose capacity has never been exercised is an assumption. AWS recommends defined and tested recovery strategies and warns against ad-hoc recovery or dependence on control-plane operations during a crisis 4.

4.4 4. Disaster-recovery patterns and tradeoffs

4.4.1 Backup and restore

Back up data and configuration to a separate failure domain. During recovery, redeploy infrastructure, restore data, and redirect traffic. This is the least complex and generally lowest-cost option, but it has the highest RTO and often the highest RPO. Use infrastructure as code so the environment can be recreated consistently. AWS Backup can store backups across AZs and copy them across Regions for additional resilience 17.

For S3 workloads, versioning can address accidental deletion or overwrite. For databases, automated backups and point-in-time recovery can reduce RPO, but restoration time must be measured rather than assumed. Backup and restore is a poor answer when the requirement says “serve immediately in a second Region.”

4.4.2 Pilot light

Keep the data layer and core replication infrastructure running in the recovery Region. Application compute is minimized or not deployed and is created/scaled during failover. Pilot light costs less than a fully running standby, but its RTO includes provisioning and scaling actions. It also depends on deployment automation and capacity availability.

Pilot light is appropriate when the business can tolerate tens of minutes of recovery and wants lower ongoing cost than warm standby. It is not the same as a database-only backup: core infrastructure and deployment capability should already be defined and tested 4.

4.4.3 Warm standby and hot standby

A warm standby is a scaled-down, fully functional environment that can serve traffic immediately at reduced capacity. During recovery, scale it up and redirect traffic. It usually offers a lower RTO than pilot light at higher cost. If provisioned at full production capacity, it is commonly called **hot standby** 4.

Do not assume Auto Scaling will instantly create all capacity during a Regional event. Scaling depends on control-plane availability, quota, subnet IP space, instance capacity, image readiness, and application warm-up. For stringent RTOs, pre-provision enough baseline capacity and test promotion.

4.4.4 Multi-Region active/active

Each Region serves production traffic, and traffic is routed among healthy Regional endpoints. This can deliver near-zero recovery time for infrastructure failures because there is no cold failover, but it is the most complex pattern. Data writes must be synchronized or partitioned, conflicts must be handled, and the remaining Regions must have enough capacity to absorb traffic after one Region is lost. Replication alone does not protect against corruption, bad deployments, or malicious deletion; retain backups and point-in-time recovery 4.

DynamoDB global tables can support a multi-active data plane for suitable key-value/document workloads 13. Aurora Global Database can provide cross-Region read scaling and disaster recovery with asynchronous replication, but the primary writer and failover model must be considered 12. For relational active/active writes, application conflict resolution and transaction semantics are usually the hard part.

4.4.5 Choosing the pattern from the scenario

1. Identify the failure scope: instance, task, AZ, Region, dependency, or operator/data error.
2. Write down RTO and RPO. If the question does not state them, infer from words such as “minutes,” “near zero,” “no data loss,” or “cost-effective.”
3. Determine whether the recovery environment must process traffic immediately. If yes, warm standby or active/active is more likely than pilot light.
4. Determine whether the workload can tolerate asynchronous replication and conflict resolution. If not, avoid an active/active write design.
5. Check whether the proposed recovery depends on a control plane that may be impaired. Pre-provision capacity when the RTO is strict.
6. Include traffic routing, data replication, credentials, configuration, quotas, observability, and rollback. A copied database without a usable application stack is not a recovery architecture.
7. Prefer the simplest pattern that meets the business objective. Extra Regions and replicas are not automatically the correct answer.

4.5 5. Common architectures and scenario reasoning

4.5.1 Highly available web application

Requirement: tolerate an AZ failure and scale for variable HTTP traffic.

Use Route 53 to an ALB spanning multiple AZs. Run a stateless EC2 Auto Scaling group or ECS service in private subnets across those AZs. Store session state externally, use an AZ-resilient database, and put static assets in S3/CloudFront. Add CloudWatch alarms and queue asynchronous work. The important reasoning is that each tier is redundant and independently scalable; simply adding two EC2 instances in one AZ is not sufficient.

4.5.2 Burst-processing application

Requirement: accept unpredictable uploads and process them asynchronously without losing jobs.

Upload the object to S3, emit an event to EventBridge or SNS, deliver to an SQS queue, and run consumers on Lambda or ECS. Configure retries and a dead-letter queue. Scale consumers on queue depth/age. Make

processing idempotent using an object version, event ID, or job key. This architecture absorbs bursts and prevents a temporary processor outage from rejecting user uploads.

4.5.3 Relational read-heavy application

Requirement: keep writes highly available and reduce read pressure.

Use RDS Multi-AZ or Aurora for writer availability. Add read replicas or Aurora Replicas for read scaling, and optionally ElastiCache for hot data. Route writes to the writer/cluster endpoint and reads to a reader endpoint or replica-aware layer. Do not select a read replica as a substitute for Multi-AZ failover, and account for replica lag when the application requires read-after-write consistency.

4.5.4 Regional disaster recovery

Requirement: recover a critical application in another Region within minutes with small data loss.

Replicate data continuously or near-continuously, maintain a warm standby sized for initial traffic, deploy infrastructure using IaC, and use Route 53 failover or Global Accelerator to shift traffic. Test the complete runbook and ensure the recovery Region has quotas, IP space, images, keys, secrets, and capacity. A backup-and-restore design may not meet a minutes-level RTO; multi-site active/active may be unnecessarily complex if warm standby meets it.

4.5.5 Global application with active writes

Requirement: users write locally in multiple Regions and service must survive Regional loss.

Consider DynamoDB global tables for an access pattern that fits multi-Region, multi-active semantics. Design keys and conflict handling explicitly, understand consistency mode, and retain backups for corruption or deletion. If the workload requires strict cross-Region relational transactions, do not assume a generic active/active pattern will work; evaluate whether a single writer plus read replicas or a partitioned ownership model better matches the business rules.

4.6 6. Exam pitfalls and “almost right” answers

- **Two instances are not automatically highly available.** They must be spread across AZs, behind a health-aware traffic director, and backed by resilient dependencies.
- **Multi-AZ is not multi-Region.** Multi-AZ protects against an AZ failure; it does not by itself protect against a Regional outage.
- **Read replicas are not Multi-AZ standby.** Read replicas primarily scale reads and replicate asynchronously. Multi-AZ is the usual writer-failover answer.
- **Backups are not replication.** A backup can recover from corruption and operator error; replication can propagate bad changes. Use both when the threat model requires both.
- **A cache is not the source of truth.** The origin must remain correct after cache loss or eviction.
- **SQS, SNS, and EventBridge are not interchangeable.** Choose queue buffering, pub/sub fanout, or rule-based event routing based on delivery and consumer behavior.
- **At-least-once delivery requires idempotency.** Retries and redeliveries are normal; do not design side effects that cannot safely repeat.
- **DNS failover is not an instantaneous switch.** Resolver and client caches can delay a change.
- **Global Accelerator is not CloudFront.** Global Accelerator provides static anycast entry and endpoint failover/acceleration; CloudFront is primarily an HTTP content delivery and caching service.

- **Auto Scaling does not fix stateful local storage.** Move state to a managed data layer or design explicit replication.
- **A load balancer health check can be too shallow or too deep.** A shallow check misses dependency failures; a check that calls expensive dependencies can amplify an outage.
- **More Regions can reduce rather than improve resilience** if cross-Region dependencies, inconsistent deployments, or unsolved data conflicts become new failure modes 3 4.
- **"Serverless" does not mean no limits.** Lambda concurrency, API throttles, queue throughput, database partitions, IP addresses, and account quotas still require planning.
- **Multi-site active/active is not automatically best.** It is operationally and data-wise complex; select it only when the business requires its recovery or global-serving properties.

4.7 7. Compact decision checklist

When selecting an answer, ask:

1. **Failure domain:** What must survive—instance, AZ, Region, or data corruption?
2. **State:** Is the compute stateless? Where does session and durable state live?
3. **Traffic:** Is the protocol HTTP layer 7, TCP/UDP layer 4, or static content? Which routing policy matches the goal?
4. **Scaling:** Can components scale independently? What metric reveals demand or backlog?
5. **Data:** Is the requirement failover, read scaling, multi-Region writes, or point-in-time recovery?
6. **Messaging:** Does the workload need buffering, fanout, or event filtering/routing?
7. **Recovery:** What RTO/RPO does the strategy achieve in practice, and what must be tested?
8. **Operations:** Are quotas, retries, idempotency, deployment consistency, observability, and rollback covered?
9. **Tradeoff:** Is the option more complex or costly than necessary for the stated requirement?

4.8 References

4.8.1 Source-use note

The supplied slide deck was treated as supplementary study material, not as an authoritative source for current service behavior. The report uses the supplied official exam guide for scope and weighting and current AWS documentation for service behaviors and tradeoffs. Stable numeric limits were included only when directly relevant (for example, SQS long polling up to 20 seconds); no claim here should be read as exhaustive coverage of all SAA-C03 services, quotas, or regional feature availability.

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5 Domain 3: Design High-Performing Architectures

The official SAA-C03 outline assigns **24% of the exam** to Domain 3 and divides it into five tasks: scalable storage, elastic compute, high-performing databases, scalable networks, and data ingestion/transformation. The attached guide is the controlling scope for this report 30; the current AWS certification page confirms the same task boundaries and knowledge/skill statements.1 The supplementary slide deck 31 is useful for hybrid-migration context, but AWS service documentation is the authority for current names and behavior.

A performance answer is not automatically the fastest individual service. It is the design that meets the stated latency, throughput, concurrency, availability, growth, and operational constraints with an appropriate amount of management. AWS's Performance Efficiency pillar emphasizes using managed and serverless technologies, deploying globally when latency requires it, experimenting with alternatives, and selecting technology according to access patterns rather than habit.2

5.1 1. A repeatable scenario method

When a question asks for a high-performing architecture, identify the bottleneck before selecting a service.

1. **Translate the requirement into a measurable dimension.** Distinguish latency (time to respond), throughput (bytes or requests per second), concurrency (in-flight work), capacity (how much data or compute can be handled), and elasticity (how quickly capacity follows demand). "Fast" without a workload pattern is not enough.
2. **Classify the workload.** Ask whether it is read-heavy or write-heavy, synchronous or asynchronous, steady or spiky, interactive or batch, stateful or stateless, regional or global, and structured or unstructured.
3. **Choose the primitive that matches the access pattern.** Object, file, and block storage solve different problems; relational, key-value, document, and in-memory databases have different scaling models; ALB, NLB, CloudFront, and Global Accelerator operate at different layers.
4. **Remove coupling.** Put a queue between producers and workers when work need not be completed in the request path. Use independent target groups, services, partitions, streams, or tables so one component can scale without scaling the entire system.
5. **Scale on a useful signal.** CPU is sometimes useful, but queue backlog per worker, requests per target, concurrency, latency, or a domain-specific utilization metric can be a better trigger. Set minimum, maximum, cooldown, and burst behavior deliberately.
6. **Check the failure and consistency implications.** A cache can be stale, a read replica can lag, asynchronous work changes user-visible completion semantics, and a single-AZ storage class trades resilience for latency or cost. Do not optimize one dimension by silently violating another requirement.
7. **Prefer the least operationally complex service that meets the need.** Managed services and serverless options reduce provisioning work, but they do not remove quota, connection, cold-start, partition-key, or network-placement considerations.

A strong exam answer normally changes only the bottleneck: add CloudFront for global static-content latency, add a cache for repeated reads, add read replicas for database reads, add a queue and workers for asynchronous bursts, or choose a provisioned-performance storage tier for sustained I/O. Replacing every component with a larger instance is usually less elastic and less independent.

5.2 2. Task 3.1 — High-performing and scalable storage

5.2.1 Storage types and access patterns

- **Object storage** stores whole objects addressed by a key and is accessed through APIs. Amazon S3 is the default choice for durable, massively scalable unstructured data, static content, backups, data lakes, and analytics input. It is not a mounted POSIX file system and is not a replacement for a low-latency block device.

- **File storage** presents a shared file system and is appropriate when multiple clients need file and directory semantics. Amazon EFS is a managed, elastic NFS file system for Linux workloads. Amazon FSx provides purpose-built file systems such as Windows File Server and high-performance Lustre; select based on protocol and workload rather than choosing EFS for every file problem.
- **Block storage** provides raw volumes attached to compute. Amazon EBS is appropriate for boot volumes, transactional databases, and applications that expect a disk device. Performance is described mainly in IOPS and throughput and depends on both the volume and the attached instance.

The exam's core tradeoff is **access semantics versus performance and scale**. An application that needs shared file locking and a mounted path should not be moved to S3 merely because S3 scales highly. Conversely, an archive or object collection should not be placed on a fleet of EBS volumes just to imitate a file system.

5.2.2 Amazon S3 performance design

Amazon S3 is object storage for data lakes, websites, backup/restore, archives, and big-data analytics. General-purpose buckets are the normal default; specialized storage classes address access frequency and latency. S3 Express One Zone is a single-AZ, low-latency option for latency-sensitive object workloads, so it should be selected only when the workload can accept that Availability Zone resilience model.[3]

For high throughput, use **parallel requests** and reusable HTTP connections. Large objects can be uploaded with multipart upload and downloaded using concurrent parts or byte-range GETs. The right concurrency is workload-dependent; measure CPU, network, and client-side bottlenecks rather than assuming that more parallel requests always help. AWS's S3 performance guidance also recommends SDK retry behavior with exponential backoff for transient throttling and latency, and a fresh connection/DNS lookup for selected latency-sensitive retry cases.[4]

For a hot read working set, put **CloudFront** in front of S3 so popular objects are served from edge caches. For an application that needs application-aware object lookup, an in-memory cache such as ElastiCache can avoid repeated S3 GETs, but the application must implement population, invalidation, and stale-data policy. Caching improves latency and reduces origin request volume; it does not change S3's durability or make mutable data automatically consistent with the cache.[4]

S3 Transfer Acceleration is a specialized choice for geographically distant clients uploading to or downloading from a regional S3 bucket. It uses AWS edge locations as an optimized ingress path. It is not the same as CloudFront caching and should be justified by measured cross-continent transfer performance.

Exam pitfalls:

- Do not choose S3 for a workload that requires in-place block updates, a boot disk, or POSIX file locking.
- Do not treat a key prefix as a database index. Model keys and metadata for the access pattern, and use a database when you need queryable attributes and transactional behavior.
- Do not assume CloudFront eliminates origin reads for every request. Cache-key design, TTL, invalidation, and dynamic request behavior determine the hit rate.
- Do not use S3 One Zone or S3 Express One Zone when the scenario requires multi-AZ durability or regional resilience without saying why the risk is acceptable.

5.2.3 Amazon EBS

EBS is persistent block storage attached to EC2. SSD-backed volumes are suited to transactional, small-I/O workloads where IOPS and latency dominate. HDD-backed `st1` and `sc1` are throughput-oriented and

fit sequential big-data, log-processing, or infrequently accessed data; they are not the default for random transactional I/O. AWS's volume comparison identifies `gp3` as the general-purpose SSD choice when a workload needs independently configurable storage, IOPS, and throughput, and `io2` Block Express for demanding, sustained IOPS and very low-latency workloads.[5]

A useful exam distinction is **volume performance versus instance performance**. The instance must be EBS-optimized and have sufficient EBS bandwidth to consume the provisioned volume performance. A larger volume is not automatically the right answer when the requirement is more IOPS or throughput; with `gp3`, performance can be provisioned separately from size within the service's current limits.[5]

EBS is AZ-scoped. For resilience, use snapshots, replication, a Multi-AZ database service, or an architecture that can recreate/attach storage in the target AZ. An EBS volume is not a shared multi-AZ file system. Multi-Attach has specific restrictions and should not be used as a generic shared-disk solution.

Scenario reasoning: Choose `gp3` for a typical web server or medium database needing predictable SSD performance; choose `io2` when sustained, high IOPS and low latency are explicit; choose `st1` for large sequential log processing; choose S3 when the data is an object collection or analytics lake rather than an attached disk.

5.2.4 Amazon EFS and hybrid storage

Amazon EFS is a shared, elastic NFS file system. Its **General Purpose** performance mode has the lowest per-operation latency and is the recommended mode for most new file systems. EFS throughput can be Elastic, Provisioned, or Bursting. Elastic throughput fits spiky or difficult-to-forecast workloads; Provisioned throughput fits a known performance requirement; Bursting fits workloads whose throughput should scale with the amount of data in the file system.[6]

EFS is a strong fit for shared application content, home directories, container workloads that need a common file system, and concurrent access from multiple AZs. It is not a low-latency local scratch disk and should not be selected when the application needs a single EC2 instance's block-device semantics. EFS's storage classes also distinguish frequently accessed Standard data from lower-latency-insensitive IA and Archive data; lifecycle movement can change first-byte latency.

Hybrid storage means the location and access path are part of the design. AWS DataSync is a managed, high-speed service for moving or replicating file/object data between on-premises, AWS storage, and supported other-cloud storage. It uses parallel, multithreaded transfers, encryption, and integrity validation, and can target S3, EFS, or FSx.[7] AWS Storage Gateway supplies a hybrid appliance pattern: S3 File Gateway exposes file shares backed by S3, Volume Gateway provides cached or stored volumes, and Tape Gateway supports virtual tape workflows.[8]

Use DataSync for scheduled or one-time online transfer and replication; use Storage Gateway when on-premises applications need an ongoing storage protocol interface and local cache; use Direct Connect or VPN when the network path itself must be private or predictable. A transfer service does not by itself solve application consistency, bandwidth, or cutover sequencing.

5.2.5 Storage decision checklist

- Need API-addressed objects, very large scale, static content, backup, or data lake? Start with S3.
- Need shared files and NFS semantics across compute? Start with EFS or a purpose-built FSx file system.
- Need a disk for an EC2 operating system or transactional engine? Start with EBS.
- Need repeated global reads? Add CloudFront; for application-controlled hot data, consider ElastiCache.

- Need high, sustained random I/O? Use a provisioned-performance EBS type and verify instance bandwidth.
- Need online hybrid movement? Use DataSync; need a persistent on-premises file/volume/tape interface? Consider Storage Gateway.

5.3 3. Task 3.2 — High-performing and elastic compute

5.3.1 Compute selection

Choose compute by control, execution duration, workload shape, and scaling model.

- **EC2** provides the most control over OS, networking, instance family, local storage, and specialized hardware. Use it when the workload needs persistent processes, custom agents, unusual OS/kernel behavior, predictable instance-level performance, or software that cannot fit Lambda/container constraints. Select instance family by measured CPU, memory, network, storage, or accelerator need; do not select by vCPU count alone.
- **Amazon ECS** is AWS's managed container orchestration service; it is a good fit for containerized services where the team wants AWS-native orchestration and does not need Kubernetes API compatibility.
- **Amazon EKS** is managed Kubernetes control-plane orchestration. Choose it when Kubernetes portability, ecosystem, or existing operational expertise outweighs the added control-plane and platform complexity.
- **AWS Fargate** runs ECS tasks without the customer managing EC2 hosts. Each task has an isolation boundary and the task definition specifies CPU, memory, networking, and IAM. Fargate is attractive for stateless services and variable workloads, but task sizing, startup time, networking, and service quotas still matter.9
- **Lambda** is event-driven, short-lived serverless compute. It scales execution environments with in-flight requests, so it is a strong fit for APIs, event processing, scheduled jobs, and asynchronous transformations. It is a poor fit for long-running processes, arbitrary privileged OS control, or workloads whose connection and cold-start behavior cannot be designed around.
- **AWS Batch** is for queued, containerized batch, simulation, analytics, and compute-intensive work. It provisions and schedules compute capacity across EC2, Fargate, and related options, so the job is not forced into the request/response model.10
- **Amazon EMR** is for distributed big-data frameworks such as Spark and Hadoop. It is appropriate when the workload benefits from cluster-oriented distributed processing and control of the processing environment, not for a simple point query against S3.11

The "serverless" label does not mean "unlimited" or "zero latency." Lambda has account and function concurrency controls, Fargate has task and resource quotas, and a serverless database still has capacity, partition, or connection behavior. Size the resource and test the p95/p99 path.

5.3.2 Decouple before scaling

Amazon SQS is a durable, hosted queue that decouples producers and consumers. Producers can return quickly while workers process the backlog at their own rate; consumers can be scaled independently. Standard queues provide at-least-once delivery, so consumers should be idempotent. Use visibility timeout to hide a message during processing and delete it only after successful work. Use a dead-letter queue for repeated failures and poison messages.12

Amazon SNS is a publish/subscribe topic for fanout to multiple subscribers. A common architecture is **SNS topic → multiple SQS queues**, which gives each consumer an independent durable backlog and retry behavior. Use SQS alone for one logical worker pool; use SNS plus queues when several independent consumers need the same event. Do not put an interactive user request behind a queue unless the user experience can represent asynchronous completion.

Queues absorb bursts but do not remove work. Monitor age of oldest message, backlog per worker, visibility-timeout expirations, and DLQ depth. Scale workers on a backlog-per-worker metric when processing time is the actual bottleneck; CPU alone can be misleading for I/O-bound workers.

5.3.3 Auto Scaling and signals

EC2 Auto Scaling maintains a fleet's desired capacity and can replace unhealthy instances. Application Auto Scaling applies comparable policies to services such as ECS, DynamoDB, Aurora, Lambda provisioned concurrency, and ElastiCache. Target tracking sets a CloudWatch metric and a target value; AWS creates/maintains alarms and adds or removes capacity to keep the metric near the target. Scale-out is intentionally more aggressive than scale-in, and cooldowns prevent oscillation.[13]

For web services, useful signals include ALB requests per target, latency, queue depth, and saturation. For ECS, target tracking can use CPU or memory, step scaling can react differently to different alarm breaches, scheduled scaling handles known peaks, and an SQS backlog-per-task metric is suitable for asynchronous workers.[14]

Do not scale on an aggregate metric that hides the hot component. If one service handles search and another handles checkout, place them in independent target groups/services and scale them separately. If a dependency is the bottleneck, scaling the callers can worsen the incident; use backpressure, connection limits, caching, or rate limiting.

5.3.4 Lambda concurrency and resource sizing

Lambda **concurrency** is the number of in-flight requests, not requests per second. A practical relationship is:

$$\text{concurrency} \approx \text{average requests per second} \times \text{average duration in seconds}$$

Each concurrent invocation needs an execution environment. Lambda automatically creates and reuses environments until concurrency limits or downstream limits are reached. **Reserved concurrency** sets a function's maximum and protects a portion of account capacity for that function; it can also starve other functions if allocated carelessly. **Provisioned concurrency** pre-initializes environments to reduce cold-start latency; it increases cost and must be sized and scheduled for the latency-sensitive path.[15]

Lambda memory is also a CPU allocation lever, not only a memory setting. Increase memory when execution is CPU-bound or latency-sensitive, then benchmark total duration and cost. Protect databases from Lambda fan-out with RDS Proxy, a queue, connection reuse, or a concurrency cap. A function that can scale faster than its database is not a high-performing end-to-end design.

5.3.5 Common compute architectures

Stateless web tier: Route users through Route 53 and optionally CloudFront; use ALB across multiple AZs; place stateless EC2 instances in an Auto Scaling group, or ECS/Fargate tasks behind the ALB. Store

session state in a suitable external store rather than local instance memory. Scale using requests per target or a workload-specific metric.

Asynchronous worker tier: API or event producers publish to SNS or send to SQS; a worker fleet on ECS/Fargate, Lambda, or EC2 consumes messages; failed messages go to a DLQ; outputs are written to S3/DynamoDB/RDS. Scale workers by backlog per worker and keep processing idempotent.

Batch analytics: Land source data in S3; use Glue Catalog and Glue ETL or EMR for transformation; query with Athena; visualize with QuickSight. Use AWS Batch when the unit of work is independent, queued, and compute-intensive rather than a distributed SQL/Spark job.

Pitfalls: scaling instances while requests are stateful; adding workers without increasing the downstream database capacity; choosing Kubernetes when ECS/Fargate meets the requirement; using Lambda for a long-running or highly stateful process; ignoring container image size, task startup, subnet IP capacity, or ENI/service quotas.

5.4 4. Task 3.3 — High-performing database solutions

5.4.1 Match data model and access pattern

Database selection begins with the data model and query shape, not brand preference. AWS groups choices into relational and purpose-built non-relational models including key-value, document, in-memory, graph, time-series, and wide-column. Relational services fit structured tables, joins, transactions, and less predictable ad-hoc queries. Key-value and document stores fit known access patterns and horizontal scale; in-memory stores fit hot, ephemeral data; wide-column databases fit high-throughput, low-latency workloads with a suitable partition model.16

- **Amazon RDS** is managed relational database infrastructure for familiar engines such as PostgreSQL, MySQL, MariaDB, Oracle, SQL Server, and Db2. Use it when the engine compatibility and SQL model matter.
- **Amazon Aurora** is AWS's cloud-optimized relational option compatible with MySQL or PostgreSQL editions. It is a fit when the relational model is required but higher managed throughput, cluster storage, reader scaling, or Aurora-specific features are useful.
- **Amazon DynamoDB** is a serverless key-value/document database for predictable primary-key access at very high scale. It can deliver single-digit millisecond performance when the table and key design avoid hot partitions; it is not a drop-in replacement for joins or arbitrary SQL.
- **Amazon ElastiCache** is a managed in-memory cache/data store using Valkey, Memcached, or Redis OSS. Use it to reduce repeated database reads and provide very low-latency access to a working set; design invalidation, eviction, persistence, and failover explicitly.17

Engine selection within RDS/Aurora is a compatibility and workload decision. A homogeneous migration preserves the same engine and usually reduces application change. A heterogeneous migration changes engine or data model and can improve managed cost/performance, but requires schema, SQL, extension, indexing, and operational changes. The supplementary slides describe AWS DMS and schema conversion as migration aids; treat the current AWS DMS documentation as the authority for supported sources, targets, and conversion behavior.

5.4.2 Capacity planning

For RDS/Aurora, plan for CPU, memory, storage throughput, IOPS, connections, read/write mix, and query latency. Provisioned IOPS is appropriate when the workload needs sustained, predictable I/O rather than

a burstable general-purpose volume. Scale vertically when a single writer or instance is the constraint; scale reads horizontally with replicas; partition or redesign when writes exceed a single-node model.

For DynamoDB, choose **on-demand** when traffic is unknown, spiky, or operational simplicity is more important than a preplanned rate. Choose **provisioned** when traffic is steady and predictable and capacity can be forecast. On-demand automatically manages throughput; provisioned gives capacity and cost predictability but requires correct sizing and, where needed, auto scaling.[18]

DynamoDB performance depends heavily on partition-key distribution. A monotonically increasing or overly popular key can create a hot partition even when aggregate table capacity appears adequate. Use a high-cardinality, evenly distributed key and design indexes around real access patterns. Avoid scans for request-path reads; model items so the required query is a keyed query.

5.4.3 Replication and reads

A read replica is a read-only copy used to offload read traffic. It does not make writes scale and it can lag, so read-after-write behavior must be specified. Route read-only queries to replicas only when eventual consistency is acceptable, and keep strongly consistent or write-dependent reads on the writer where required.

Aurora Replicas are read-only instances in an Aurora cluster. The reader endpoint distributes read connections across replicas; replicas also provide failover targets. The current Aurora guide describes up to 15 replicas per cluster and notes that replica lag is usually low but can increase during heavy writes, so the exact lag should be measured for the workload rather than assumed.[19]

For a global application, local read replicas or an Aurora global database can reduce cross-Region read latency, but cross-Region replication and failover imply replication lag, write-region choices, conflict behavior, and data-residency considerations. Multi-AZ standby is primarily a high-availability/failover pattern, not a read-scaling endpoint.

5.4.4 Caching patterns

Caching is a performance optimization, not the system of record. Common patterns are:

- **Cache-aside:** The application reads the cache first; on a miss it reads the database and populates the cache. This is flexible and common, but invalidation and stampede control are application responsibilities.
- **Read-through:** A cache layer loads missing values for the application. It simplifies callers but depends on the selected service/client behavior.
- **Write-through:** Writes update the cache and persistent store together or in a managed sequence. It can reduce stale reads but adds write latency and failure coordination.
- **TTL and negative caching:** Expire data deliberately, and optionally cache “not found” results briefly to protect the database. TTL is a consistency tradeoff.

Use CloudFront for HTTP/content edge caching, ElastiCache for application data and sessions, and database read replicas for durable database read scaling. Do not choose ElastiCache as the only copy of irreplaceable data unless the selected mode and durability characteristics explicitly satisfy the requirement.

5.4.5 Connections and proxies

Short-lived compute can create connection storms against a relational database. Amazon RDS Proxy pools and reuses database connections, queues or throttles excess client connections, and can preserve

application connections during some database failovers. It protects the database from connection oversubscription; it does not create more database CPU, memory, storage I/O, or write capacity.[20]
Use connection pooling in long-lived services as well, but avoid holding connections idle unnecessarily. Monitor active connections, database CPU/memory, query latency, pool saturation, and proxy borrow latency. If queries pin proxy sessions because of session state, multiplexing benefits may be reduced.

5.4.6 Database scenario patterns and pitfalls

Read-heavy relational application: Aurora/RDS writer plus read replicas, reader endpoint, and ElastiCache for repeated hot reads. Keep writes and read-after-write traffic on the writer. Scale the writer and storage independently from readers.
High-scale key-value workload: DynamoDB with a well-distributed partition key, Query operations instead of Scan, on-demand capacity for unknown/spiky traffic or provisioned capacity for predictable steady traffic. Use DAX or ElastiCache only when the access pattern benefits from caching and the consistency semantics are acceptable.
Lambda-to-relational integration: Lambda functions consume events or API requests; reserved concurrency protects the account and downstream database; RDS Proxy pools connections; a queue absorbs bursts if the work is asynchronous. This is better than allowing unlimited concurrent invocations to open one connection each.
Pitfalls: treating Multi-AZ standby as a read replica; using read replicas for writes; assuming replica reads are immediately consistent; picking DynamoDB before identifying primary-key queries; using cache TTLs without a stale-data decision; scaling app servers while a single relational writer is saturated; and quoting a service quota as if it were universal when quotas vary by Region, account, engine, or configuration.

5.5 5. Task 3.4 — High-performing and scalable networks

5.5.1 Layer the network

A common regional topology is Route 53 → CloudFront or a regional entry point → public subnets containing ALB/NLB → private application subnets → private database/cache subnets. Use multiple AZs, route tables appropriate to each tier, security groups as stateful workload-level controls, and network ACLs only for the additional subnet boundary controls required by the design. Reserve CIDR space for future subnets, AZ expansion, VPC peering/TGW attachments, and hybrid routes.
Resource placement is a performance decision. Put compute close to the database and storage it uses to avoid unnecessary cross-AZ or cross-Region hops, but retain the required AZ distribution for resilience. Cross-AZ traffic can add latency and data-transfer cost; do not sacrifice fault isolation simply to avoid every cross-AZ byte. Keep chatty components together where appropriate and make the failure-domain tradeoff explicit.

5.5.2 CloudFront versus Global Accelerator

Amazon CloudFront is a content delivery network. It caches and serves HTTP/HTTPS static or dynamic content from edge locations nearest to viewers, reducing viewer latency and origin load. It is the natural choice for cacheable web assets, media, downloads, and HTTP application acceleration. CloudFront cache keys, TTLs, invalidations, origin protocol, and dynamic request behavior determine actual benefit.[21]
AWS Global Accelerator provides static anycast IP addresses and enters the AWS global network at an edge location, then routes traffic to healthy regional endpoints. It is useful for global TCP/UDP or non-

cacheable applications, stable IP requirements, fast regional failover, and reducing exposure to variable internet paths. Standard accelerators can use ALB, NLB, EC2, or Elastic IP endpoints across Regions; Global Accelerator does not cache application content.22

A useful exam rule is: **CloudFront for content and HTTP caching; Global Accelerator for global network entry, static IPs, and endpoint routing.** They can be combined when a workload needs both content caching and global application failover, but do not choose CloudFront merely because users are global if the requirement is a non-HTTP, non-cacheable protocol.

5.5.3 Load balancing

Elastic Load Balancing distributes traffic across healthy targets in one or more AZs and scales as traffic changes. ALB operates at Layer 7 and supports host/path/header/query routing, HTTP/HTTPS, target groups, health checks, and targets including EC2, IP addresses, containers, and Lambda. The default target-group algorithm is round robin; least outstanding requests is available where appropriate.23

Choose **ALB** for HTTP/HTTPS content-based routing, microservices, containers, and application-layer features. Choose **NLB** for Layer 4 TCP/UDP/TLS, very high connection rates, static IP/EIP needs, or protocols that are not HTTP. Choose **Gateway Load Balancer** for transparent virtual appliances. A load balancer distributes and health-checks traffic; it does not by itself make a stateful application scalable.

For Fargate tasks using `awsvpc`, use IP targets because tasks have their own elastic network interfaces rather than sharing an EC2 instance network identity.9

5.5.4 Hybrid and private connectivity

- **AWS Site-to-Site VPN** uses encrypted IPsec tunnels over the internet. It is fast to provision and useful for backup connectivity or lower-volume hybrid traffic, but internet path performance is less predictable.
- **AWS Direct Connect** provides a private connection from a colocation or provider location to AWS. It can offer more predictable throughput and network behavior than internet connectivity. It is not encrypted by default; add MACsec where supported or VPN encryption over the connection when the requirement demands it. Design redundant connections and locations for resilience.24
- **AWS PrivateLink** uses interface endpoints/private connectivity to reach supported AWS services or endpoint services without an internet gateway, NAT device, public IP, Direct Connect, or Site-to-Site VPN for the service path. It is the right abstraction for private service consumption across VPC/account boundaries, not a general replacement for VPC-to-VPC routing.25

For hybrid designs, choose VPN when agility and encryption over the internet are the priority, Direct Connect when predictable private bandwidth/latency is important, and both when Direct Connect is the primary path with VPN backup or encrypted overlay. Direct Connect alone does not automatically provide application-level encryption.

5.5.5 Network scaling pitfalls

- One public subnet or one AZ limits both scale and failure tolerance.
- A single NAT Gateway can become a throughput, AZ-dependency, or egress-cost bottleneck; place NAT per AZ when the availability and traffic pattern justify it, or use VPC endpoints for supported AWS services.
- A large VPC CIDR does not help if subnet ranges, secondary CIDRs, ENIs, or endpoint addresses are exhausted. Plan IP capacity for autoscaling, Fargate tasks, Lambda VPC access, and future tiers.

- ALB path routing cannot repair an application that stores session state locally. Externalize state or use an explicit affinity strategy with its tradeoffs.
- Global Accelerator does not cache, and CloudFront does not replace a database read replica.
- Cross-Region placement can reduce user latency but adds replication, consistency, routing, and operational complexity. Deploy globally only when the measured latency or resilience requirement justifies it.

5.6 6. Task 3.5 — Data ingestion and transformation

5.6.1 Classify the ingestion pattern

Start with **frequency, latency, volume, ordering, replay, and format**:

- One-time or scheduled file movement favors DataSync or a managed batch transfer pattern.
- Near-real-time events favor Kinesis Data Streams, Amazon MSK when Kafka compatibility is required, or an event/queue service according to the delivery and ordering model.
- Continuous delivery of data to S3 or analytics destinations without custom consumers favors a managed delivery stream such as Kinesis Data Firehose.
- Large independent jobs favor AWS Batch; distributed Spark/Hadoop transformations favor EMR; serverless ETL and cataloging favor Glue.
- Interactive SQL over data in S3 favors Athena; dashboards and visualization can use Amazon QuickSight after the data is curated.

5.6.2 Kinesis Data Streams

Kinesis Data Streams collects and processes large, continuous streams such as logs, telemetry, clickstreams, and market data. Producers can write records immediately rather than waiting for batches, and multiple independent applications can consume the same stream for different purposes such as aggregation, archiving, and alerting. The stream's partitioning and consumer design determine parallelism and ordering; scale for records/bytes and hot partition-key behavior rather than only aggregate average rate.[26]
A common architecture is producers → Kinesis Data Streams → independent consumers. One consumer writes raw data to S3, another updates a real-time DynamoDB aggregate, and another invokes a stream-processing job. Retain raw events in durable storage for replay. Use a partition key that distributes load while preserving ordering only where ordering is required. Do not use SQS when multiple independent consumers need to read the same retained stream history; do not use Kinesis when the business only needs a simple work queue with one consumer per message.

5.6.3 Data lakes, Glue, Athena, and EMR

Amazon S3 is the durable landing zone. Separate raw, transformed, and curated prefixes or buckets; choose formats that suit query and lifecycle needs; and secure access with IAM, bucket policies, VPC endpoints, and Lake Formation where fine-grained data-lake permissions are required.
AWS Glue is serverless data integration. It discovers/catalogs sources, runs ETL/ELT jobs, cleans and transforms data, and can handle batch or streaming integrations. Its Data Catalog supplies metadata for Athena, EMR, and other analytics services. Glue jobs can scale workers with workload, but transformation design, partitioning, small-file management, shuffles, and schema quality still determine performance.[27]
Amazon Athena runs serverless SQL directly over S3 and scales queries in parallel. It is a strong choice for ad-hoc or interactive analysis without provisioning a cluster. Organize and partition data, project only

required columns, and prefer columnar formats such as Parquet when the query engine and pipeline support them; converting CSV to Parquet reduces scanned bytes and improves analytical efficiency, but it adds a transformation stage.28

Amazon EMR runs distributed processing frameworks on clusters. Use it when Spark/Hadoop-style distributed computation, custom libraries, repeated multi-step jobs, or cluster-level tuning are required. EMR can read input from S3 and write results back to S3; do not assume HDFS data survives cluster termination unless it is deliberately copied to durable storage.29

5.6.4 Transfer and secure ingestion

DataSync is appropriate for online movement and scheduled replication from on-premises NFS/SMB or other supported systems to S3, EFS, or FSx. Storage Gateway is appropriate when an on-premises application must continue using file, volume, or tape protocols while data is integrated with AWS storage. Direct Connect/VPN/private endpoints are connectivity controls; they do not replace the transfer orchestration and verification provided by DataSync.

Secure ingestion by using private VPC endpoints where supported, least-privilege IAM roles, encryption in transit and at rest, validation/checksums, and separate producer/consumer permissions. A public endpoint is not automatically wrong, but the question must justify it when private access is a stated requirement. For data lakes, restrict writes to landing locations and separate transformation roles from query/read roles.

5.6.5 Common data architectures and pitfalls

Batch data lake: source files → DataSync → S3 raw zone → Glue crawler/catalog → Glue ETL to Parquet partitioned data → Athena → QuickSight. This is the default pattern for recurring file ingestion and SQL analytics without a persistent cluster.

Streaming analytics: producers → Kinesis Data Streams → stream consumers/Glue streaming or EMR → S3 raw archive and DynamoDB/OpenSearch/warehouse aggregates → dashboard/alerts. Keep a durable raw copy for replay and independent consumers for independent scaling.

High-volume compute pipeline: events/files → SQS or Kinesis → AWS Batch or EMR workers → S3 outputs → Athena. Choose the queue when each work item should be processed once by one worker pool; choose the stream when multiple consumers need ordered/replayable events.

Pitfalls: using Athena as a low-latency OLTP database; sending millions of tiny files to S3 and expecting efficient analytical scans; using CSV when columnar output would materially reduce scans; launching a permanent EMR cluster for a sporadic job when Glue/Athena/EMR Serverless or Batch fits; failing to define replay and late-event handling; and scaling ingestion without sizing the destination, partition keys, or consumer lag.

5.7 7. Integrated exam scenarios

5.7.1 Scenario A — Global read-heavy website

Requirements: global users, cacheable images and JavaScript, dynamic API requests, low latency, multi-AZ regional service.

Design: S3 for static objects behind CloudFront; Route 53 for DNS; ALB across AZs for dynamic HTTP; stateless ECS/Fargate or EC2 Auto Scaling targets; ElastiCache for hot application reads; Aurora/RDS writer plus read replicas for durable read scaling. Use CloudFront because content is cacheable; use Global

Accelerator instead if the main requirement is global routing of non-cacheable TCP/UDP or stable IPs. Do not put user sessions only on individual instances.

5.7.2 Scenario B — Flash-sale order processing

Requirements: unpredictable burst, no loss of orders, database must not be overwhelmed.
Design: API tier accepts/validates the request and publishes an idempotent command to SQS; workers on ECS/Fargate or Lambda consume; scale on backlog per worker; use DLQ and retry policy; use DynamoDB for key-value order state or a relational writer when transactions/joins are required; protect RDS with RDS Proxy and concurrency limits. The response should communicate accepted/queued status rather than pretending the order is complete before asynchronous processing.

5.7.3 Scenario C — Large hybrid file migration

Requirements: move active on-premises files to EFS/FSx and archive cold data to S3 with integrity verification.
Design: DataSync for parallel, encrypted, validated transfer and recurring synchronization; Storage Gateway only if the existing applications need an ongoing file/volume/tape interface during transition; Direct Connect or VPN based on required private connectivity and predictable bandwidth. Preserve application consistency and plan cutover; a faster transfer channel alone does not make the migration transactionally safe.

5.7.4 Scenario D — Unknown-scale key-value API

Requirements: single-digit millisecond key lookups, unpredictable growth, no joins, no capacity planning.
Design: DynamoDB on-demand with a high-cardinality partition key and Query/GetItem access patterns. Add ElastiCache only for a measured hot working set or latency target that the database cannot meet. Avoid scans, hot keys, and a relational database merely because the team knows SQL. If the requirement adds multi-row transactions and ad-hoc joins, revisit Aurora/RDS.

5.7.5 Scenario E — Real-time telemetry plus daily reporting

Requirements: records must be visible in seconds, several consumers need the data, daily reports use durable history.
Design: producers → Kinesis Data Streams; independent consumers update a real-time aggregate and write raw records to S3; Glue transforms raw data to partitioned Parquet; Athena queries the curated zone; QuickSight visualizes results. Retain raw data for replay. Use a partition key that spreads load while preserving only the ordering actually required.

5.8 8. Final pitfalls to memorize

- **Performance is end-to-end.** A fast frontend that saturates the database is not a high-performing architecture.
- **Scale horizontally when the workload can be partitioned.** Use target groups, queues, replicas, stream consumers, or partitions rather than only larger servers.
- **Read replicas scale reads, not writes.** Multi-AZ standby is primarily failover, not read capacity.
- **A cache is not a database.** Decide TTL, invalidation, consistency, eviction, and miss behavior.
- **S3 is object storage.** Use EBS for a disk, EFS/FSx for shared files, and DataSync/Storage Gateway for hybrid movement or protocol integration.

- **CloudFront caches content; Global Accelerator routes network traffic.** They solve different global-latency problems.
- **ALB is Layer 7; NLB is Layer 4.** Pick by protocol and routing features, not by the generic phrase "load balance."
- **Queues absorb bursts but require idempotency and DLQs.** Standard SQS delivery is at least once.
- **Lambda can scale faster than a downstream database.** Use reserved/provisioned concurrency, RDS Proxy, pooling, or queues as appropriate.
- **Athena queries S3; Glue transforms/catalogs; EMR runs distributed frameworks; Batch schedules independent jobs.** Do not substitute them without matching the processing model.
- **Avoid unstable memorization.** Quotas, default concurrency, regional support, and feature availability can change. On the exam, use documented behavior and the scenario's stated numbers first; verify current quotas for production design.

5.9 References

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6 Domain 4: Design Cost-Optimized Architectures

AWS Certified Solutions Architect – Associate (SAA-C03)

6.1 Exam scope and the meaning of "cost-optimized"

Domain 4 is **20% of the SAA-C03 exam**. The official guide divides it into four task statements: cost-optimized storage, compute, database, and network architectures. The guide explicitly calls out cost allocation tags, AWS Cost Explorer, AWS Budgets, the AWS Cost and Usage Report (AWS CUR), storage tiering and lifecycle policies, EC2 purchasing options, scaling and utilization, database capacity and engine choices, NAT gateways, connectivity, routing, data transfer, and CDNs. It is a scope outline rather than an exhaustive list of services. 1

AWS defines a cost-optimized workload as one that **fully utilizes its resources, achieves the required outcome at the lowest possible price point, and still meets functional requirements**. Cost optimization is therefore not the same as selecting the cheapest service in isolation. It is a tradeoff among required availability, durability, latency, throughput, operational effort, resilience, licensing, data-transfer paths, and the workload's demand pattern. 2

A reliable exam method is:

1. **Translate the requirement into a demand pattern.** Is use steady, spiky, unpredictable, interruptible, latency-sensitive, read-heavy, write-heavy, archival, or continuously active?
2. **Select the resource or managed service that matches the access and performance requirement.** Avoid paying for idle capacity, over-provisioned IOPS, unnecessary replication, or a more operationally expensive design than the requirement needs.
3. **Choose the pricing model only after sizing the workload.** Commitments reduce unit cost but create utilization risk.
4. **Include secondary charges.** Storage retrieval, requests, snapshots, NAT data processing, cross-AZ or cross-Region transfer, public IPv4, endpoint hours, support, and licensed software can change the answer.

5. **Preserve the stated reliability and performance objectives.** A single-AZ or interruptible design is not cost-optimized if the requirement says multi-AZ, durable, always available, or predictable.
6. **Measure and iterate.** Tag ownership and environment, allocate costs to accounts or teams, use the right cost-analysis tool, and revisit the design as usage changes.

The supplementary presentation supplied with the task reinforces these exam ideas: choose the right size and type, turn off unused resources, replace servers with managed services where appropriate, use caching to avoid repeated retrieval, and distinguish Cost Explorer, Budgets, and CUR. Its explanatory material is useful as study context, but the current AWS documentation cited below controls where terminology or behavior differs.

6.2 Cost visibility and financial controls

6.2.1 Cost allocation and account structure

A resource tag is a key/value label. After activation, **cost allocation tags** allow AWS to organize costs in allocation reports and filter views in Cost Explorer. AWS supports AWS-generated and user-defined cost allocation tags; the relevant tags must be activated before appearing in Cost Explorer or cost allocation reports. Tags can represent application, owner, environment, or cost center. They do not replace a multi-account strategy: separate AWS accounts can provide stronger workload isolation, chargeback boundaries, service-control-policy boundaries, and clearer ownership, while consolidated billing can aggregate charges and apply organization-wide benefits. 3

Exam reasoning: if the question asks how to attribute spend to teams or applications, choose **cost allocation tags and an appropriate reporting view**. If it asks for organizational isolation or separate billing boundaries, think **AWS Organizations and consolidated billing**, not tags alone. Tags also require governance: standardize keys and values, enforce them where possible, and account for untagged shared services.

6.2.2 Which AWS cost tool to use

- **AWS Cost Explorer:** interactive investigation of historical cost and usage. It supports filters and groupings such as service and account, trend analysis, forecasts, and RI reports. Current documentation says the console can show up to 13 months of historical data and forecast up to 18 months; treat these as current product behavior rather than timeless architectural limits. Cost Explorer is the natural answer for "Which service, Region, account, or usage type caused the increase?" 5
- **AWS Budgets:** proactive thresholds and notifications for actual or forecasted cost and usage. It also supports RI and Savings Plans utilization and coverage budgets, with notifications through email or Amazon SNS. A budget alert is not an instantaneous hard spending cap; billing data can lag resource use, and actions must be configured deliberately. 6
- **AWS Cost and Usage Report (AWS CUR):** the most detailed report, delivered to an S3 bucket. It can break costs down by hour, day, or month and by product, resource, and user-defined tags. Use CUR for durable, queryable chargeback or FinOps analysis with Athena, Redshift, or another reporting pipeline —not for a quick console investigation. 4
- **AWS Pricing Calculator:** a planning tool for modeling an architecture before deployment, testing service and payment options, and exporting estimates. It is an estimate based on the inputs supplied, not a guarantee of the final bill; validate assumptions such as hours, requests, data transfer, storage growth, and data retrieval. 18

- **AWS Trusted Advisor and Compute Optimizer:** useful follow-up signals for idle, underutilized, or oversized resources, but a recommendation must still be checked against performance, resilience, licensing, and business requirements.

A frequent distractor is to choose Budgets for detailed historical attribution or CUR for a one-click alert. Match the tool to the question: **investigate, alert, or analyze at line-item depth**.

6.3 Task 1: Design cost-optimized storage solutions

6.3.1 Match the storage type to the access pattern

- **Amazon S3 (object):** objects accessed through APIs, suitable for data lakes, static content, logs, backups, and archival. It scales without pre-provisioning block capacity. Choose a class based on access frequency, retrieval latency, resiliency, minimum duration, and retrieval/transition costs.
- **Amazon EBS (block):** persistent block volumes attached to EC2, commonly for boot volumes and databases that need block semantics. Cost depends on provisioned capacity and, for relevant types, provisioned IOPS and throughput.
- **Amazon EFS (file):** shared file semantics for multiple compute clients. Select it when shared, POSIX-like file access matters; do not use it merely because it is familiar if an object or block interface fits better.
- **Amazon FSx:** managed file systems for specific file-system capabilities and performance requirements. The specialized capability may justify its cost; it is not automatically cheaper than S3 or EFS.
- **Instance store:** local, ephemeral block storage. It can be inexpensive for caches, scratch space, or reconstructable temporary data, but data is lost when the instance lifecycle or host failure removes it. Do not select it for durable primary data.
- **Hybrid transfer/storage services:** AWS DataSync is for managed bulk data movement and synchronization; AWS Storage Gateway presents cloud-backed storage integrations to on-premises applications; AWS Transfer Family provides managed SFTP, FTPS, or FTP endpoints. Select based on the migration or protocol requirement rather than assuming that all are interchangeable.

6.3.2 S3 storage classes and lifecycle decisions

S3 Standard is for frequently accessed objects. **S3 Intelligent-Tiering** is a strong default when access is unknown, changing, or unpredictable: it automatically moves eligible objects between access tiers and has monitoring/automation charges but no retrieval charges for its access tiers. **S3 Standard-IA** is for long-lived, infrequently accessed data that still needs millisecond access and has retrieval charges. **S3 One Zone-IA** is cheaper and single-AZ, so it is appropriate only when data is recreatable or the single-AZ resilience tradeoff is acceptable. 8

S3 Glacier Instant Retrieval serves rarely accessed archives requiring millisecond access. S3 Glacier Flexible Retrieval is for archives retrieved in minutes to hours, and S3 Glacier Deep Archive is for data rarely accessed and acceptable to restore over hours. Glacier classes have retrieval fees and minimum storage durations. The current S3 documentation lists 30-day minimums for Standard-IA and One Zone-IA, 90 days for Glacier Instant Retrieval and Flexible Retrieval, and 180 days for Deep Archive; use the current pricing page when a question asks for an exact price or when service terms may change. 8

S3 Lifecycle rules automate **transition actions** and **expiration actions**. A typical lifecycle is Standard for the hot period, then Intelligent-Tiering or an IA class, then Glacier, then expiration after the required retention period. Include transition-request costs, retrieval charges, minimum-duration charges, object size, and the risk that frequent access makes a cheaper storage class more expensive overall. Lifecycle

rules also apply to existing objects when added; versioned buckets require explicit treatment of noncurrent versions and delete markers, or storage can continue to grow. 9

Other exam-relevant S3 choices:

- **Batch uploads versus many small operations:** aggregate or batch data where the application permits it, because request overhead and per-object metadata can matter. Preserve required object-level semantics.
- **S3 versioning:** improves recovery and rollback but retains previous versions. Add lifecycle rules for noncurrent versions when policy permits; versioning is not a free backup strategy.
- **Requester Pays:** use when the data owner wants requesters to bear request and data-transfer charges. It changes who pays; it does not make access free.
- **S3 gateway VPC endpoint:** private access from a VPC to S3 without an internet gateway or NAT device and with no additional endpoint charge. It is Region-specific and cannot be used by on-premises networks, peered VPCs in another Region, or through a transit gateway; those paths may require an interface endpoint or another design. 15

6.3.3 EBS sizing and snapshot cost

Choose the least expensive EBS volume type that meets the measured IOPS, throughput, latency, durability, and boot requirements. Current documentation describes **gp3** as general-purpose SSD with independently configurable IOPS and throughput, **io2 Block Express** for demanding, consistent IOPS and low latency, and **st1/sc1** HDD types for throughput-oriented workloads where low storage cost matters. Provisioned IOPS is wasteful if the workload cannot use it; a larger volume is not automatically a substitute for required IOPS or throughput. 10

EBS snapshots are incremental, but billing is based on the stored snapshot data. Deleting a snapshot does not necessarily remove all blocks retained by later snapshots. Set backup frequency and retention to the recovery-point objective, delete obsolete volumes, and review snapshot retention. A snapshot is not the same as a cross-Region disaster-recovery copy; copy and retention requirements can add storage and transfer cost.

6.3.4 Storage scenario reasoning

Scenario: "A compliance archive is almost never read, must be retained for years, and can take hours to restore." Choose an S3 archival class such as Glacier Deep Archive, then account for minimum duration, retrieval, and expiration rules. Do not choose Standard-IA merely because it is cheaper than Standard; the archive's access latency and low frequency point to Glacier.

Scenario: "Access is unpredictable and the application cannot reliably forecast when data will be read." Choose S3 Intelligent-Tiering rather than guessing Standard-IA or Glacier. If the data is recreatable and loss of one AZ is acceptable, One Zone-IA may be a valid optimization; it is not valid for the only copy of irreplaceable data.

Scenario: "Private EC2 instances download large amounts of objects from S3." Add an S3 gateway endpoint and route the S3 prefix list to it. This removes the NAT path and its processing charge. Do not answer with an interface endpoint unless the topology requires access from on-premises, another Region, or a transit gateway.

6.4 Task 2: Design cost-optimized compute solutions

6.4.1 Size and scale for demand

Use utilization evidence to right-size instance families and sizes. Separate baseline from burst demand: Auto Scaling can keep a small baseline and add capacity as load rises, while scheduled scaling handles known patterns. Containers and serverless can reduce the cost of idle infrastructure when the application can scale by task, request, or event. AWS guidance explicitly recommends considering managed services, serverless, containers, and event-driven designs, while also evaluating license cost and operational effort. [17]

A cost-optimized design does not blindly replace every EC2 workload with Lambda. Lambda is attractive for event-driven, variable, short-duration work, but runtime, concurrency, package, networking, execution, and observability characteristics must fit. Fargate can remove node management for containers, but always-on high utilization may be less expensive on EC2 with the right commitment. A managed service can cost more per unit while reducing labor, failure risk, and idle management overhead; include the total workload cost, not only the hourly line item.

6.4.2 EC2 purchase options

The current EC2 purchasing guidance distinguishes these choices: [7]

- **On-Demand:** pay for running capacity without a long-term commitment. Use for new, short-lived, irregular, or uncertain workloads and as the flexible baseline while usage is being measured.
- **Savings Plans:** commit to a consistent spend amount in USD per hour for one or three years. Use when compute spend is predictable but instance configuration may change; flexibility is the key tradeoff.
- **Reserved Instances:** commit to a consistent instance configuration, including Region and instance type, for one or three years. Use when the configuration is stable and continuously used; the commitment creates underutilization risk if the workload changes.
- **Spot Instances:** use spare EC2 capacity at a lower price, with the possibility of interruption. Use only for fault-tolerant, restartable, checkpointed, or distributed work such as batch processing, rendering, or stateless workers. Do not use Spot alone for a stateful, single-instance production database.
- **Dedicated Hosts or Dedicated Instances:** select for compliance, tenancy, or licensing constraints, not as the default cost optimization. Dedicated Hosts can help use existing server-bound licenses, but dedicated capacity can cost more.
- **Capacity Reservations:** reserve capacity in a specific Availability Zone when capacity assurance is the requirement. A capacity reservation is not simply a cheaper purchasing model.

Do not choose a three-year commitment just because it has the lowest advertised unit rate. First measure steady-state usage, validate business duration, consider growth and instance-family changes, and monitor utilization and coverage. Commitments apply to eligible usage; they do not eliminate storage, data transfer, load balancer, or other service charges.

6.4.3 Load balancing and compute architecture

Select the load balancer that matches the protocol and routing requirement. Application Load Balancer is appropriate for HTTP/HTTPS content-based routing; Network Load Balancer is for high-performance TCP/UDP/TLS-style flows and static-IP or connection requirements; Gateway Load Balancer is for inline virtual appliances. A cheaper-looking single load balancer is not a cost optimization if it violates

availability or protocol requirements. Conversely, deploying multiple layers without a requirement adds hourly and data-processing charges.

Common compute patterns:

- **Steady web tier:** ALB, multi-AZ Auto Scaling, and a right-sized baseline. Use Savings Plans or RIs only after steady usage is established.
- **Spiky asynchronous workload:** queue or event source, Lambda or a scalable worker fleet, and backpressure. Pay for execution or workers when work exists rather than keeping peak servers running.
- **Batch workload:** Spot-capable Auto Scaling groups or a managed batch scheduler, checkpointing, retry, and an On-Demand fallback for deadlines.
- **Dev/test:** scheduled start/stop, smaller instances, ephemeral environments, and automated cleanup. A budget alert alone does not stop forgotten resources.

6.5 Task 3: Design cost-optimized database solutions

6.5.1 Choose the data model and service before capacity

Start with access patterns, consistency, query shapes, write/read rates, retention, latency, and operational requirements. A relational engine is appropriate when transactions, joins, relational constraints, or SQL compatibility are central. Amazon Aurora or Amazon RDS can reduce database administration compared with self-managed databases, but the instance, storage, I/O, backup, and Multi-AZ components still incur costs. DynamoDB is appropriate for key-value or document access at scale when the access pattern can be modeled around partition keys and indexes.

Do not select DynamoDB solely because it is serverless or select RDS solely because it is familiar. A relational workload forced into a non-relational model can increase development cost and operational risk; an always-on relational instance for a highly variable key-value workload can waste capacity.

6.5.2 Capacity modes and commitments

DynamoDB **on-demand mode** uses pay-per-request pricing and removes capacity planning, making it a strong fit for unpredictable or newly launched workloads. **Provisioned mode** charges for the read/write capacity provisioned, whether or not all of it is consumed, and can be cost-effective for steady, predictable traffic when capacity is forecast accurately. 12

For a predictable, continuously running RDS workload, Reserved DB Instances can reduce the DB instance hourly cost in exchange for a one- or three-year commitment tied to eligibility such as Region, engine, and instance family. They do not discount every part of the bill: storage, backups, and I/O remain separate charges. Choose reservations only after confirming utilization and expected engine/instance stability. 11

Read replicas can reduce pressure on a primary and improve read scalability, but they add instance, storage, replication, and possibly cross-Region transfer cost. Caching with ElastiCache can reduce repeated database reads and improve latency, but it introduces cache nodes, invalidation logic, and an availability design. Add a cache when the hit rate and reduced database load justify it—not as an automatic response to every slow query.

6.5.3 Database backup and retention

Match snapshots, automated backups, point-in-time recovery, export, and cross-Region copies to the recovery objectives and retention policy. Long retention and many cross-Region copies can dominate database cost. Delete test databases, unused snapshots, and abandoned read replicas. For a production

system, “cheapest” cannot mean removing backups or reducing retention below the stated compliance and recovery requirement.

6.5.4 Database scenario reasoning

Scenario: “Traffic is unpredictable, the data model is simple, and the application needs automatic scaling.” DynamoDB on-demand is usually the intended answer. Provisioned capacity is a distractor unless traffic is steady and forecastable.

Scenario: “A relational database runs continuously with stable engine, Region, and instance family.” RDS Reserved DB Instances may lower the instance portion of cost. Check the question for storage, I/O, backup, and Multi-AZ charges that remain.

Scenario: “The primary database is read-heavy and repeatedly serves the same data.” First consider query/index tuning and a cache; add read replicas when read scaling or isolation is required. Do not use cross-Region replicas only for cost unless the question justifies their resilience or locality value.

6.6 Task 4: Design cost-optimized network architectures

6.6.1 NAT gateways and endpoints

A NAT gateway has an hourly availability charge and a per-gigabyte data-processing charge. AWS recommends keeping high-volume resources in the same AZ as the NAT gateway or deploying one per AZ when that avoids cross-AZ transfer; it also recommends endpoints when most traffic is to AWS services that support them. 13

The exam tradeoff is not simply “one NAT is cheapest.”

- **One centralized NAT gateway:** fewer hourly NAT gateways, but private-subnet traffic from other AZs may incur cross-AZ transfer and creates a cross-AZ dependency.
- **One NAT gateway per AZ:** more hourly gateway charges, but localizes traffic, improves AZ isolation, and can be cheaper for high-volume traffic or required for resilient egress.
- **NAT instance:** may have lower direct cost at very low and controlled traffic volumes, but requires patching, scaling, failover, and management. A NAT gateway is the managed, scalable default when the question prioritizes operational simplicity and availability.
- **Gateway VPC endpoints:** use for S3 and DynamoDB from VPC route tables; no additional endpoint charge. They can eliminate NAT processing for those destinations.
- **Interface VPC endpoints:** AWS PrivateLink powered, with endpoint hourly and per-GB charges. Use for supported services and private connectivity from VPCs, and when topology requirements make a gateway endpoint unavailable.

Private connectivity does not automatically mean lower cost. Compare endpoint hourly/per-GB cost with NAT processing and data-transfer paths, and check whether DNS and route tables actually send traffic through the endpoint.

6.6.2 Data transfer and placement

Model the direction and volume of every major flow: client-to-edge, edge-to-origin, AZ-to-AZ, VPC-to-VPC, Region-to-Region, VPC-to-on-premises, and private-to-public. AWS guidance recommends CloudFront, caching, Direct Connect, Site-to-Site VPN, VPC endpoints, NAT placement, and data-transfer modeling to reduce transfer cost. 14

- Keep chatty tiers in the same AZ when doing so does not violate availability goals, or use a design that minimizes cross-AZ chatter. Do not collapse a required multi-AZ architecture merely to avoid transfer charges.
- Keep large, repeated object delivery at the edge. CloudFront caches content at edge locations and, for AWS origins such as S3, the origin-to-CloudFront transfer is free; viewers still incur CloudFront request and edge egress charges. 16
- Use compression, caching, and appropriate payloads before scaling network links. A CDN is most useful when content is cacheable and viewers are distributed; it is not a universal substitute for an API-origin architecture.
- Use **Site-to-Site VPN** for encrypted connectivity over the internet where rapid setup and flexible, lower fixed commitment matter. Use **Direct Connect** for a dedicated, more consistent connection and high or predictable throughput; include port, provider, redundancy, and data-transfer charges. A VPN is not automatically cheaper when traffic volume is large, and Direct Connect is not justified for a small temporary connection.
- Use **VPC peering** for simple, direct VPC-to-VPC connectivity at small scale or specific paths. Use **Transit Gateway** as a hub-and-spoke option when many VPCs and networks would otherwise require a mesh; include attachment and data-processing charges and avoid unnecessary inspection or hairpin paths. Transit Gateway attachments require AZ-specific attachment subnets and route-table configuration. 19

6.6.3 Network architecture patterns

Public static site: S3 as an origin plus CloudFront, with caching and an appropriate DNS name. Avoid an always-on EC2 web tier when the content is static. Keep the bucket private and use the supported origin access mechanism when security is part of the requirement.

Private application with AWS-service access: private subnets for application instances, NAT only for destinations that require general egress, and gateway endpoints for S3/DynamoDB. Add interface endpoints for other supported AWS APIs when the volume and topology justify their cost.

Multi-AZ application: ALB across AZs, application capacity in each AZ, and a database/cache design that meets availability. Decide whether local NAT gateways are justified by traffic volume and failure isolation. The cost answer must include the reliability requirement.

Multi-VPC organization: Transit Gateway can simplify routes and centralize shared services, but direct peering may be less expensive for a small number of stable connections. At scale, compare route-management labor, attachments, inspection, inter-AZ paths, data processing, and failure domains.

6.7 Common pitfalls and distractors

- **"Cheapest per hour" is always correct.** It is wrong when the option violates availability, performance, durability, or recovery requirements.
- **Confusing storage price with total storage cost.** Retrieval, requests, minimum durations, snapshots, replication, and transfer can dominate.
- **Using One Zone-IA for the only copy of critical data.** Lower price comes with single-AZ resilience.
- **Choosing Glacier for data that must be read immediately.** Archive retrieval behavior and fees matter.

- **Overprovisioning EBS IOPS, DynamoDB capacity, EC2 size, or database replicas.** Select to measured demand and scale rather than peak capacity that is idle most of the time.
- **Buying commitments before measuring.** Savings Plans, RIs, and Reserved DB Instances lower committed usage cost but can be wasted when utilization or configuration changes.
- **Using Spot for non-interruptible state.** A low price does not compensate for interruption if the architecture cannot retry or recover.
- **Putting every private subnet behind one centralized NAT.** This may add cross-AZ transfer and a failure dependency. Conversely, one NAT per AZ may be unnecessary for very low traffic; compare cost and resilience.
- **Treating a VPC endpoint as free in every case.** Gateway endpoints for S3/DynamoDB have no additional endpoint charge, but interface endpoints have hourly and data-processing charges; standard service and transfer charges still apply.
- **Using CloudFront without cacheability.** Dynamic, uncacheable traffic can still incur request and data-transfer charges without much origin reduction.
- **Assuming a budget is a hard cap.** Alerts can lag and actions need configuration. Budgets are controls and signals, not a substitute for cleanup and governance.
- **Ignoring shared and untagged resources.** A tag strategy must include allocation rules for NAT, Transit Gateway, shared endpoints, logging, support, and other platform costs.
- **Using a cost-only answer for a performance or resilience question.** The official guide tests selecting the appropriate solution under requirements, not minimizing the bill at any expense.

6.8 A compact decision checklist

Before selecting an answer, ask:

1. What is the required outcome and what constraints are explicit?
2. Is the demand steady, bursty, unpredictable, or interruptible?
3. Is the data hot, infrequent, archival, recreatable, or shared concurrently?
4. What is the correct object, block, file, relational, key-value, container, or serverless interface?
5. Which capacity can scale, sleep, expire, or be deleted automatically?
6. Are requests, retrieval, I/O, snapshots, replicas, NAT processing, endpoints, cross-AZ transfer, cross-Region transfer, or licensed software significant?
7. Would a managed service reduce total cost of ownership even if its unit price is higher?
8. Is a one- or three-year commitment justified by measured, stable utilization?
9. Which flow should use a cache, CloudFront, gateway endpoint, interface endpoint, VPN, Direct Connect, peering, or Transit Gateway?
10. Which tool will prove the decision: Pricing Calculator before build, Cost Explorer for investigation, Budgets for alerts, CUR for detailed allocation?

6.9 References

7 AWS service decision matrix and cross-domain architecture patterns

This section is an exam-oriented decision aid, not a catalog of every AWS feature. The SAA-C03 guide says candidates are assessed on designing secure, resilient, high-performing, and cost-optimized solutions, and its service/concept list is explicitly non-exhaustive and subject to change.[1] The practical skill is to translate requirements into a small set of justified service choices, then check how those choices interact across compute, storage, data, networking, integration, security, operations, and recovery.

7.1 1. A repeatable service-selection method

Start with the workload rather than a favorite service. In a scenario, extract these requirements in order:

1. **Interface and execution model.** Is traffic synchronous HTTP/gRPC, asynchronous work, a stream, a scheduled job, or a file/object arrival? Is the workload short-lived and event-driven, long-running, stateful, or batch?
2. **Data model and access pattern.** Decide whether the primary data is object, block, shared file, relational, key-value/document, graph, time series, or a cache. Identify read/write ratio, transaction boundaries, query shape, ordering, and consistency.
3. **Scale and elasticity.** Identify steady versus spiky demand, horizontal versus vertical scaling, concurrency, burst absorption, and whether capacity can be provisioned ahead of time.
4. **Availability and failure scope.** Note the required RTO/RPO, single-AZ versus Regional failure assumptions, whether a standby must serve reads, and whether multi-Region active/active is actually required.
5. **Performance and locality.** Separate throughput, latency, connection count, and geographic proximity. A CDN improves content delivery; it does not make a transactional database globally writable.
6. **Security and connectivity.** Determine public/private exposure, identity model, encryption and key ownership, inspection requirements, on-premises paths, and cross-account boundaries.
7. **Operations and cost.** Prefer managed services when they meet requirements, but account for request, data-transfer, idle-capacity, NAT, replication, retrieval, and operational costs. A serverless label is not a guarantee of lowest cost.

This is the same style of tradeoff analysis encouraged by the AWS Well-Architected Framework: use the framework to understand the advantages and disadvantages of decisions, evaluate an architecture against the qualities expected of the workload, and improve it rather than treating review as an audit.[2] The six pillars are operational excellence, security, reliability, performance efficiency, cost optimization, and sustainability.[3]

7.1.1 Requirement-to-service matrix

Requirement signal	Usually start with	Why it fits	Re-check before choosing
Unpredictable short tasks triggered by events or HTTP	AWS Lambda	Per-invocation serverless execution	15-minute invocation limit, cold-start sensitivity,

Requirement signal	Usually start with	Why it fits	Re-check before choosing
		and native event integrations	concurrency, package/runtime constraints, VPC access, and downstream throttling 5 9
Long-running or custom-runtime container	Amazon ECS with AWS Fargate, or ECS/EKS on EC2	Container packaging, persistent process model, and more control than a function	Need for Kubernetes APIs, privileged access, GPUs, host networking, daemon workloads, predictable utilization, and image-start latency 4 5
Full OS control, specialized instance, licensed software, or legacy migration	Amazon EC2 with EC2 Auto Scaling	Broad instance/storage/network choices and control over guest OS	Patching, AMI lifecycle, placement, capacity, scaling signals, and operational burden 4
Durable object, data lake, static assets, backups, or global distribution origin	Amazon S3	API-accessed object storage with lifecycle and storage-class choices	Object semantics are not a POSIX file system; request patterns, consistency needs, versioning, replication, and retrieval charges 6 14
Low-latency durable block device for one host or compatible container task	Amazon EBS	Block device semantics and provisioned performance attached to compute	Attachment/topology, AZ scope, throughput/IOPS, snapshot/restore, and whether shared access is really needed 6
Shared NFS/SMB-style file access from many compute clients	Amazon EFS or an appropriate Amazon FSx service	Managed file semantics and shared read/write access	Protocol, performance mode, Windows/OpenZFS/

Requirement signal	Usually start with	Why it fits	Re-check before choosing
			ONTAP/Lustre needs, throughput, regional/AZ design, and cost 6
SQL schema, joins, ACID transactions, and existing relational engine	Amazon RDS or Amazon Aurora	Managed relational engines and managed HA/backup capabilities	Engine compatibility, write/read scaling, connection pressure, licensing, Multi-AZ behavior, and cross-Region DR 7 12 13
Massive key-value access with known access patterns and horizontal scale	Amazon DynamoDB	Managed distributed key-value model with single-digit-millisecond design target and global-table option	Partition-key distribution, item/access-pattern design, transaction/strong-read needs, indexes, hot keys, and eventual consistency 7 9
Frequently reused, ephemeral, low-latency data	Amazon ElastiCache	In-memory cache/session/leaderboard layer that reduces primary-store load	Cache invalidation, persistence/failover mode, memory limits, eviction, and whether loss is acceptable 7
Buffer work and let consumers scale independently	Amazon SQS	Durable pull queue, consumer-controlled rate, and decoupling	Visibility timeout, idempotency, DLQ, ordering/FIFO requirement, retry behavior, and backlog age 8 9
Push the same notification to many subscribers	Amazon SNS	Push-based pub/sub and fanout to endpoints or SQS/Lambda	Subscriber retry/slow-consumer isolation; use SNS-to-SQS when each consumer needs durable independent processing 8
Route business or AWS events by	Amazon EventBridge	Event bus, rule matching, content	At-least-once delivery,

Requirement signal	Usually start with	Why it fits	Re-check before choosing
content across many targets/accounts		filtering, and integration with AWS/SaaS sources	asynchronous/ eventual behavior, ordering/latency needs, schema evolution, and replay/retention requirements 8 9
Layer-7 HTTP routing, host/path rules, or WebSocket-capable application entry	Application Load Balancer or API Gateway	ALB routes to targets using application rules; API Gateway adds managed API front-door features	Authentication, throttling, transformation, private APIs, WebSocket/HTTP API needs, and cost/ latency 4 11
High-throughput TCP/UDP/TLS, static IP, or very low transport overhead	Network Load Balancer	Layer-4 connection distribution and static IP options	It does not replace application-aware routing or WAF-centric HTTP controls 4 11
Insert scalable virtual appliances such as firewalls or IDS/IPS	Gateway Load Balancer	Service insertion for third-party appliances	Appliance routing, asymmetric paths, inspection availability, and operational ownership 11
Cache HTTP content at edge locations	Amazon CloudFront	CDN caching, edge TLS/WAF integration, and origin protection	Cacheability, invalidation, dynamic request behavior, origin failover, and cache-key correctness 10 15
Improve global TCP/ UDP application path to Regional endpoints	AWS Global Accelerator	Anycast static IPs and network-path optimization to healthy Regional endpoints	It is not a cache; origin still serves requests and data-transfer cost/health routing still matter 10 11

Requirement signal	Usually start with	Why it fits	Re-check before choosing
Resolve names and steer users by health, latency, geography, or weighted policy	Amazon Route 53	DNS hosting, health checks, and routing policies	DNS caching/TTL means failover is not instantaneous; it does not proxy application traffic 10
Private multi-VPC/ enterprise network hub	AWS Transit Gateway	Centralized hub for VPC and VPN/DX attachments	Route domains, transitive routing, inter-Region design, inspection, cost, and blast radius 10
Private service exposure to selected consumers without network-wide routing	AWS PrivateLink	Interface endpoint/ service-provider model that keeps traffic private and limits exposure	Endpoint cost, AZ coverage, DNS, provider acceptance, and whether full bidirectional network reachability is required 10
Encrypted rapid hybrid connection	AWS Site-to-Site VPN	IPsec tunnels over the internet, quick to deploy and flexible	Internet variability, throughput, HA tunnels, route propagation, and encryption overhead 10
Predictable private hybrid bandwidth and latency	AWS Direct Connect	Dedicated connectivity into AWS	Provisioning lead time, redundancy, encryption (often add VPN), DX location/ provider, and routing design 10

7.2 2. Cross-domain decision matrix

7.2.1 Compute and execution

Amazon EC2 is the control-oriented choice: select an instance family and size, manage the guest operating system, and use EC2 Auto Scaling to add or remove instances. It is appropriate when the scenario names an OS requirement, custom kernel/agent, special instance family, sustained workload, host-level control, or a migration that cannot be refactored. Put instances behind an ELB and distribute them across multiple AZs; Auto Scaling should react to a meaningful workload metric, not merely a guess at CPU utilization.

The operational tradeoff is patching, AMIs, capacity planning, and more moving parts than managed runtimes.4

AWS Lambda is a function execution model, not a generic replacement for a server. It is strong for short, stateless, event-driven work, request transformation, scheduled jobs, and fanout processing. Standard invocations can run for at most 15 minutes; the current AWS decision guide also distinguishes Lambda's event-driven execution from Fargate's long-running container model.5 Keep state in a durable store such as S3, DynamoDB, or a database. When Lambda consumes SQS, design for retries and at-least-once effects: make handlers idempotent, use a DLQ or failure destination where appropriate, and prevent a fast producer from overwhelming a relational database. RDS Proxy can help absorb connection pressure from ephemeral consumers, but it does not remove database capacity limits.9

Amazon ECS is AWS-native container orchestration. Choose **AWS Fargate** when the application is already containerized, needs any language/runtime available in a container, benefits from task-level scaling, and the team wants AWS to manage the underlying servers. Choose ECS on **EC2** when sustained utilization, specialized instances, host access, privileged workloads, daemon processes, or cost control through instance commitment outweigh server-management overhead. Choose **Amazon EKS** when Kubernetes compatibility, its API/ecosystem, or portability is a hard requirement—not simply because it is available. Fargate has no hard application execution-time limit in the guide's comparison and is suited to long-running containers; Lambda is better for short, event-driven tasks.5

AWS Batch is a scheduler for queued batch jobs, not a front end for synchronous interactive requests. It fits rendering, simulation, scientific jobs, and large parallel workloads whose compute environment can scale down after the queue drains. **Elastic Beanstalk** accelerates conventional web application deployment while retaining an EC2-based runtime; it is useful when the scenario emphasizes application deployment simplicity rather than a function or container architecture.4

7.2.2 Storage and data lifecycle

Choose by **semantics first**. Block storage presents volumes to compute and is appropriate for filesystems, boot disks, and database I/O. File storage presents a shared hierarchy over a protocol such as NFS or SMB. Object storage presents objects through an API and is the default for durable, independently addressable content, data lakes, static assets, and backup/archive workflows.6

Amazon S3 is the general-purpose object layer for decoupled architectures: upload an object, emit an event, process it asynchronously, and store derived results. Use S3 versioning for recovery from overwrites/deletes, lifecycle rules to transition or expire data, and appropriate storage classes for access frequency. S3 Standard-IA and One Zone-IA are for long-lived infrequent access and have retrieval/minimum-duration considerations; One Zone-IA is cheaper but not resilient to loss of its AZ. Glacier classes require restore before access, so they are archive choices rather than primary online storage.14 Do not use S3 as if it were a low-latency POSIX filesystem or a transaction coordinator.

Amazon EBS is persistent block storage associated with compute topology. It is a fit for a database or filesystem needing low-latency block I/O from a compatible instance/task. Snapshots are useful for backup and replication workflows, but snapshot restore and volume initialization are not equivalent to a ready-to-serve multi-AZ database. **Amazon EFS** is shared elastic file storage for multiple clients. **Amazon FSx** is the family choice when the application needs a particular managed filesystem: Windows File Server, Lustre, OpenZFS, or NetApp ONTAP.6

For hybrid storage, **AWS Storage Gateway** provides cloud-backed storage access patterns from on premises; **AWS DataSync** moves files/objects online; and Snow Family devices are relevant when bandwidth or isolation makes an offline transfer more appropriate. The decision guide distinguishes online migration for frequently changing/time-critical data from offline transfer when bandwidth is constrained and data can be static in transit.6

7.2.3 Databases and caching

Use a **relational database** when the scenario requires ACID transaction boundaries, referential integrity, joins, a mature SQL engine, or an existing relational application. **Amazon RDS** provides managed common engines and **Aurora** provides AWS-optimized relational clusters. **Amazon Aurora** is still a relational choice; it does not become a NoSQL database merely because it scales differently. Select engine compatibility (such as MySQL or PostgreSQL), schema/query behavior, licensing, connection count, read/write ratio, and recovery objectives before comparing instance size.

Do not confuse RDS high availability with read scaling. A traditional RDS Multi-AZ DB instance deployment has a standby for failover that does **not** serve read traffic. A Multi-AZ DB cluster has two standby instances that can also serve read traffic.12 A read replica is for read scaling and may be promoted for recovery, but it is not the same mechanism or guarantee as synchronous Multi-AZ failover. Cross-Region replicas/global database features improve DR options, but add replication lag, regional cost, and application failover complexity.13

Use **DynamoDB** when the access patterns can be designed around a partition key and the workload needs managed horizontal scale, high request concurrency, and predictable key-value/document access. Design the key distribution before creating secondary indexes; a single hot key can defeat an otherwise scalable table. Use transactions or strongly consistent reads only when the business requirement justifies their cost/latency. DynamoDB global tables support multi-Region replication and active/active-style writes, but conflict handling and business semantics still belong in the design.7 13

Choose **Amazon ElastiCache** for a cache, session store, leaderboard, or other low-latency ephemeral layer. Treat the cache as disposable unless the selected mode explicitly provides the durability required. A cache is not a substitute for durable source-of-truth data, and the design must state invalidation, expiration, stampede protection, and behavior during cache loss.

For purpose-built workloads, the AWS database decision guide maps document databases to JSON-like documents, graph databases to relationship traversal, wide-column services to Cassandra-like high-throughput access, and time-series services to timestamp-oriented data. It also emphasizes that a workload may combine multiple database types—for example, relational order/inventory transactions, key-value high-volume reads, and a document catalog—rather than forcing one engine to satisfy incompatible access patterns.7

7.2.4 Networking, edge, and load balancing

A typical Regional application uses a **VPC** with public subnets for internet-facing entry points and private subnets for application/data tiers. Internet gateways provide internet routing for public subnets; a NAT gateway lets private-subnet resources initiate outbound IPv4 connections without accepting unsolicited inbound connections. NAT gateways are an availability and cost decision: place them for the required AZ failure model, and avoid routing every internal AWS API call through NAT when VPC endpoints can keep traffic private and reduce egress/NAT processing.

Use **Elastic Load Balancing** to distribute traffic across healthy targets in one or more AZs; ELB scales capacity and performs health checks.11 Select the type from the protocol and routing requirement:

- **ALB**: HTTP/HTTPS, host/path/header routing, target groups, and application-aware behavior. It is the normal choice for web APIs and microservices where rules matter.
- **NLB**: Layer-4 TCP/UDP/TLS, high connection/throughput needs, static IP requirements, or protocols that are not HTTP. It does not provide ALB-style path routing.
- **GWLB**: transparent service insertion for virtual network appliances such as firewalls or intrusion-prevention systems. It is not a generic application load balancer.

Amazon API Gateway is an API front door with API-specific authentication/authorization, throttling, stages, usage plans, and integration choices. It often fronts Lambda, but it can also front other HTTP integrations.18 **AWS AppSync** is the purpose-built managed GraphQL/API layer when GraphQL data aggregation and subscriptions are the requirement. Do not choose API Gateway simply because an application has HTTP; choose it when API-management functions are part of the requirement.

CloudFront caches and delivers content from edge locations, optionally protects origins with WAF, and can serve dynamic HTTP requests.16 **Global Accelerator** provides static anycast IPs and routes traffic over the AWS global network to healthy Regional endpoints; it does not cache content.17 Use Route 53 for DNS naming and policy-based routing, but remember that DNS responses are cached according to TTL and are not an instant request proxy.10 11

For multi-VPC design, use **Transit Gateway** as a Regional network hub when many VPCs and hybrid attachments need transitive routing. Use **VPC peering** for simpler point-to-point connectivity. Use **PrivateLink** when a consumer needs private access to a provider service without exposing the provider's whole network; its endpoint model gives consumers access to selected services without requiring an internet gateway, NAT, Direct Connect, or VPN.19 Use Transit Gateway/VPN/DX when the requirement is broader network reachability. For on-premises, Site-to-Site VPN is quick and encrypted over the internet; Direct Connect is a dedicated connectivity option with more predictable network characteristics and more provisioning/redundancy considerations.10

7.2.5 Integration, messaging, and workflows

A reliable service boundary usually combines the service that **routes** work with the service that **buffers** work:

- **SQS** is pull-based durable queuing. Consumers control their processing rate, so it is the default decoupling buffer for a worker pool. Design visibility timeout, retries, idempotency, dead-letter handling, and backlog alarms.
- **SNS** is push-based pub/sub. It is a good fanout source for notifications or for publishing one event to multiple subscriptions. When consumers must process independently and durably, subscribe SQS queues rather than pushing all work directly to a single synchronous endpoint.
- **EventBridge** is an event bus. Use it when content-based rule matching, AWS/SaaS event sources, cross-account routing, or loose producer/consumer coupling matters. It is not a FIFO task queue and should not be used to hide a requirement for strict ordering or durable work backlog.

AWS's current decision guide summarizes the distinction as SQS pull/persistence, SNS push/fanout, and EventBridge event-bus routing; all require consumers to tolerate the documented delivery semantics, commonly at-least-once.8 Lambda event-source mappings pull from queues/streams, whereas services such as S3 and API Gateway can push events directly.9

Use **AWS Step Functions** for explicit multi-step orchestration, retries, timeouts, branching, and compensation. A queue is a buffer; Step Functions is a workflow state machine. Do not embed a long synchronous chain of service calls in one Lambda when a workflow needs durable state and per-step retry. For a very low-latency synchronous requirement, asynchronous events may be the wrong pattern because network hops introduce variable latency and eventual consistency.9

7.3 3. Common cross-domain architectures

7.3.1 Pattern A: Highly available three-tier web application

Flow: Route 53 → CloudFront/WAF (if global/cacheable) → ALB in public subnets → EC2 Auto Scaling or ECS service across at least two AZs → RDS/Aurora Multi-AZ in private subnets. Store images/static assets in S3 and use ElastiCache for hot reads. Use NAT or VPC endpoints for required outbound access, CloudWatch for metrics/logs, and IAM roles rather than embedded credentials.

Why it works: each tier has an appropriate failure boundary and scale mechanism. ALB health checks remove bad targets; Auto Scaling adjusts compute; the database provides transactional durability and HA; S3 removes static content from application servers. **Pitfalls:** putting the database in a public subnet; using a Multi-AZ standby as a read replica; scaling only the web tier while the database or connection pool saturates; caching private/user-specific responses without a correct cache key; and assuming CloudFront removes all origin load for dynamic, uncached requests.11 12

7.3.2 Pattern B: Serverless API and event-driven processing

Flow: Route 53/CloudFront → API Gateway → Lambda → DynamoDB; S3 upload → EventBridge or SNS → SQS queue → Lambda/ECS worker; failures go to a DLQ and metrics alert on age/backlog. Use Step Functions for multi-step business workflows and CloudWatch/X-Ray for tracing and operational visibility.

Why it works: API handling, asynchronous processing, and persistence scale independently. The queue absorbs bursts and lets a slow downstream payment/reporting worker catch up without blocking the upload/API request. **Pitfalls:** synchronous Lambda-to-Lambda chains, unbounded concurrency against RDS, non-idempotent handlers, assuming event order, losing events because only a direct push was configured, and treating eventual consistency as immediate read-after-write across every component.8 9

7.3.3 Pattern C: Containerized microservices with selective platform control

Flow: CloudFront/ALB or API Gateway → ECS services on Fargate, with SQS/EventBridge for asynchronous interactions; ECR for images; DynamoDB/Aurora/ElastiCache selected per bounded context; S3 for objects. Use ECS on EC2 for specialized or consistently busy workloads and EKS only when Kubernetes is a firm requirement.

Why it works: services can scale and deploy independently, but the architecture remains operationally simpler than introducing Kubernetes without a need. **Pitfalls:** making every service synchronous; sharing one database schema and calling it microservices; using a distributed transaction without a saga/compensation strategy; storing container-local state; and ignoring per-service quotas, connection pools, and observability.4 5 15

7.3.4 Pattern D: Data lake and analytics path

Flow: producers/files → S3 landing zone → EventBridge/SQS or Kinesis for notifications/streaming → Glue catalog/ETL → Athena for ad-hoc SQL and/or EMR for distributed processing → Redshift for

governed warehouse workloads → QuickSight for dashboards. Apply Lake Formation/IAM/KMS controls and S3 lifecycle/storage classes.

Selection logic: S3 is the durable object foundation; Athena queries objects without a database server; Glue discovers/transforms/catalogs; Kinesis is for managed streaming ingestion; EMR is for managed big-data frameworks and deeper cluster control; Redshift is for repeated warehouse analytics and joins at warehouse scale. **Pitfalls:** using a relational OLTP database as an analytics lake; querying millions of tiny unpartitioned objects; forgetting schema evolution and data-quality controls; putting sensitive data in an unrestricted bucket; and treating streaming ingestion as an automatic substitute for batch where the business only needs periodic reports.1 6

7.3.5 Pattern E: Multi-account and hub-and-spoke foundation

Flow: AWS Organizations/Control Tower-style account separation → shared networking/security/logging accounts → Transit Gateway attachments for VPCs → centralized inspection with GWLB/firewall where required → centralized CloudTrail/Config/security findings → workload accounts with least-privilege IAM roles and KMS key policies.

Why it works: account boundaries reduce blast radius and separate billing/security ownership; a hub simplifies routing; centralized controls improve auditability. **Pitfalls:** assuming Transit Gateway automatically solves overlapping CIDR blocks; placing all workloads in one account for convenience; granting administrator access instead of using roles/SCP guardrails; centralizing inspection without designing return paths; and using a network hub when PrivateLink would provide narrower service-level exposure.10

7.3.6 Pattern F: Hybrid application and migration

Flow: on-premises network → redundant Direct Connect (often with VPN backup) → Transit Gateway → private VPC application/data tiers; use DataSync, DMS, Storage Gateway, or Snow Family as the data-movement tool that matches volume, change rate, and bandwidth.

Selection logic: rehost when time/risk dominates; replatform when a managed service such as RDS removes operational work without changing the application contract; refactor when cloud-native elasticity and decoupling justify redevelopment; retain/retire/repurchase/relocate when business constraints dictate. The supplied supplementary deck presents the seven migration strategies and emphasizes Site-to-Site VPN and Direct Connect for hybrid connectivity; use current AWS documentation to validate implementation details.10

Pitfalls: treating Direct Connect as automatically encrypted; using one circuit/tunnel as a single point of failure; moving a database before sizing replication and cutover; and assuming a VPN has the same predictable throughput or latency as dedicated connectivity.10

7.3.7 Pattern G: Regional or multi-Region disaster recovery

Choose DR from business-defined RTO/RPO rather than from a product name:

- **Backup and restore:** lowest standing cost; restore time and data loss depend on backup frequency and automation.
- **Pilot light:** core data/replication is ready, application capacity is started during recovery.
- **Warm standby:** a scaled-down but running environment reduces recovery time at higher cost.
- **Active/active:** traffic and writes run in multiple Regions; highest complexity and usually the highest cost, but can provide very low recovery time and near-zero data loss when the data model supports it.

For database choices, AWS guidance maps RDS cross-Region read replicas to pilot-light/warm-standby patterns, Aurora Global Database to hot/warm standby options, and DynamoDB global tables to active/active-style replication with sub-second RPO in the guidance table.[13] These are reference characteristics, not a universal SLA: validate Region/service support, automation, failover routing, replication lag, conflict behavior, and application correctness. A backup is not a DR architecture until restore, DNS/traffic switch, secrets, dependencies, and validation are tested.

7.4 4. Security and operations are cross-cutting, not a final layer

- **Identity:** IAM users are not an application credential store. Use roles, temporary credentials, federation, least privilege, resource policies where supported, and separate administrative/workload roles. Cross-account access commonly uses a role assumption path.
- **Data protection:** encrypt in transit with TLS and at rest with service encryption; use AWS KMS when key policy, audit, or customer-control requirements matter. Keep key administration separate from data administration where the threat model requires it. Store database/API credentials in Secrets Manager or another purpose-built secret store, not AMIs or source code.
- **Network/application protection:** security groups are stateful instance/ENI controls; network ACLs are subnet-level stateless controls. WAF filters web requests and common application attacks; Shield addresses DDoS protection. Put the control at the traffic layer that matches the threat; a security group is not a SQL-injection control.
- **Observability:** CloudWatch metrics/logs/alarms reveal saturation and failure; CloudTrail records API activity; Config evaluates resource configuration; X-Ray helps trace distributed requests. Alarms should reflect user impact and queue/database saturation, not just host CPU.
- **Resilience:** remove single points of failure, deploy across AZs, health-check targets, automate replacement, back up state, and test recovery. Managed service does not mean application failover is automatic: clients need retry/backoff, timeout, connection refresh, and idempotency logic.

7.5 5. Scenario reasoning and exam pitfalls

1. **"The workload is spiky and runs for seconds."** Prefer Lambda, but check concurrency and downstream capacity. If it needs a custom container or can run for hours, compare Fargate/Batch instead.[5]
2. **"A single queue consumer is too slow during bursts."** Increase consumer parallelism or scale a worker fleet from queue depth/age; do not simply increase web-server size. Add a DLQ and idempotency.
3. **"Users worldwide need lower latency for images and videos."** Put objects in S3 and deliver through CloudFront. Global Accelerator is not the first answer because it does not cache content.[16] [17] [14]
4. **"A TCP service needs static IPs and high connection rates."** Consider NLB. An ALB is selected for HTTP-aware routing, not merely because it is a load balancer.[11]
5. **"A private service must be consumed by many VPCs without exposing all networks."** PrivateLink is narrower than full Transit Gateway connectivity. If the requirement is transitive access to many networks, evaluate Transit Gateway.[19] [10]

6. **“The database must survive an AZ failure and scale reads.”** Use the correct combination: Multi-AZ for failover, read replicas or a Multi-AZ cluster/Aurora read endpoint for read scaling, and a cache for hot/reusable reads. Do not call the ordinary standby a read replica.12
7. **“The object workload has unknown access patterns.”** S3 Intelligent-Tiering is a candidate; if objects are infrequently accessed and retrieval latency can be minutes or hours, Glacier classes may be more cost-effective. Include retrieval and minimum-duration costs.14
8. **“The application requires shared files from multiple instances.”** Choose EFS/FSx based on protocol and filesystem feature needs, not EBS on one instance. EBS is block storage and is not a general shared NFS service.6
9. **“The question says strict ordering and exactly-once business effect.”** Do not infer this from EventBridge. Evaluate SQS FIFO/SNS FIFO where supported, and still make consumers idempotent because business-level exactly-once effects require application design.8
10. **“The requirement is 99.99% availability.”** Ask what failure scope is included, then use multi-AZ targets, health checks, automated replacement, and tested recovery. A multi-Region architecture may be unjustified if the stated RTO/RPO can be met within one Region.
11. **“The source data is on premises and bandwidth is limited.”** Compare DataSync/online transfer with Snow Family/offline transfer. The correct choice depends on change rate, downtime, security, and bandwidth—not simply data size.6
12. **“The service is managed, so security is automatic.”** Managed services reduce undifferentiated infrastructure work; customers still configure IAM, network exposure, encryption, logging, backup, retention, and application controls.2

7.5.1 Fast elimination checklist

Before selecting an answer, eliminate options that:

- violate the stated execution-time, protocol, data-model, or consistency requirement;
- put a required private tier on a public route or expose a database directly to the internet;
- use a standby for read scaling, a cache as durable truth, a CDN for uncachable transactions, or a queue for workflow state;
- introduce an expensive multi-Region or always-on fleet when the requirement only calls for Regional HA or burst capacity;
- ignore retries, duplicate delivery, DLQs, hot keys, connection exhaustion, NAT/DX redundancy, or AZ placement;
- solve a local problem with a broader network/security exposure than needed; or
- assert a numerical limit without checking the current service documentation.

7.6 References

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8 Scenario reasoning, exam traps, and practice-question blueprint

This section is a reasoning playbook, not a service catalog. The SAA-C03 exam asks the candidate to select the **best architecture for stated requirements**, not merely a service that can work. AWS describes the

exam as validating design against the AWS Well-Architected Framework and the ability to design secure, resilient, high-performing, and cost-optimized solutions. The current guide describes 65 total questions (50 scored and 15 unscored), multiple-choice and multiple-response formats, no penalty for guessing, and four scored domains weighted 30% security, 26% resilience, 24% performance, and 20% cost optimization. 1

8.1 1. The governing idea: requirements beat service familiarity

AWS Well-Architected is a decision framework: it is meant to expose the pros and cons of architectural decisions and provide a consistent way to identify improvements. It is not an audit checklist with one universal design. 2 A question may therefore have several technically possible answers, while only one best satisfies the stated constraints.

A reliable reading order is:

1. **Extract the hard constraints.** Mark words such as *must*, *cannot*, *only*, *within*, *without*, *automatically*, *minimum cost*, *no operational overhead*, *single-digit millisecond*, *cross-Region*, *private*, *ordered*, *durable*, *replay*, *exactly once*, *RTO*, and *RPO*.
2. **Classify the workload.** Is it synchronous or asynchronous? Stateful or stateless? Read-heavy or write-heavy? Predictable or spiky? Interactive, batch, streaming, or archival? Internet-facing, private, hybrid, or edge?
3. **Translate the requirements into AWS primitives.** Examples: a queue for buffering, a topic for fan-out, an event bus for content-based routing, object storage for durable blobs, a load balancer for target health and distribution, a read replica for read scaling, or a Multi-AZ deployment for failover.
4. **Eliminate violations before comparing advantages.** A low-cost answer is wrong if it cannot meet RTO. A highly available answer is wrong if it violates a private-network requirement. A fast answer is wrong if it loses ordering or durability.
5. **Use the smallest managed design that meets every requirement.** Favor managed services and automation when the prompt emphasizes reduced administration, but do not add components that the requirement does not justify.
6. **Resolve the final tie with the requested priority.** "Most cost-effective" is not the same as "lowest price"; "highest availability" is not the same as "good enough"; "least operational effort" is not the same as "most configurable."

A useful scratchpad has five lines: **data**, **traffic**, **failure**, **security**, and **economics**. Fill each line with one or two facts from the stem. This prevents a vivid but secondary detail from dominating the decision.

8.1.1 Requirement words and the design signal they usually imply

- **"Decouple," "buffer," "retry," or "consumer may be temporarily unavailable"** → a durable queue, commonly Amazon SQS, rather than a direct synchronous call.
- **"Notify many independent subscribers"** → Amazon SNS fan-out, often SNS to multiple SQS queues when each consumer needs independent buffering and retry.
- **"Route events based on content, SaaS events, or cross-account event sharing"** → Amazon EventBridge.
- **"Private subnets need outbound internet updates, but the internet must not initiate connections"** → a public NAT gateway in a public subnet, with private-subnet routes through it. A NAT gateway does not make private instances internet-addressable. 5

- **"Private access to S3 or DynamoDB without NAT"** → a gateway VPC endpoint; for other endpoint services, an interface endpoint or another PrivateLink endpoint type may be appropriate. 19
- **"Host/path/header routing for HTTP"** → Application Load Balancer (ALB), which operates at Layer 7 and routes through listener rules to healthy target groups. 6
- **"TCP/UDP, static IP, or Layer-4 behavior"** → investigate Network Load Balancer rather than selecting ALB by default; the exam guide explicitly includes load-balancer selection as a design skill. 1
- **"Failover between Regions"** → identify whether the required mechanism is DNS failover, CloudFront origin failover, Global Accelerator, database replication, or a complete DR strategy. These are not interchangeable.
- **"No schema, key-value access at scale"** → DynamoDB; then check partition-key distribution, consistency, and capacity mode rather than assuming a relational database.
- **"Relational transactions, joins, or an existing relational engine"** → RDS or Aurora, then decide Multi-AZ, read replicas, engine, and migration path.
- **"RTO/RPO"** → select a recovery strategy, not just a backup product. Recovery time and recovery point are business requirements that constrain architecture.

8.2 2. Common architectures and how to reason through them

8.2.1 2.1 Highly available three-tier web application

A recurring pattern is Route 53 → CloudFront or an ALB → stateless application instances/tasks across at least two Availability Zones → a Multi-AZ relational database or a purpose-built NoSQL store. Static assets can be served from S3 through CloudFront. The application tier should avoid local state so Auto Scaling can replace instances and distribute traffic. An Auto Scaling group maintains desired capacity, replaces unhealthy instances, and can distribute instances across specified Availability Zones. 14

Reasoning checkpoints:

- **Availability:** two AZs are not automatically enough; ask whether the database, cache, load balancer targets, and dependencies also span failure domains.
- **Security:** public entry points should not imply public database access. Keep data stores private and use security-group references or tightly scoped rules.
- **Scaling:** scale the bottleneck. Add read replicas or caching for read pressure; add queue workers for asynchronous work; do not scale every tier because the web tier is busy.
- **State:** store sessions in a shared service such as a database or cache if session affinity would prevent flexible scaling. Sticky sessions may solve a narrow problem but often reduce resilience and load distribution.
- **Cost:** CloudFront and caching may reduce origin load and data-transfer cost, but caching dynamic data requires explicit cache-key and invalidation correctness.

8.2.2 2.2 Private application and data tiers

A typical private architecture places the ALB in public subnets, application tasks or instances in private subnets, and the database in isolated/private subnets. The route table, security group, network ACL, DNS, and endpoint choices all have to agree. A security group is a stateful virtual firewall attached to resources; response traffic for an allowed connection is automatically allowed. A network ACL is a stateless, subnet-level filter with ordered inbound and outbound rules, so return traffic must be allowed separately. 3 4

The question often tests one missing dependency: an instance can have a correct security group and still fail because the subnet route table has no route, the NACL blocks the ephemeral response port, DNS is wrong, or the private subnet cannot reach a required AWS API. If the prompt asks to remove NAT cost for S3/DynamoDB access, a gateway endpoint is usually the relevant direction; a NAT gateway is a general IPv4 egress service, not a private path to every service.

8.2.3 2.3 Decoupled order or job processing

For a bursty producer and slower worker, put a durable queue between them. Consumers poll the queue and can scale independently. Amazon SQS is pull-based and persists messages until consumed or expired; standard queues favor high throughput with at-least-once delivery, while FIFO queues provide ordering and deduplication semantics with different throughput and design constraints. 10

For fan-out, publish once to SNS and subscribe multiple queues or endpoints. Give each queue its own consumer scaling, retry, visibility timeout, and dead-letter handling. For event-driven routing, use EventBridge rules to match event content and route to targets. The key trap is treating all three as generic "messaging": SQS is a queue and buffer, SNS is push-based pub/sub, and EventBridge is an event bus with rule-based routing.

A worker must be idempotent because at-least-once delivery can invoke it more than once. The visibility timeout hides a received message temporarily; it is not a transaction lock that guarantees completion. Extend or tune it for long jobs, and use a dead-letter queue or failure destination where appropriate. Do not claim exactly-once end-to-end processing merely because one component offers FIFO or deduplication.

8.2.4 2.4 Global content delivery and traffic steering

CloudFront caches content at edge locations and can use an origin group with a primary and secondary origin. On a cache miss, CloudFront sends the request to the primary and fails over only for configured failure status codes or connection/timeout conditions. Its documented origin failover applies to `GET`, `HEAD`, and `OPTIONS`, not arbitrary write methods such as `POST` and `PUT`. 8 This is a classic trap: "CloudFront origin failover" is not a general database write failover mechanism.

Route 53 selects among DNS answers using a routing policy. Simple, failover, latency, geolocation, geoproximity, IP-based, multivalue, and weighted policies solve different problems. Latency routing chooses the Region with the best measured latency; weighted routing shifts a specified proportion; failover is active-passive when paired with health checks. 7 DNS answers are cached according to TTL, so DNS failover is not the same as an instantaneous connection switch.

Use Global Accelerator when the requirement emphasizes static anycast IPs, rapid global traffic steering, or preserving a performant network path to regional endpoints. Do not choose it merely because the word "global" appears; CloudFront is the stronger clue for cacheable HTTP content.

8.2.5 2.5 Relational database resilience and read scaling

A standard RDS Multi-AZ DB instance deployment has a standby for failover; that standby does not serve read traffic. The current RDS documentation also describes Multi-AZ DB cluster deployments in which standby instances can serve reads, so read the wording carefully rather than applying an old blanket rule to every Multi-AZ variant. 11

Use the following decision sequence:

- **Need automatic failover and resilience, not more reads?** Multi-AZ is the first direction.

- **Need read scaling?** Read replicas, Aurora reader instances, caching, or a read-optimized data model may fit; a classic Multi-AZ standby is not the read-scaling answer.
- **Need a managed relational engine with high read throughput and cross-Region capabilities?** Investigate Aurora and its replication options, but verify the exact latency, failover, and cost requirement.
- **Need flexible schema and key-value/document access at very high scale?** Consider DynamoDB instead of forcing a relational design.

A read replica is not a backup substitute. Replication lag, promotion behavior, application connection handling, and write/read consistency matter. A question that says "read-heavy reporting must not affect the primary" points toward a read replica or separate analytics path; "database must survive an AZ failure with automatic failover" points toward Multi-AZ.

8.2.6 2.6 DynamoDB access-pattern design

DynamoDB questions are often data-model questions disguised as service-selection questions. The partition key must distribute traffic; a single hot key can throttle a table even when the aggregate table capacity appears sufficient. A sort key supports ordered/range access within a partition. Avoid answering with a scan when the requirement asks for predictable, scalable keyed queries.

Eventually consistent reads are the default and cost less than strongly consistent reads. Strongly consistent reads are available for tables and local secondary indexes, but not global secondary indexes or streams. 12 On-demand capacity removes capacity planning and charges per request; provisioned capacity suits a predictable workload where capacity can be forecast and cost must be governed. 13

Do not infer that "NoSQL" means "no durability," "no transactions," or "no consistency choices." Instead, map the required access pattern and consistency boundary to table/index design. If a prompt requires multi-Region active-active writes, examine global-table consistency and conflict behavior; do not substitute a read replica or a simple cross-Region copy.

8.2.7 2.7 Disaster recovery: match strategy to RTO, RPO, and budget

AWS groups DR approaches from lower-cost backup and restore through pilot light, warm standby, and multi-site active-active. Backup and restore requires restoring data and redeploying infrastructure; infrastructure as code reduces recovery errors. Pilot light keeps core data-replication infrastructure available while application capacity is reduced or not running. Warm standby keeps a scaled-down but functional environment ready. Multi-site active-active runs production capacity in more than one Region and is the most complex/costly option. 15

Reasoning rules:

- **RPO** asks how much data loss is acceptable. Frequent or continuous replication is needed for a small RPO.
- **RTO** asks how quickly service must return. A backup that still requires infrastructure deployment and data restoration cannot meet a near-zero RTO.
- **Region failure** requires more than an AZ-spread deployment. Check DNS, application artifacts, secrets, quotas, data replication, and a tested promotion/failover procedure.
- **Human error or corruption** is not solved by replication alone. Versioning, point-in-time recovery, immutable backups, and restore testing address different failure modes.
- **Failover must be operable.** The whitepaper cautions that recovery designs should minimize dependence on control-plane actions during the outage and should be tested regularly. 15

The trap is selecting the strongest DR pattern when the question asks for cost optimization, or selecting backup and restore when the stated RTO requires a warm environment. State the tradeoff explicitly in practice answers.

8.2.8 2.8 Hybrid and migration scenarios

The supplementary slides are useful for recognizing the migration vocabulary: retain, relocate, rehost, replatform, repurchase, refactor, and retire. They also emphasize the distinction between migration and hybrid operation. Treat those slides as study orientation, not as the authority for current service names or limits; the current AWS exam guide and service documentation take precedence.

For a hybrid scenario, first determine the traffic direction and reliability requirement. Site-to-Site VPN is commonly selected for encrypted connectivity over the internet and faster initial setup; Direct Connect provides a dedicated network connection with more predictable performance but normally needs separate resilience planning. PrivateLink is a private service-consumer/provider pattern, not a general replacement for VPC routing or a site-to-site network. 13

For migration:

- **Rehost:** move with minimal application change when speed and low transformation effort dominate.
- **Replatform:** keep the basic architecture but adopt a managed platform such as RDS.
- **Refactor:** redesign around cloud-native services when long-term agility or elasticity justifies the investment.
- **Data transfer:** choose online transfer when bandwidth and duration are acceptable; choose an offline/edge device only when the current, supported AWS offering and data-volume constraints justify it.
- **Database migration:** identify homogeneous versus heterogeneous engine changes, ongoing replication needs, cutover downtime, and schema-conversion work.

A migration question often hides a business constraint: “minimal downtime” points to continuous replication and a planned cutover; “no code changes” points away from refactoring; “data must remain on-premises” points toward hybrid or an edge/on-premises service rather than a public-Region-only design.

8.3 3. Security and networking traps that eliminate otherwise plausible answers

8.3.1 Security groups versus network ACLs

- Security groups are **stateful**, resource-attached allow rules. They do not use deny rules in the normal model.
- NACLs are **stateless**, subnet-level allow/deny rules evaluated in rule-number order. Return traffic needs an explicit matching rule.
- A security group allowing port 443 does not overcome a route-table or NACL failure.
- Opening SSH/RDP to `0.0.0.0/0` is generally a security anti-pattern unless the stem explicitly requires unrestricted access; AWS recommends restricting administrative source ranges. 3

8.3.2 Public subnet versus public resource

A subnet is “public” because its route table has a route to an internet gateway; an instance is internet-reachable only if it also has an appropriate public address and security path. A private subnet can have outbound internet access through NAT without accepting unsolicited inbound connections. Do not call an

instance public solely because it is in a subnet labeled public, and do not put a database in a public subnet to let an application reach it.

8.3.3 NAT gateway versus VPC endpoint

NAT is for outbound translation and costs per gateway-hour and data processing. A gateway endpoint provides private access for supported services such as S3 and DynamoDB, without sending that traffic through NAT. Interface endpoints use private network interfaces and DNS for supported endpoint services. The best answer depends on the destination and whether general internet egress is required.

8.3.4 IAM least privilege and credential type

AWS recommends temporary credentials with IAM roles for workloads, federation and temporary credentials for human users, MFA where root or IAM-user access is required, and permissions narrowed toward least privilege. 9 Common distractors include embedding long-lived access keys in application code, using the root user for routine work, granting `AdministratorAccess` to an application, or treating an SCP as a permission grant. An SCP sets an organization-level maximum; it does not itself grant a principal permission.

When a question asks for cross-account access, determine which side owns the resource, whether a resource policy is supported, and whether an IAM role with a trust policy is the intended mechanism. "Same account" and "cross account" are often the actual discriminator.

8.4 4. Compute, storage, and performance traps

8.4.1 Lambda, Fargate, and EC2

Lambda is designed for short-lived, stateless invocations; the current documentation lists a 15-minute maximum for a standard invocation and notes that memory allocation also controls proportional CPU. 16 A long-running process, custom host-level control, persistent local state, or a workload that exceeds Lambda's execution model points toward containers or EC2.

Fargate runs ECS tasks without managing the underlying servers. Each task has an isolation boundary, and Fargate services can use ALB, NLB, or GWLB with IP targets when using `awsvpc`. 17 Choose Fargate when containers are needed but host management should be minimized; choose EC2-backed ECS when host customization, specialized capacity, or economics make it worthwhile. Do not choose Lambda merely because "serverless" appears in the stem if the application is a continuously running container.

8.4.2 Object, block, and file storage

- **S3:** object storage for durable, API-accessed objects and static content; choose storage class from access pattern, latency, retrieval, resilience, and cost.
- **EBS:** block storage attached to compute; use for a server's filesystem or database volume when block semantics are required.
- **EFS:** managed shared file storage for multiple clients needing a file-system interface.
- **S3 Standard-IA, One Zone-IA, Glacier classes, and Intelligent-Tiering:** these are not interchangeable. S3 documentation emphasizes access pattern, retrieval behavior, minimum-duration or retrieval charges, Availability Zone placement, and durability/availability differences. 18

A frequent trap is choosing Glacier for data that must be read immediately, choosing One Zone-IA for data that cannot tolerate an AZ loss, or choosing EBS when multiple independent instances need concurrent

shared files. Another is using versioning as a backup without considering delete markers, retention, replication, and restore testing.

8.4.3 Caching is a consistency decision

Caching can reduce latency and database load, but it creates freshness and invalidation requirements. Before selecting ElastiCache or a CDN, identify the cache key, TTL, invalidation mechanism, session ownership, and behavior after an origin or cache failure. “Lowest latency” alone is not enough if users must see an immediately committed value.

8.5 5. Exam traps and how to reject them

1. **“Most secure” distractor:** an answer may add a WAF, KMS, or private subnet but fail the actual requirement such as cross-account authorization, least privilege, or encryption in transit.
2. **“Highly available” distractor:** two EC2 instances in one AZ are not AZ resilience; a Multi-AZ database standby is not necessarily a read replica; a backup in the same Region is not regional DR.
3. **“Serverless” distractor:** serverless does not mean unlimited, synchronous, stateful, or free. Check Lambda duration, concurrency, payload, connection, and retry behavior when relevant. Cite quotas only when the question makes them decisive; quotas can change.
4. **“Auto Scaling” distractor:** Auto Scaling changes capacity; it does not make a stateful application stateless, repair a bad data model, or replace a queue.
5. **“Read replica” distractor:** replication for read scale is not automatic failover, and replica lag can violate read-after-write expectations.
6. **“Multi-AZ” distractor:** Multi-AZ is a resilience/failover concept; do not use it as a generic answer to every performance problem.
7. **“Cross-Region” distractor:** copying data cross-Region does not automatically copy routes, IAM, secrets, AMIs, quotas, DNS, or application configuration.
8. **“Private” distractor:** a private subnet still needs routes to reach AWS services, on-premises networks, or the internet. A private IP does not by itself provide service connectivity.
9. **“Exactly once” distractor:** at-least-once delivery plus idempotent processing is the normal safe design. Ordering, deduplication, and end-to-end business effects are separate properties.
10. **“Lowest cost” distractor:** a single NAT gateway across AZs can be cheaper but may create an AZ dependency and cross-AZ traffic; the exam guide explicitly calls out the choice between a single shared NAT gateway and one per AZ as a cost/resilience tradeoff. 1
11. **Service-name drift:** older study material may use retired names or old limits. Validate current names and behaviors in AWS documentation. The exam guide itself says its service list is non-exhaustive and subject to change. 1
12. **Overengineering:** a distractor may use a more sophisticated architecture that meets the requirements but violates “least operational effort” or “most cost-effective.” Prefer the simplest design that satisfies every explicit constraint.

8.6 6. Scenario-solving algorithm for timed questions

Use this seven-pass method; it is intentionally mechanical.

8.6.1 Pass 1 — Identify the verb

Is the question asking to **select**, **improve**, **reduce cost**, **increase availability**, **minimize administration**, **migrate**, **secure**, or **troubleshoot**? The verb sets the optimization target.

8.6.2 Pass 2 — Extract the non-negotiables

Write down one line each for security, availability, performance, operations, cost, and data semantics. If a line is absent, do not invent a requirement.

8.6.3 Pass 3 — Determine the failure boundary

Is the failure an instance, AZ, Region, service dependency, network path, operator error, data corruption event, or traffic spike? Choose redundancy at the same or larger boundary.

8.6.4 Pass 4 — Determine the data contract

Ask whether data needs object, block, file, relational, key-value, stream, queue, or event semantics. Then ask about durability, ordering, replay, consistency, and retention.

8.6.5 Pass 5 — Choose the managed building block

Prefer the AWS service whose native behavior directly answers the requirement. Do not assemble EC2-based infrastructure to imitate a managed feature unless the stem requires custom control.

8.6.6 Pass 6 — Test the distractors

For each option, ask: **What exact phrase in the stem does this fail?** This is faster and safer than comparing every feature.

8.6.7 Pass 7 — Re-read the objective phrase

Before submitting, re-read the final clause: "with the least operational overhead," "at the lowest cost," "without changing the application," "with automatic failover," or "with the fewest changes." Many questions are won by this last phrase.

8.7 7. Practice-question blueprint

The blueprint should measure reasoning, not memorization. A practice bank can mirror the official scored-domain weights while acknowledging that an individual exam form is not a fixed count by domain. For a 20-item internal set, a reasonable proportional allocation is approximately 6 security, 5 resilience, 5 performance, and 4 cost items; rebalance if the bank has a particular learning objective.

8.7.1 Blueprint dimensions

Each item should specify:

- **Primary domain:** security, resilience, performance, or cost optimization.
- **Scenario archetype:** web tier, private networking, database, storage, messaging, serverless, global delivery, DR, hybrid, or migration.
- **Requirement tension:** cost versus resilience, latency versus consistency, operational effort versus control, or availability versus complexity.
- **Response type:** one best answer or two-or-more correct answers. For multiple response, state how many answers are required in the answer key, but do not make "all of the above" the test.
- **Evidence clue:** the exact phrase that should drive the answer.
- **Distractor rationale:** which plausible but incorrect assumption the distractor exploits.
- **Confidence check:** one reason the selected answer meets all hard constraints.

8.7.2 Recommended item mix

- **Security:** least-privilege IAM and roles; public/private subnet controls; security group versus NACL; encryption and key policy; cross-account/resource policy; private service access.
- **Resilience:** Multi-AZ versus read replicas; stateless scaling; queue retries and DLQs; Route 53 failover; CloudFront origin failover; backup/restore versus pilot light/warm standby; regional dependencies.
- **Performance:** ALB versus NLB; CloudFront versus Global Accelerator; cache placement; DynamoDB partition keys and consistency; S3/EBS/EFS selection; Lambda memory/concurrency; Fargate versus EC2; decoupling for independent scaling.
- **Cost optimization:** storage class/lifecycle selection; NAT versus endpoints; Savings Plans/Spot where interruption is acceptable; single versus per-AZ NAT with resilience caveat; provisioned versus on-demand DynamoDB; right-sizing and managed-service tradeoffs.

8.7.3 Item-writing templates

1. **Best architecture:** "A workload has [traffic pattern], [failure requirement], and [security constraint]. Which design best satisfies all three with the least operational effort?"
2. **Two correct controls:** "A private application must [requirement]. Which two changes are required?" Combine one network control with one identity/data control.
3. **Troubleshooting:** "The application can reach the load balancer but not the database. Which two checks should be performed first?" Test route, security group, NACL, DNS, and target health without making every option correct.
4. **Tradeoff selection:** "The business accepts [RPO/RTO] and wants the lowest ongoing cost. Which DR strategy is appropriate?"
5. **Migration:** "An existing relational database must move with minimal downtime and a heterogeneous engine target. Which migration sequence fits?" Test replication, schema conversion, and cutover rather than naming a generic transfer service.
6. **Data semantics:** "Messages must be processed in order for each customer, but customers can process independently. Which queue and message-group design fits?"
7. **Performance diagnosis:** "Read latency is high while writes are acceptable. Which change improves reads without changing the primary write path?" Test caching/read replicas/index/access-pattern choices.
8. **Cost diagnosis:** "Instances in private subnets need S3 access but no general internet access. Which change removes unnecessary NAT processing?" Test gateway endpoint and route-table knowledge.
9. **Multi-response security:** "Select two actions that reduce credential exposure for an EC2 workload." Correct answers should involve an instance role/temporary credentials and least-privilege policy; long-lived access keys should be a distractor.
10. **Global routing:** "Users must reach the lowest-latency healthy Region, but the application is not cacheable." Test Route 53 latency routing or Global Accelerator rather than CloudFront caching.

8.7.4 Answer-key format

For each practice item, record:

- **Correct option(s).**
- **Requirement mapping:** quote or paraphrase the decisive stem facts.
- **Why it works:** name the AWS behavior, not just the service.

- **Why each distractor fails:** one sentence per option, tied to a violated requirement.
- **Trap label:** e.g., "Multi-AZ is not read scaling," "NACL is stateless," "NAT is outbound only," or "CloudFront failover is not write failover."
- **Confidence and uncertainty:** identify any fact that should be checked against current documentation rather than memorized as a number.

8.7.5 Scoring and remediation

A correct answer without a requirement explanation is weak evidence of mastery. Score review in three layers:

1. **Selection:** Was the answer correct?
2. **Reasoning:** Did the explanation identify the hard constraint and AWS behavior?
3. **Transfer:** Can the learner solve the same pattern when the service names or presentation order changes?

Remediate by error type, not by service name. "Chose a read replica for failover" is a resilience-concept error; "forgot the NAT route" is a network-path error; "selected Glacier for immediate reads" is an access-pattern error. Reattempt a new scenario after explaining the trap, rather than repeating the same memorized question.

8.8 8. Final exam-day checklist

Before choosing an answer, ask:

- What is the **one non-negotiable requirement**?
- Which failure boundary matters: instance, AZ, Region, network, or data?
- Is the workload synchronous, asynchronous, or event-driven?
- What are the data semantics: object/file/block, relational/key-value, ordering, consistency, replay?
- Does the answer actually provide private connectivity, or only private IP addressing?
- Is the proposed redundancy for **availability**, **read scaling**, **backup**, or **regional DR**?
- Does "serverless" remove servers only, or does it also change duration, state, retry, and quota behavior?
- Which answer meets the requirement with the least operational effort and no unnecessary components?
- What exact phrase makes each distractor wrong?

The goal is not to recognize a familiar diagram. It is to show that the selected architecture follows from the stated business requirements, AWS service behavior, and Well-Architected tradeoffs. The official guide explicitly frames the test around design and improvement rather than a comprehensive list of every possible service, so practice should focus on transferable reasoning patterns and current documentation checks. 1

8.9 References

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9 Deep-dive service chapters

9.1 IAM, IAM Identity Center, Organizations, and SCPs

Use temporary credentials for people and workloads whenever possible. Separate authentication, authorization, and governance: IAM Identity Center or federation helps establish workforce identity;

permission sets and IAM policies authorize; Organizations, OUs, SCPs, and Control Tower establish account-level guardrails.

An IAM role has a trust policy and permissions policies. Cross-account access generally requires a role in the trusting account plus permission for the caller to assume it. A permissions boundary or SCP limits maximum permissions but does not grant access. Explicit denies win.

Exam method: identify the principal, target resource, account boundary, session type, and policy layer. If the requirement says central workforce access across many accounts, favor IAM Identity Center. If it says prevent a member account from disabling logging or using an unapproved Region, favor an SCP. If it says an application needs AWS access, use an execution role rather than an access key embedded in code.

9.1.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.2 VPC networking and security controls

A VPC design is the interaction of CIDR ranges, subnets, route tables, gateways or endpoints, security groups, NACLs, DNS, and service policies. A public subnet has a route to an Internet Gateway; a private subnet may still have egress through NAT; an isolated subnet has no route to external networks.

Security groups are stateful and attached to ENIs. NACLs are stateless, subnet-level, ordered filters. Route tables decide where traffic goes before the destination security controls and service policies decide whether it is accepted. Use tier-specific security groups and references between tiers.

For private AWS API access, use an S3/DynamoDB gateway endpoint or an interface endpoint for supported services. For general outbound Internet access from private IPv4 resources, use NAT. For enterprise network connectivity, compare VPN, Direct Connect, Transit Gateway, peering, and PrivateLink by encryption, routing scope, throughput, and operational complexity.

9.2.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.3 S3, EBS, EFS, FSx, and Storage Gateway

S3 is API-based object storage for static content, backup, archives, logs, and data lakes. EBS is AZ-scoped block storage for a host or compatible task. EFS is shared elastic NFS storage. FSx is a family of managed file systems selected for protocol or workload requirements such as Windows File Server, Lustre, OpenZFS, or ONTAP.

S3 lifecycle, versioning, replication, encryption, object ownership, access points, and storage classes must be selected together. Intelligent-Tiering suits uncertain access. IA and Glacier classes trade lower

storage cost for retrieval and minimum-duration considerations. One Zone classes are not the default for irreplaceable data.

Use DataSync for online file/object movement and scheduled synchronization. Use Storage Gateway when on-premises applications need a persistent cloud-backed file, volume, or tape interface. Use Snow Family only when physical transfer or edge compute is justified by bandwidth, isolation, or location constraints.

9.3.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.4 EC2, Auto Scaling, ELB, Lambda, ECS, EKS, Fargate, Batch, and EMR

Choose EC2 for OS and host control, specialized hardware, legacy compatibility, or persistent processes. Choose Lambda for short, event-driven, stateless tasks. Choose ECS/Fargate for long-running containers with lower host management; choose ECS on EC2 for host-level control or steady high utilization; choose EKS when Kubernetes compatibility is a hard requirement.

Elastic Load Balancing distributes traffic and health-checks targets. ALB is Layer 7 HTTP-aware; NLB is Layer 4 TCP/UDP/TLS-oriented; GWLB inserts virtual appliances. Auto Scaling should use demand-relevant metrics such as request count, latency, or queue depth, not just CPU by habit.

Batch schedules queued jobs and can scale compute down after the queue drains. EMR fits distributed data-processing frameworks. For every elastic design, check quotas, startup time, warm-up, downstream capacity, idempotency, and the persistence location for state.

9.4.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.5 RDS, Aurora, DynamoDB, ElastiCache, and database patterns

Use RDS or Aurora for SQL, joins, ACID transactions, existing relational engines, and referential integrity. Use DynamoDB for key-value/document access patterns designed around partition keys and high horizontal scale. Use ElastiCache for hot, ephemeral, low-latency data, sessions, or read reduction.

RDS Multi-AZ is primarily writer availability and failover in the classic DB-instance model. Read replicas are for read scaling and may lag. Aurora Replicas are read-capable and can be promoted. Do not answer a read-scaling requirement with a classic standby.

DynamoDB design begins with access patterns: partition-key distribution, sort-key range access, indexes, consistency, capacity mode, and hot-key avoidance. A cache is not a source of truth unless the design

explicitly provides the required durability. RDS Proxy can help connection pressure but cannot make an undersized database unlimited.

9.5.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.6 SQS, SNS, EventBridge, Step Functions, and API Gateway

SQS is a durable pull queue and load-leveling buffer. SNS is push-based pub/sub and fanout. EventBridge is rule-based event routing across many sources and targets. Use SNS to multiple SQS queues when consumers need independent retry and backlog handling.

At-least-once delivery means consumers must be idempotent. Configure visibility timeout, dead-letter handling, retention, and alarms for backlog age. FIFO behavior addresses ordering and deduplication needs but introduces throughput and design tradeoffs.

Step Functions provides explicit workflow state, retries, timeouts, branching, and auditability. API Gateway is a managed API front door with throttling and integrations. Do not use a queue as a workflow state machine or keep long-running work in a synchronous HTTP request when asynchronous processing is acceptable.

9.6.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.7 Route 53, CloudFront, Global Accelerator, and traffic steering

Route 53 resolves names and applies routing policies such as simple, weighted, latency, geolocation, geoproximity, multivalue, and failover. DNS answers are cached by resolvers, so failover is not instantaneous. CloudFront caches HTTP content near viewers and can protect origins.

Global Accelerator provides static anycast IPs and traffic steering for Regional endpoints without being a cache. Use it for global TCP/UDP or dynamic applications where path optimization and rapid endpoint changes matter. Use CloudFront when cacheable content and edge delivery are the dominant requirement. CloudFront origin failover is not a database write-failover mechanism. Analyze the request method, cacheability, origin health signals, and data replication separately.

9.7.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.8 KMS, ACM, Secrets Manager, Cognito, WAF, Shield, GuardDuty, Macie, CloudTrail, and Config

KMS manages encryption keys and policy-controlled cryptographic operations. Encryption does not authorize access; IAM, key policies, resource policies, and application permissions still matter. ACM manages certificates for supported AWS integrations and is Regional except for CloudFront certificates, which use us-east-1.

Secrets Manager stores and rotates application secrets. Cognito user pools provide application identity and tokens; identity pools can exchange identity for temporary AWS credentials. WAF filters web requests, Shield helps DDoS protection, GuardDuty detects threats, Macie discovers sensitive S3 data, CloudTrail records API activity, and Config evaluates resource configuration and compliance.

Use layered controls: identity, network, application edge, encryption, secret handling, logging, detection, and response. The shared responsibility model determines who patches or configures each layer.

9.8.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.9 Migration and hybrid architecture

Use the seven Rs as a decision vocabulary: retain, retire, relocate, rehost, repurchase, replatform, and refactor. Rehost minimizes application change; replatform makes targeted managed-service changes; refactor changes the architecture for cloud-native benefits.

DMS migrates and continuously replicates databases; SCT helps convert schemas for heterogeneous migrations. Application Migration Service supports lift-and-shift server migration with continuous block replication. DataSync moves file/object data. Storage Gateway provides ongoing hybrid storage access. VPN is faster to establish over the Internet; Direct Connect is dedicated private connectivity with planning and redundancy requirements.

Migration questions often hide a downtime or consistency requirement. Match the service to data type, source/target, cutover method, bandwidth, change rate, and rollback plan. Treat the supplied migration slides as supplementary and verify current service names and retirement notices against AWS documentation.

9.9.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

9.10 Cost tools, pricing models, and FinOps controls

Cost Explorer investigates historical usage and trends. Budgets alerts on actual or forecast spend and commitment coverage. CUR delivers detailed line items to S3 for queryable allocation and chargeback. Pricing Calculator estimates an architecture before deployment.

On-Demand is flexible; Savings Plans commit to spend with flexibility; Reserved Instances fit stable configurations; Spot fits interruptible work. Storage classes and lifecycle rules must include retrieval, request, transition, minimum-duration, replication, and transfer charges.

Cost optimization means meeting the required outcome at the lowest total workload cost. Include management effort, idle capacity, NAT processing, cross-AZ and cross-Region transfer, public IPv4, endpoint hours, and support or license costs. Never sacrifice an explicit availability, durability, security, or RTO requirement solely to lower the bill.

9.10.1 Rapid review

What problem does it solve? Identify the workload constraint before naming the service. **What does it replace?** State the simpler or more operationally expensive alternative. **What can go wrong?** Check failure scope, consistency, quota, policy, and cost. **What is the exam clue?** Rewrite the scenario in one requirement sentence.

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10 Hands-on lab and review checklists

10.1 Lab 1: Build a secure three-tier VPC

Objective. Create public, private, and isolated subnets across two AZs. Configure route tables, an internet-facing ALB, private application targets, and a private database. Verify security-group references and test that the database has no public path.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.

Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.

Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.2 Lab 2: Compare NAT and VPC endpoints

Objective. From a private workload, access S3 through a gateway endpoint and compare the route path with general Internet egress through NAT. Review endpoint policies, DNS, route-table associations, and cost implications.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.

Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.

Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.3 Lab 3: Cross-account role access

Objective. Create a role in a second account with a trust policy and narrow permissions. Assume it from the first account. Add an explicit deny and observe policy evaluation.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.4 Lab 4: S3 lifecycle and recovery

Objective. Create versioned objects, lifecycle transitions, noncurrent-version rules, and a recovery test. Record how retention and retrieval requirements change the storage class choice.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.5 Lab 5: EC2 Auto Scaling and ALB

Objective. Deploy a stateless application behind an ALB and ASG across two AZs. Scale on request count or CPU, terminate an instance, and observe replacement and target health.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.6 Lab 6: SQS worker with DLQ

Objective. Publish jobs to SQS, process them with a worker, make the handler idempotent, force a failure, and observe visibility timeout and dead-letter behavior.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.7 Lab 7: SNS fanout

Objective. Publish one event to an SNS topic with multiple SQS subscribers. Stop one consumer and verify that its queue retains independent backlog without blocking other consumers.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.8 Lab 8: EventBridge routing

Objective. Create an event rule that routes only matching events to a target. Test filtering, retries, and duplicate-safe processing.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.9 Lab 9: DynamoDB access-pattern design

Objective. Write down query patterns first, then choose a partition and sort key. Test a hot-key anti-pattern, add an index only when needed, and compare eventual and strong reads.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.10 Lab 10: RDS availability versus read scaling

Objective. Create a test RDS deployment and a read replica. Document which endpoint serves writes, what fails over, and what scales reads. Do not infer behavior from the word Multi-AZ alone.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.

Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.

Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.11 Lab 11: CloudFront cache behavior

Objective. Serve an S3 object through CloudFront. Test cache hits, invalidation, TTLs, cache keys, and origin behavior. Explain why dynamic writes are not solved by CDN caching.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.

Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.

Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.12 Lab 12: Route 53 routing policies

Objective. Model weighted, latency, and failover policies with health checks. Record DNS TTL effects and explain why DNS failover is not an instantaneous connection switch.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.

Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.

Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.13 Lab 13: KMS and Secrets Manager

Objective. Encrypt an object or secret, inspect key policy and IAM policy, rotate the secret, and prove that encryption and authorization are separate controls.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.

Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.

Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.14 Lab 14: CloudTrail and Config evidence

Objective. Enable an audit trail and a configuration rule. Generate an API event and a configuration drift event. Identify which service answers which investigation question.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.15 Lab 15: Hybrid connectivity comparison

Objective. Draw a VPN and Direct Connect design with redundant paths. List encryption, lead time, routing, failure, bandwidth, and cost tradeoffs.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.16 Lab 16: Migration decision workshop

Objective. For a legacy application, compare retain, rehost, replatform, repurchase, refactor, relocate, and retire. Select a strategy based on downtime, skills, risk, and business value.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.17 Lab 17: DR tabletop

Objective. Given four RTO/RPO pairs, select backup and restore, pilot light, warm standby, or active/active. Include DNS, data replication, secrets, quotas, artifacts, and runbook testing.

Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.18 Lab 18: Cost investigation

Objective. Tag resources, activate cost allocation tags, inspect Cost Explorer, create a budget, and describe when CUR is required. Identify NAT, transfer, snapshots, retrieval, and idle-capacity costs.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.19 Lab 19: Well-Architected review

Objective. Review an architecture using all six pillars. Record one risk, one measurable question, one recommended improvement, and one verification test per pillar.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

10.20 Lab 20: Timed scenario set

Objective. Complete 65 mixed questions in 130 minutes. Mark uncertain items, answer every item, and classify each miss by data, traffic, failure, security, economics, or service semantics.
Before you start. Define the expected result, failure boundary, security boundary, and cost boundary.
Evidence to capture. Save a small architecture sketch, the key configuration choice, one test result, and one tradeoff you would explain in an exam answer.
Review questions. What requirement made this design appropriate? What simpler design would fail? What is the single most likely distractor? How would the answer change if the failure scope expanded from an AZ to a Region?

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11 Practice question bank and answer rationales

11.1 Question 1

Scenario. A company needs workers to absorb bursts while producers return immediately. Workers may be temporarily unavailable. Which design is best?

Answer. Amazon SQS between producers and workers, with visibility timeout, idempotent consumers, and a DLQ.

Why. SQS is a durable pull queue and load-leveling buffer. A direct call is tightly coupled; SNS is fanout; EventBridge is content-based event routing.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.2 Question 2

Scenario. A private EC2 fleet must access S3 without NAT processing charges. Which option is best?
Answer. An S3 gateway VPC endpoint associated with the private subnet route tables.
Why. Gateway endpoints provide private regional access to S3 without the NAT path. An interface endpoint may be needed for other topology requirements.
Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.3 Question 3

Scenario. A web application must survive an AZ failure with automatic database failover. It does not need more read capacity. Which option is best?

Answer. A Multi-AZ relational database deployment with application compute and load balancing across multiple AZs.

Why. The requirement is writer availability and AZ fault tolerance, not read scaling. A read replica alone is the wrong primary mechanism.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.4 Question 4

Scenario. A global application serves cacheable images with low viewer latency. Which service should be added?

Answer. Amazon CloudFront with an appropriate origin and cache policy.

Why. CloudFront is the CDN for cacheable HTTP content. Global Accelerator is for network-path steering, not edge caching.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.5 Question 5

Scenario. A team needs workforce console access across many AWS accounts using an existing corporate directory. Which design is best?

Answer. Federate the directory with IAM Identity Center and assign permission sets to groups across accounts.

Why. This centralizes authentication and account access. Duplicating IAM users in each account creates lifecycle and credential-management overhead.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.6 Question 6

Scenario. A workload has relational joins, ACID transactions, and an existing PostgreSQL application. Which direction is best?

Answer. Amazon RDS for PostgreSQL or Aurora PostgreSQL, selected after checking compatibility, resilience, performance, and cost requirements.

Why. DynamoDB is not a drop-in answer for relational joins and transaction semantics.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.7 Question 7

Scenario. A workload is unpredictable, short-lived, and event-driven. The team wants no server management. Which compute is the best first choice?

Answer. AWS Lambda, provided the invocation duration, concurrency, runtime, and downstream behavior fit.

Why. Lambda is event-driven and serverless. Fargate is better for long-running containers; EC2 adds more operations.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.8 Question 8

Scenario. An application needs HTTP host- and path-based routing to healthy targets. Which load balancer is the best fit?

Answer. Application Load Balancer.

Why. ALB operates at Layer 7 and supports application-aware listener rules. NLB is the Layer-4 choice for TCP/UDP/TLS behavior.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.9 Question 9

Scenario. A compliance archive is rarely accessed and can take hours to restore. Which storage strategy is best?

Answer. An S3 Glacier archival class with lifecycle and retention rules sized to the business policy.

Why. The access latency and low frequency point to archival storage. Include retrieval and minimum-duration costs.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.10 Question 10

Scenario. A third-party SaaS provider needs limited cross-account access without long-lived keys. Which design is best?

Answer. A dedicated cross-account IAM role with a narrow permissions policy and an ExternalId condition in the trust policy.

Why. Temporary role credentials reduce key exposure, and ExternalId helps prevent confused-deputy access.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.11 Question 11

Scenario. A team wants read scaling for a relational database while preserving a Multi-AZ writer failover design. Which option should be considered?

Answer. Read replicas or Aurora reader instances, with application read routing and lag considerations.

Why. A classic Multi-AZ standby is for failover, not read scaling.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.12 Question 12

Scenario. A process has branching, retries, timeouts, parallel steps, and audit requirements. Which service best models it?

Answer. AWS Step Functions.

Why. Step Functions makes workflow state and retry behavior explicit. A queue alone is not a workflow state machine.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.13 Question 13

Scenario. A private subnet needs outbound package downloads, but inbound Internet connections must not be allowed. Which design is best?

Answer. A NAT gateway in a public subnet with a private-subnet default route to the NAT gateway, plus restrictive security controls.

Why. NAT supports outbound-initiated connections; it does not assign public reachability to private instances.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.14 Question 14

Scenario. A business needs rapid recovery from a Regional outage with small RTO/RPO and accepts higher operating cost. Which direction is most appropriate?

Answer. A multi-Region strategy with replicated data, pre-provisioned capacity, health-based routing, and tested failover.

Why. Multi-AZ protects against AZ faults; a Region outage requires a Regional recovery design. The exact active/passive or active/active pattern depends on consistency and cost.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.15 Question 15

Scenario. A workload has steady, predictable EC2 usage for three years and stable instance configuration. Which pricing model should be evaluated first?

Answer. Reserved Instances or a suitable Savings Plan, after validating utilization and flexibility needs.

Why. Commitments reduce unit cost when utilization is reliable. On-Demand is flexible but typically costs more for steady long-lived use.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.16 Question 16

Scenario. An application needs shared NFS file semantics across compute in multiple AZs. Which service is the best starting point?

Answer. Amazon EFS, or a specific FSx service if the protocol or workload requires it.

Why. S3 is object storage and EBS is block storage; neither is a general shared NFS file system.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.17 Question 17

Scenario. A system emits one event to many independent consumers, each with its own retry and backlog. Which pattern is best?

Answer. SNS topic fanout to a separate SQS queue for each consumer.

Why. SNS provides fanout and each queue isolates consumer failure and scaling.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.18 Question 18

Scenario. A team needs detailed line-item cost allocation in an analytics pipeline. Which tool is the best fit?
Answer. AWS Cost and Usage Report delivered to Amazon S3 for querying.
Why. CUR is the detailed, durable report. Cost Explorer is for interactive analysis; Budgets is for alerting.
Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.19 Question 19

Scenario. A service needs to expose a private endpoint to selected consumer VPCs without full network reachability. Which service should be considered?

Answer. AWS PrivateLink.

Why. PrivateLink exposes a service through interface endpoints and limits the access path. Transit Gateway provides broader network connectivity.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.20 Question 20

Scenario. A serverless function consumes SQS and can be retried after a timeout. What is the key application requirement?

Answer. The handler must be idempotent and delete the message only after durable processing succeeds.

Why. SQS commonly provides at-least-once delivery; retries and duplicates are expected design conditions.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.21 Question 21

Scenario. A company needs workers to absorb bursts while producers return immediately. Workers may be temporarily unavailable. Which design is best? Consider also the stated operational simplicity and cost constraint.

Answer. Amazon SQS between producers and workers, with visibility timeout, idempotent consumers, and a DLQ.

Why. SQS is a durable pull queue and load-leveling buffer. A direct call is tightly coupled; SNS is fanout; EventBridge is content-based event routing.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.22 Question 22

Scenario. A private EC2 fleet must access S3 without NAT processing charges. Which option is best? Consider also the stated operational simplicity and cost constraint.

Answer. An S3 gateway VPC endpoint associated with the private subnet route tables.

Why. Gateway endpoints provide private regional access to S3 without the NAT path. An interface endpoint may be needed for other topology requirements.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.23 Question 23

Scenario. A web application must survive an AZ failure with automatic database failover. It does not need more read capacity. Which option is best? Consider also the stated operational simplicity and cost constraint.

Answer. A Multi-AZ relational database deployment with application compute and load balancing across multiple AZs.

Why. The requirement is writer availability and AZ fault tolerance, not read scaling. A read replica alone is the wrong primary mechanism.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.24 Question 24

Scenario. A global application serves cacheable images with low viewer latency. Which service should be added? Consider also the stated operational simplicity and cost constraint.

Answer. Amazon CloudFront with an appropriate origin and cache policy.

Why. CloudFront is the CDN for cacheable HTTP content. Global Accelerator is for network-path steering, not edge caching.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.25 Question 25

Scenario. A team needs workforce console access across many AWS accounts using an existing corporate directory. Which design is best? Consider also the stated operational simplicity and cost constraint.

Answer. Federate the directory with IAM Identity Center and assign permission sets to groups across accounts.

Why. This centralizes authentication and account access. Duplicating IAM users in each account creates lifecycle and credential-management overhead.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.26 Question 26

Scenario. A workload has relational joins, ACID transactions, and an existing PostgreSQL application. Which direction is best? Consider also the stated operational simplicity and cost constraint.

Answer. Amazon RDS for PostgreSQL or Aurora PostgreSQL, selected after checking compatibility, resilience, performance, and cost requirements.

Why. DynamoDB is not a drop-in answer for relational joins and transaction semantics.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

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Scenario. A workload is unpredictable, short-lived, and event-driven. The team wants no server management. Which compute is the best first choice? Consider also the stated operational simplicity and cost constraint.

Answer. AWS Lambda, provided the invocation duration, concurrency, runtime, and downstream behavior fit.

Why. Lambda is event-driven and serverless. Fargate is better for long-running containers; EC2 adds more operations.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.28 Question 28

Scenario. An application needs HTTP host- and path-based routing to healthy targets. Which load balancer is the best fit? Consider also the stated operational simplicity and cost constraint.

Answer. Application Load Balancer.

Why. ALB operates at Layer 7 and supports application-aware listener rules. NLB is the Layer-4 choice for TCP/UDP/TLS behavior.

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11.29 Question 29

Scenario. A compliance archive is rarely accessed and can take hours to restore. Which storage strategy is best? Consider also the stated operational simplicity and cost constraint.

Answer. An S3 Glacier archival class with lifecycle and retention rules sized to the business policy.

Why. The access latency and low frequency point to archival storage. Include retrieval and minimum-duration costs.

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11.30 Question 30

Scenario. A third-party SaaS provider needs limited cross-account access without long-lived keys. Which design is best? Consider also the stated operational simplicity and cost constraint.

Answer. A dedicated cross-account IAM role with a narrow permissions policy and an ExternalId condition in the trust policy.

Why. Temporary role credentials reduce key exposure, and ExternalId helps prevent confused-deputy access.

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Answer. Read replicas or Aurora reader instances, with application read routing and lag considerations.

Why. A classic Multi-AZ standby is for failover, not read scaling.

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Scenario. A private subnet needs outbound package downloads, but inbound Internet connections must not be allowed. Which design is best? Consider also the stated operational simplicity and cost constraint.
Answer. A NAT gateway in a public subnet with a private-subnet default route to the NAT gateway, plus restrictive security controls.
Why. NAT supports outbound-initiated connections; it does not assign public reachability to private instances.
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Scenario. A business needs rapid recovery from a Regional outage with small RTO/RPO and accepts higher operating cost. Which direction is most appropriate? Consider also the stated operational simplicity and cost constraint.

Answer. A multi-Region strategy with replicated data, pre-provisioned capacity, health-based routing, and tested failover.

Why. Multi-AZ protects against AZ faults; a Region outage requires a Regional recovery design. The exact active/passive or active/active pattern depends on consistency and cost.

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Answer. A Multi-AZ relational database deployment with application compute and load balancing across multiple AZs.

Why. The requirement is writer availability and AZ fault tolerance, not read scaling. A read replica alone is the wrong primary mechanism.

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Scenario. A global application serves cacheable images with low viewer latency. Which service should be added? Consider also the stated operational simplicity and cost constraint.

Answer. Amazon CloudFront with an appropriate origin and cache policy.

Why. CloudFront is the CDN for cacheable HTTP content. Global Accelerator is for network-path steering, not edge caching.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.45 Question 45

Scenario. A team needs workforce console access across many AWS accounts using an existing corporate directory. Which design is best? Consider also the stated operational simplicity and cost constraint.

Answer. Federate the directory with IAM Identity Center and assign permission sets to groups across accounts.

Why. This centralizes authentication and account access. Duplicating IAM users in each account creates lifecycle and credential-management overhead.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

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Answer. Amazon RDS for PostgreSQL or Aurora PostgreSQL, selected after checking compatibility, resilience, performance, and cost requirements.

Why. DynamoDB is not a drop-in answer for relational joins and transaction semantics.

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Answer. AWS Lambda, provided the invocation duration, concurrency, runtime, and downstream behavior fit.

Why. Lambda is event-driven and serverless. Fargate is better for long-running containers; EC2 adds more operations.

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Scenario. An application needs HTTP host- and path-based routing to healthy targets. Which load balancer is the best fit? Consider also the stated operational simplicity and cost constraint.

Answer. Application Load Balancer.

Why. ALB operates at Layer 7 and supports application-aware listener rules. NLB is the Layer-4 choice for TCP/UDP/TLS behavior.

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11.49 Question 49

Scenario. A compliance archive is rarely accessed and can take hours to restore. Which storage strategy is best? Consider also the stated operational simplicity and cost constraint.

Answer. An S3 Glacier archival class with lifecycle and retention rules sized to the business policy.

Why. The access latency and low frequency point to archival storage. Include retrieval and minimum-duration costs.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.50 Question 50

Scenario. A third-party SaaS provider needs limited cross-account access without long-lived keys. Which design is best? Consider also the stated operational simplicity and cost constraint.

Answer. A dedicated cross-account IAM role with a narrow permissions policy and an ExternalId condition in the trust policy.

Why. Temporary role credentials reduce key exposure, and ExternalId helps prevent confused-deputy access.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

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Scenario. A team wants read scaling for a relational database while preserving a Multi-AZ writer failover design. Which option should be considered? Consider also the stated operational simplicity and cost constraint.

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Why. A classic Multi-AZ standby is for failover, not read scaling.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.52 Question 52

Scenario. A process has branching, retries, timeouts, parallel steps, and audit requirements. Which service best models it? Consider also the stated operational simplicity and cost constraint.

Answer. AWS Step Functions.

Why. Step Functions makes workflow state and retry behavior explicit. A queue alone is not a workflow state machine.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

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Scenario. A private subnet needs outbound package downloads, but inbound Internet connections must not be allowed. Which design is best? Consider also the stated operational simplicity and cost constraint.
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Why. NAT supports outbound-initiated connections; it does not assign public reachability to private instances.
Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.54 Question 54

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Why. PrivateLink exposes a service through interface endpoints and limits the access path. Transit Gateway provides broader network connectivity.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

11.120 Question 120

Scenario. A serverless function consumes SQS and can be retried after a timeout. What is the key application requirement? Consider also the stated operational simplicity and cost constraint.

Answer. The handler must be idempotent and delete the message only after durable processing succeeds.

Why. SQS commonly provides at-least-once delivery; retries and duplicates are expected design conditions.

Exam habit. Identify the hard constraint first, eliminate answers that violate it, then choose the smallest managed design that satisfies the remaining requirements.

12 Flashcards and final revision sheets

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Answer: It limits the maximum permissions available in an account or OU; it does not grant permissions.

12.0.2 Flashcard 2

Prompt: Security group versus NACL?

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12.0.3 Flashcard 3

Prompt: Authentication versus authorization?

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Prompt: RTO versus RPO?

Answer: RTO is acceptable recovery time; RPO is acceptable data-loss interval.

12.0.5 Flashcard 5

Prompt: Multi-AZ versus multi-Region?

Answer: Multi-AZ is usually the simpler AZ-failure design; multi-Region adds Regional protection, latency, or residency benefits at greater complexity.

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Prompt: SQS versus SNS versus EventBridge?

Answer: SQS is pull queue/buffer; SNS is push pub/sub fanout; EventBridge is rule-based event routing.

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Answer: CloudFront caches HTTP content at edges; Global Accelerator steers network traffic to healthy Regional endpoints.

12.0.9 Flashcard 9

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Answer: Classic Multi-AZ is writer failover; read replicas are primarily for read scaling and may lag.

12.0.10 Flashcard 10

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Answer: S3 is object; EBS is block; EFS is shared elastic file storage.

12.0.11 Flashcard 11

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Answer: Short-lived, event-driven, stateless work that fits runtime and concurrency constraints.

12.0.12 Flashcard 12

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12.0.177 Flashcard 177

Prompt: What is least privilege?

Answer: Grant only the actions, resources, and conditions needed for the task.

12.0.178 Flashcard 178

Prompt: What is defense in depth?

Answer: Use multiple independent layers such as identity, network, application, encryption, logging, and detection.

12.0.179 Flashcard 179

Prompt: What is a dead-letter queue?

Answer: A destination for messages that repeatedly fail processing so they can be isolated and investigated.

12.0.180 Flashcard 180

Prompt: What is a read replica?

Answer: An asynchronously replicated database copy used primarily to scale reads or support recovery workflows.

12.0.181 Flashcard 181

Prompt: What does an SCP do?

Answer: It limits the maximum permissions available in an account or OU; it does not grant permissions.

12.0.182 Flashcard 182

Prompt: Security group versus NACL?

Answer: Security groups are stateful and ENI/resource-oriented; NACLs are stateless, subnet-level, ordered filters.

12.0.183 Flashcard 183

Prompt: Authentication versus authorization?

Answer: Authentication identifies the principal; authorization determines allowed actions on resources.

12.0.184 Flashcard 184

Prompt: RTO versus RPO?

Answer: RTO is acceptable recovery time; RPO is acceptable data-loss interval.

12.0.185 Flashcard 185

Prompt: Multi-AZ versus multi-Region?

Answer: Multi-AZ is usually the simpler AZ-failure design; multi-Region adds Regional protection, latency, or residency benefits at greater complexity.

12.0.186 Flashcard 186

Prompt: SQS versus SNS versus EventBridge?

Answer: SQS is pull queue/buffer; SNS is push pub/sub fanout; EventBridge is rule-based event routing.

12.0.187 Flashcard 187

Prompt: ALB versus NLB?

Answer: ALB is Layer 7 HTTP-aware routing; NLB is Layer 4 TCP/UDP/TLS-oriented distribution.

12.0.188 Flashcard 188

Prompt: CloudFront versus Global Accelerator?

Answer: CloudFront caches HTTP content at edges; Global Accelerator steers network traffic to healthy Regional endpoints.

12.0.189 Flashcard 189

Prompt: RDS Multi-AZ versus read replica?

Answer: Classic Multi-AZ is writer failover; read replicas are primarily for read scaling and may lag.

12.0.190 Flashcard 190

Prompt: S3 versus EBS versus EFS?

Answer: S3 is object; EBS is block; EFS is shared elastic file storage.

12.0.191 Flashcard 191

Prompt: When use Lambda?

Answer: Short-lived, event-driven, stateless work that fits runtime and concurrency constraints.

12.0.192 Flashcard 192

Prompt: When use Fargate?

Answer: Long-running container tasks when the team wants to avoid managing EC2 worker hosts.

12.0.193 Flashcard 193

Prompt: What is a gateway endpoint?

Answer: A private VPC route for supported services such as S3 and DynamoDB, without NAT.

12.0.194 Flashcard 194

Prompt: What is a NAT gateway?

Answer: A managed outbound translation service allowing private resources to initiate Internet connections.

12.0.195 Flashcard 195

Prompt: Why idempotency?

Answer: At-least-once delivery and retries can repeat effects; idempotent processing makes duplicates safe.

12.0.196 Flashcard 196

Prompt: Cost Explorer versus Budgets versus CUR?

Answer: Explore, alert, and detailed line-item analysis respectively.

12.0.197 Flashcard 197

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12.0.200 Flashcard 200

Prompt: What is a read replica?

Answer: An asynchronously replicated database copy used primarily to scale reads or support recovery workflows.

13 Final exam-week checklist

- Re-read the current AWS SAA-C03 exam guide and confirm the exam mechanics, domain weights, and in-scope terminology.
- Practice requirement extraction: data, traffic, failure, security, economics.
- Be able to explain the difference between every common pair: Multi-AZ/read replica, SQS/SNS/EventBridge, ALB/NLB, CloudFront/Global Accelerator, NAT/gateway endpoint/interface endpoint, RDS/Aurora/DynamoDB, backup/pilot light/warm standby/active-active.
- Complete at least one timed 65-question set. Leave time to answer every question.
- For every missed question, write the violated requirement and the distractor logic.
- Avoid memorizing volatile prices and quotas without checking the current documentation.

- Prefer the simplest managed design that satisfies every explicit requirement.

14 References

The chapter reports contain inline citations and reference lists. The primary sources used include the official AWS SAA-C03 exam guide, AWS Certification domain pages, AWS Well-Architected Framework pillars, IAM and Organizations documentation, Amazon VPC and PrivateLink documentation, Amazon S3/EBS/EFS/FSx documentation, EC2/Auto Scaling/ELB/Lambda/ECS/EKS/Fargate documentation, RDS/Aurora/DynamoDB/ElastiCache documentation, SQS/SNS/EventBridge/Step Functions/API Gateway documentation, Route 53/CloudFront/Global Accelerator documentation, AWS security services documentation, AWS migration and hybrid-services documentation, and AWS cost-management documentation.